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CHEMISTRY • UPPER SIXTH | UPPER SIXTH — CHEMISTRY

CHAPTER 1

EQUILIBRIA

SUGGESTED TIME: 33 PERIODS

IN THIS CHAPTER

This chapter studies chemical equilibria in all their forms: reversible reactions and the equilibrium constant, redox equilibria and electrode potentials, and acid-base equilibria with pH calculations, titrations, buffers and solubility. It equips the learner with the quantitative tools required to predict, control and exploit chemical systems at the GCE Advanced Level.

- By the end of this chapter I can explain the dynamic nature of chemical equilibrium and state the conditions for its attainment.
- By the end of this chapter I can apply Le Chatelier's principle to predict the effect of concentration, pressure, temperature and a catalyst on an equilibrium position.
- By the end of this chapter I can write expressions for Kc and Kp, deduce their units and perform related calculations, including for heterogeneous equilibria.
- By the end of this chapter I can relate the equilibrium constant to temperature and enthalpy change and interpret its magnitude.
- By the end of this chapter I can define redox systems, balance redox equations and describe the different types of half-cells.
- By the end of this chapter I can measure and use standard electrode potentials to calculate cell emf and predict the feasibility of redox reactions.

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THIS SIGNED COURSE CONTAINS

COURSE — complete, illustrated, step by step

METHODS — and worked examples

EXERCISES — exam style + solutions

SUMMARY SHEET — the essentials to remember

CONTENTS

1. Reversibility and Dynamic Equilibrium
2. Equilibrium Position and Le Chatelier's Principle
3. The Equilibrium Law: Kc and Kp
4. Heterogeneous Equilibria and the Effect of Temperature on K
5. Redox Systems and Balancing Redox Equations
6. Types of Half-Cells and Standard Electrode Potentials
7. Electrochemical Cells, Cell emf and Uses of E^θ
8. Acid-Base Theories, pH and the Ionic Product of Water
9. pH of Strong and Weak Acids and Bases, and of Mixtures
10. Acid-Base Indicators and Titration Curves
11. Buffer Solutions
12. Salt Hydrolysis, Solubility Product and Common Ion Effect
13. Integration activity
14. Methods and worked examples
15. Exercises
16. Solutions to the exercises
17. Summary sheet

CHAPTER OBJECTIVES

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- By the end of this chapter I can measure and use standard electrode potentials to calculate cell emf and predict the feasibility of redox reactions.
- By the end of this chapter I can define acids and bases according to Arrhenius, Brønsted-Lowry and Lewis and calculate the pH of strong and weak acid/base solutions and their mixtures.

- By the end of this chapter I can interpret titration curves, select suitable indicators and explain the functioning of buffer solutions with pH calculations.
- By the end of this chapter I can explain salt hydrolysis and calculate solubility products, including the common ion effect.

PREREQUISITES

- Chemical equations, stoichiometry and the mole concept (Lower Sixth)
- Amount of substance, molar concentration and gas laws (Lower Sixth)
- Enthalpy changes of reactions and Hess's law (Lower Sixth)
- Oxidation numbers and simple redox reactions (Lower Sixth)
- Rates of reaction and collision theory (Lower Sixth)

1 REVERSIBILITY AND DYNAMIC EQUILIBRIUM

In a Douala brewery, the production of beer relies on fermentation reactions that never go to completion — the chemist must constantly push the equilibrium towards the desired product. In the same city, a car battery delivers current because of redox reactions occurring at its electrodes, and a nurse at the Laquintinie Hospital checks the pH of a patient's blood, which is kept almost constant by buffer systems. Why do some reactions stop before all reactants are consumed? How can we predict and control the yield of a reaction, the voltage of a cell, or the pH of a solution? This chapter provides the quantitative tools to answer these questions.

Until now, most of the reactions you have met — burning magnesium, neutralising an acid with a base — were treated as if they went to completion: the reactants are entirely converted into products. In reality, a very large number of chemical reactions are **incomplete**: no matter how long we wait, some reactant always remains. This is not a failure of the chemist; it is a fundamental property of matter. Such reactions are said to be **reversible**, and the state they finally reach is called a state of **chemical equilibrium**. Understanding equilibrium is the key to controlling industrial processes (the Haber synthesis of ammonia), to explaining how batteries work, and to calculating the pH of blood, of palm wine, or of the water of the Wouri river.

1.1 The Concept of Reversibility

Observation 1: water in a closed vessel.

Pour some water into a bottle, close it tightly, and leave it on the laboratory bench. The water level does not change, even after weeks. Yet water is volatile: molecules are constantly escaping from the liquid surface (evaporation). Why does the level stay constant? Because the vessel is closed: the water vapour cannot escape, and vapour molecules continually return to the liquid (condensation). After some time, evaporation and condensation occur at exactly the same rate, and the amount of liquid water remains constant. We write:

$$H_2O(l) \rightleftharpoons H_2O(g)$$

Observation 2: a saturated salt solution.

Add sodium chloride to water and stir until no more dissolves: some solid remains at the bottom. The solution is saturated. If we could watch the ions, we would see that Na^+ and Cl^- ions are still leaving the crystal (dissolution) while other ions from the solution are redepositing onto it (crystallisation), at equal rates:

$$NaCl(s) \rightleftharpoons Na^+(aq) + Cl^-(aq)$$

The mass of undissolved solid stays constant, but the solid is not the same collection of ions from one moment to the next.

Observation 3: the Haber process.

Industrially, ammonia is synthesised from nitrogen and hydrogen:

$$N_2(g) + 3\,H_2(g) \rightleftharpoons 2\,NH_3(g) \qquad \Delta H^\theta = -92\ \text{kJ mol}^{-1}$$

Even under high pressure and moderate temperature, the conversion is never complete: the mixture leaving the reactor always contains unreacted N_2 and H_2, which are recycled. The reaction is reversible — ammonia also decomposes back into nitrogen and hydrogen.

Observation 4: esterification.

Mix ethanoic acid with ethanol in the presence of a little concentrated sulphuric acid and warm. The fruity-smelling ester ethyl ethanoate forms, but the reaction stops when roughly two-thirds of the acid has reacted, because the ester simultaneously reacts with water to regenerate the acid and the alcohol:

$$CH_3COOH(l) + C_2H_5OH(l) \rightleftharpoons CH_3COOC_2H_5(l) + H_2O(l)$$

DEFINITION **Reversible reaction**

A **reversible reaction** is a chemical reaction that can proceed in both directions: the products formed can react together to regenerate the original reactants. It is represented by a **double arrow** $\rightleftharpoons$ (or $\rightleftharpoons$) between reactants and products. The reaction going from reactants to products is the **forward reaction**; the reaction going from products back to reactants is the **reverse reaction**.

WARNING **The double arrow is not decoration**

Writing $A + B \rightleftharpoons C + D$ is a statement: it claims that both $A + B \longrightarrow C + D$ and $C + D \longrightarrow A + B$ occur. Never use the double arrow for a reaction that goes essentially to completion (e.g. the combustion of methane), and never use a single arrow for a genuinely reversible reaction such as esterification. At the GCE, marks are lost for the wrong arrow.

1.2 Static Equilibrium versus Dynamic Equilibrium

A book lying on a table is in equilibrium: the downward pull of gravity is exactly balanced by the upward reaction of the table, so the book does not move. Nothing is happening — this is a **static equilibrium**. Many students wrongly imagine chemical equilibrium in the same way: everything has stopped. Nothing could be further from the truth.

DEFINITION **Dynamic equilibrium**

A system is in a state of **dynamic equilibrium** when the forward and reverse reactions continue to occur at the microscopic (molecular) level, but at equal rates, so that the macroscopic properties of the system (concentrations, pressure, colour, mass of each phase) remain constant with time.

The word dynamic is essential: the molecules never stop reacting. A useful image is that of two escalators in a busy shop in Bamenda, one going up and one going down, carrying the same number of people per minute. The number of people on each floor stays constant, yet people are continuously moving between floors. Similarly, in the closed bottle of water, molecules evaporate and condense at equal rates: the water level is constant, but the traffic of molecules never ceases.

WARNING **Equilibrium does NOT mean the reaction has stopped**

At equilibrium, the reaction has not stopped. Forward and reverse reactions both continue indefinitely at equal rates. What has stopped is the net change: the rate of formation of products exactly equals their rate of consumption. Saying "the reaction stops at equilibrium" is a classic error that costs marks.

WARNING **Constant does not mean equal**

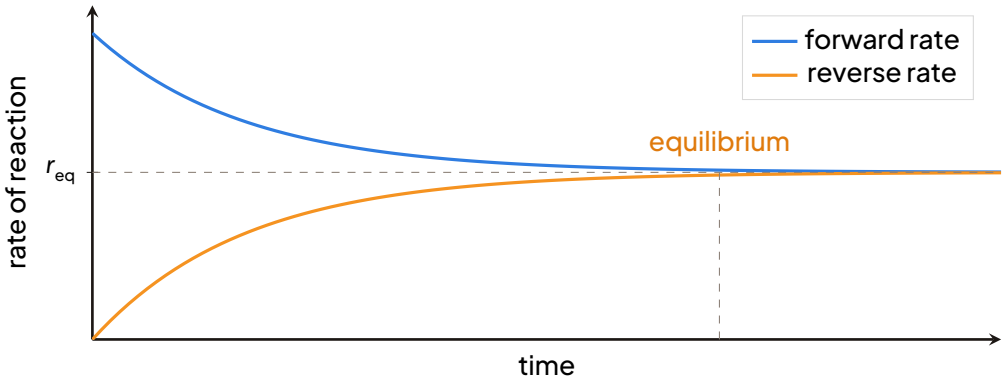
At equilibrium the concentrations of reactants and products are constant, but they are generally not equal. In the esterification of ethanoic acid, for instance, the equilibrium mixture contains appreciable amounts of all four species, in proportions fixed by the equilibrium constant — not in equal amounts. Constant ≠ equal.

1.3 Approach to Equilibrium: Rates and Concentrations

Consider the general reversible reaction A + B $\rightleftharpoons$ C + D, starting from pure A and B mixed together.

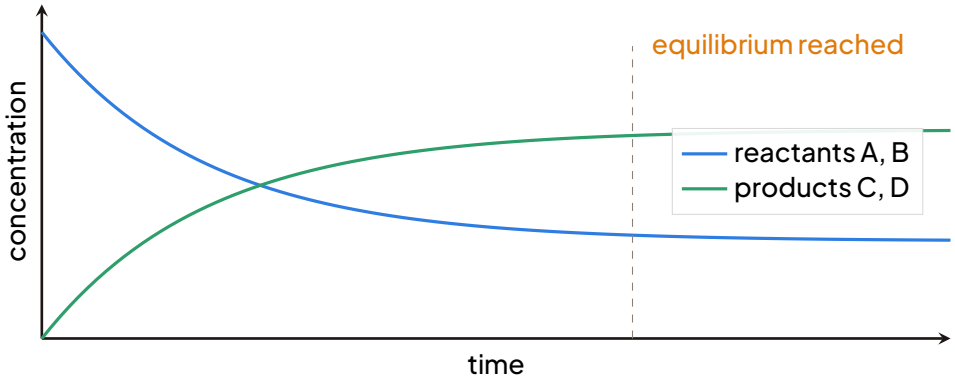
At the moment of mixing (t = 0), the concentrations of C and D are zero, so the reverse reaction cannot yet occur: the reverse rate is zero. The forward rate is at its maximum because the concentrations of A and B are highest. As time passes:

- A and B are consumed, so their concentrations fall and the **forward rate decreases**;
- C and D accumulate, so the **reverse rate increases** from zero;
- eventually the two rates become **equal**: from that instant, every molecule of C formed is matched by one destroyed, and all concentrations remain constant — dynamic equilibrium is established.

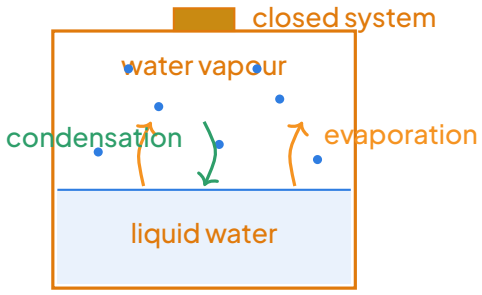


As a reversible reaction proceeds in a closed system, the forward rate falls and the reverse rate rises until they become equal: dynamic equilibrium.

The corresponding concentration–time graph shows the concentrations of reactants falling and those of products rising, then all levelling off to constant (but generally different) values.



Concentrations change rapidly at first, then level off at constant values once equilibrium is attained. Note that the equilibrium concentrations are constant but not equal.



Liquid–vapour equilibrium of water in a closed vessel: molecules evaporate and condense at equal rates, so the liquid level stays constant.

1.4 Characteristics and Conditions of Dynamic Equilibrium

PROPERTY **Characteristics of a system at dynamic equilibrium**

A chemical system at dynamic equilibrium has the following characteristics:

1. The forward and reverse reactions occur at equal rates.
2. The macroscopic properties are constant: concentrations, total pressure (for gases), colour intensity, density and mass of each phase do not change with time.
3. The equilibrium can be approached from either direction (starting from reactants or from products); the same equilibrium state is reached under the same conditions.
4. The equilibrium is dynamic: molecular-level activity continues indefinitely.

No proof of these points is required at the GCE, but you must be able to state and explain them.

PROPERTY **Conditions required for dynamic equilibrium**

Dynamic equilibrium can only be attained if all of the following conditions hold:

1. The reaction must be reversible.
2. The system must be closed: no matter (reactant, product or solvent vapour) may enter or leave, otherwise a species would be removed and the reverse reaction could not balance the forward one.
3. The temperature must be constant, because the rates of the forward and reverse reactions, and the position of equilibrium itself, depend on temperature.
4. The system must be given sufficient time for the forward and reverse rates to become

equal (a catalyst shortens this time but does not change the final state).

EXEMPLE **Why an open beaker never reaches equilibrium**

If the bottle of water is left open, the water vapour escapes into the air and is carried away. Condensation can no longer balance evaporation, the reverse process is starved of vapour, and the water level slowly falls until the bottle is dry. Equilibrium is impossible because the system is not closed. This is exactly why the esterification mixture is heated under reflux: the condenser returns the vapours to the flask, keeping the system closed so that equilibrium can be established.

WORKED EXAMPLE 1 **Identifying systems at dynamic equilibrium**

For each situation, state whether a dynamic equilibrium exists and justify your answer.

1. Ice and liquid water together in a perfectly insulated, sealed flask at 0 °C.
2. Steam escaping from a boiling open saucepan.
3. A sealed flask containing N_2, H_2 and NH_3 whose total pressure no longer changes with time.

SOLUTION

Solution.

1. Yes. The flask is closed (insulated and sealed), the temperature is constant at 0 °C, and the process $H_2O(s) \rightleftharpoons H_2O(l)$ is reversible. Ice melts and water freezes at equal rates: the masses of ice and liquid remain constant. All conditions for dynamic equilibrium are satisfied.
2. No. The saucepan is an open system: steam escapes into the kitchen and cannot condense back into the pan. The reverse process cannot balance the forward one, so no equilibrium is established — eventually all the water boils away.
3. Yes. The flask is sealed (closed system) and the reaction $N_2(g) + 3\,H_2(g) \rightleftharpoons 2\,NH_3(g)$ is reversible. The total pressure is a macroscopic property; since it is constant with time, the amounts of gas no longer change, meaning the forward and reverse rates are equal: dynamic equilibrium has been reached. (Note: constant pressure is the evidence of equilibrium here, not the cause.)

METHOD **Testing whether a system is at equilibrium**

To decide whether a reversible system has reached dynamic equilibrium:

1. Check that the system is closed and at constant temperature.
2. Identify a macroscopic property (concentration, pressure, colour, mass of a phase).
3. Measure it at two different times. If the value is unchanged, and the conditions above hold, the system is at dynamic equilibrium.
4. Remember: equal rates at the molecular level, constant properties at the macroscopic level — never "the reaction has stopped".

KEY POINTS **Key points of this section**

- A **reversible reaction** proceeds in both directions and is written with a double arrow: A + B $\rightleftharpoons$ C + D.
- At **dynamic equilibrium**, forward and reverse rates are equal, so macroscopic properties are constant — but the molecules never stop reacting.
- Constant concentrations do not mean equal concentrations.
- Equilibrium requires a reversible reaction in a closed system at constant temperature, given sufficient time.
- Equilibrium can be reached from either direction; the same state is obtained under the same conditions.

DID YOU KNOW? **Equilibrium all around us**

The fizz in a bottle of Top or Guinness kept in a Douala drinks depot is an equilibrium: carbon dioxide is dissolved in the drink under pressure, and $CO_2(g) \rightleftharpoons CO_2(aq)$ is established in the sealed bottle. When the cap is removed, the system is opened, CO_2 escapes, and the equilibrium is destroyed — the drink slowly goes flat. The same principle explains why a sealed bottle keeps its sparkle for months.

EXERCISE 1 **Checking your understanding**

1. Define a reversible reaction and give one chemical example with a balanced equation.
2. Explain, in terms of molecular behaviour, why the level of water in a tightly closed bot-

tle remains constant.

3. A student says: "At equilibrium, the concentrations of reactants and products are equal and all reaction has ceased." Identify and correct the two mistakes in this statement.
4. State the four conditions necessary for a system to attain dynamic equilibrium.

SOLUTION **Exercise 1**

1. A reversible reaction is one that can proceed in both directions: products can regenerate the reactants. Example: $CH_3COOH(l) + C_2H_5OH(l) \rightleftharpoons CH_3COOC_2H_5(l) + H_2O(l)$.
2. Water molecules continually evaporate from the liquid surface, but because the bottle is closed the vapour cannot escape; vapour molecules condense back at the same rate. Equal rates of evaporation and condensation keep the amount of liquid — and hence the level — constant.
3. Mistake 1: concentrations at equilibrium are constant, not necessarily equal. Mistake 2: the reactions have not ceased; forward and reverse reactions continue at equal rates (the equilibrium is dynamic).
4. The reaction must be reversible; the system must be closed; the temperature must be constant; sufficient time must be allowed for the forward and reverse rates to become equal.

2 EQUILIBRIUM POSITION AND LE CHATELIER'S PRINCIPLE

In the previous section we established that a reversible reaction in a closed system does not stop: it reaches a **dynamic equilibrium** in which the forward and reverse reactions continue at equal rates, so that the macroscopic composition of the mixture no longer changes. But "reaching equilibrium" does not mean "half the reactants have been converted". Some equilibria contain almost entirely products; others contain almost entirely reactants. In this section we learn to describe where an equilibrium lies, and — far more importantly for the chemist and the industrial engineer — how to push it in the direction we want.

2.1 The Equilibrium Position

DEFINITION **Equilibrium position**

The **equilibrium position** of a reversible reaction describes the relative proportions of reactants and products present in the equilibrium mixture.

- If, at equilibrium, the mixture contains mostly products, we say the equilibrium position lies to the **right** (products are favoured).
- If, at equilibrium, the mixture contains mostly reactants, we say the equilibrium position lies to the **left** (reactants are favoured).

Consider two familiar examples. The esterification of ethanol by ethanoic acid,

$$CH_3COOH(l) + C_2H_5OH(l) \rightleftharpoons CH_3COOC_2H_5(l) + H_2O(l),$$

settles at an equilibrium containing appreciable amounts of both reactants and products — the position lies roughly in the middle. By contrast, the reaction of hydrogen with oxygen to form water goes essentially to completion under suitable conditions: the equilibrium lies far to the right. The decomposition of limestone, $CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$, lies far to the left at room temperature, which is why the limestone hills around Buea do not spontaneously crumble into quicklime.

WARNING **Equilibrium position is not rate**

A common confusion: the equilibrium position tells us how far a reaction goes, not how fast it gets there. A reaction may have an equilibrium lying far to the right yet be immeasurably slow (a mixture of H_2 and O_2 at room temperature). Position is a question of energetics and balance of rates at equilibrium; speed is a question of kinetics. Keep the two ideas strictly separate.

The crucial experimental fact is this: **the equilibrium position of a given reaction is not fixed**. It depends on the conditions — concentrations, pressure (for gases) and temperature. If we change the conditions, the composition of the equilibrium mixture changes: we say the equilibrium position has shifted. The whole art of this section is to predict the direction of the shift.

2.2 Le Chatelier's Principle

EXPERIMENT **A thought experiment with the esterification equilibrium**

Suppose we have the esterification mixture above at equilibrium, and we now pour in a large excess of cheap ethanol. What happens? Immediately after the addition, the forward reaction $CH_3COOH + C_2H_5OH \longrightarrow CH_3COOC_2H_5 + H_2O$ speeds up (more ethanol molecules means more fruitful collisions), while the reverse rate is initially unchanged. The forward reaction now outruns the reverse one, so ester and water accumulate and acid is consumed, until a new equilibrium is established — one containing more ester than

before. The system has responded to the disturbance (extra ethanol) by removing some of that ethanol. It has, in a sense, "fought back" against the change we imposed.

This behaviour is completely general, and was summarised by the French chemist Henri Le Chatelier in 1884.

PROPERTY **Le Chatelier's Principle**

Statement (examinable): If a constraint (a change of concentration, pressure or temperature) is imposed on a system at dynamic equilibrium, the system adjusts itself so as to **oppose** (partially counteract) that constraint, and a new equilibrium position is established.
Equivalently: the equilibrium shifts in the direction that removes added substance, produces removed substance, reduces an imposed pressure increase, or absorbs added heat.

WARNING **The system only opposes the change — it never cancels it**

Le Chatelier's principle says the system shifts so as to reduce the effect of the disturbance, not to eliminate it. If you add ethanol, the new equilibrium contains less ethanol than immediately after your addition, but more than before the addition. The opposition is always partial.

METHOD **Predicting the shift of an equilibrium**

1. Identify the change imposed on the system (addition/removal of a species, change of pressure, change of temperature).
2. Ask: which direction opposes the change? (Which direction consumes the added species? Which direction produces more gas moles to relieve a pressure decrease? Which direction absorbs heat?)
3. State the shift (to the right or to the left) and state the consequence on the yield of the desired product.
4. Check the special cases: no gas moles change? A pure solid or liquid involved? A catalyst added? — these may cause no shift.

2.3 Effect of Changing Concentration

PROPERTY **Concentration changes**

If the concentration of a reactant is increased (or a product removed), the equilibrium shifts to the right, consuming some of the added reactant and increasing the yield of products. If the concentration of a reactant is decreased (or a product added), the equilibrium shifts to the left.

The intuition is kinetic, as in the esterification example: adding a reactant raises the forward rate above the reverse rate, so net forward reaction occurs until the two rates are equal again. Removing a product (for instance distilling off the ester as it forms) continually lowers the reverse rate, so the forward reaction keeps winning and the yield rises — a standard trick in the laboratory preparation of esters.

WORKED EXAMPLE 2 **Shifting the equilibrium in the chromate/dichromate system**

The orange dichromate ion and the yellow chromate ion coexist in the equilibrium

$$Cr_2O_7^{2-}(aq) + H_2O(l) \rightleftharpoons 2\,CrO_4^{2-}(aq) + 2\,H^+(aq) \qquad \text{(orange} \rightleftharpoons \text{yellow)}$$

Predict and explain the colour change observed when (a) dilute sodium hydroxide is added; (b) dilute hydrochloric acid is then added in excess.

SOLUTION

(a) Adding NaOH. The OH^- ions neutralise H^+ ions, so $[H^+]$ decreases. By Le Chatelier's principle, the system opposes this removal by shifting in the direction that produces H^+ — that is, to the right. More yellow CrO_4^{2-} forms: the solution turns yellow.
(b) Adding excess HCl. Now $[H^+]$ increases. The system opposes this by consuming H^+, shifting to the left. Orange $Cr_2O_7^{2-}$ re-forms: the solution turns orange.
This reversible colour change is a classic GCE qualitative observation and a beautiful visual proof that equilibria really do shift.

WARNING **Adding a pure solid or pure liquid causes no shift**

The "concentration" of a pure solid or a pure liquid is constant (it is its density in molar terms, which does not change). Adding more lumps of $CaCO_3$ to the equilibrium $CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$ does not shift the position at all — there is simply more solid sitting there. Only changes in the amounts of gases or dissolved species matter. Never write "adding more solid shifts the equilibrium" in an examination.

2.4 Effect of Changing Pressure (Gaseous Systems)

Pressure only matters when gases are involved, and it acts through the number of moles of gas

on each side of the equation. When the total pressure of an equilibrium mixture is increased (by compressing the mixture into a smaller volume), the system can relieve the constraint by shifting towards the side with **fewer moles of gas** — fewer particles means fewer collisions with the walls, hence lower pressure.

PROPERTY **Pressure changes**

For a gaseous equilibrium, increasing the total pressure (by compression) shifts the equilibrium towards the side with the **fewer moles of gas**; decreasing the pressure shifts it towards the side with more moles of gas. If both sides contain the same number of moles of gas, a pressure change causes no shift.

EXEMPLE **Applying the rule**

- $N_2(g) + 3\,H_2(g) \rightleftharpoons 2\,NH_3(g)$: 4 moles of gas on the left, 2 on the right. Increasing the pressure shifts the equilibrium to the right — more ammonia. This is why the Haber process uses high pressure.
- $N_2O_4(g) \rightleftharpoons 2\,NO_2(g)$: 1 mole versus 2 moles. Compression shifts the equilibrium to the left; the brown colour of NO_2 pales.
- $H_2(g) + I_2(g) \rightleftharpoons 2\,HI(g)$: 2 moles on each side. Changing the pressure causes no shift (although the rates change).

WARNING **Two pressure traps**

1. Pressure changes only shift equilibria involving gases with unequal numbers of moles of gas on the two sides. Solids and liquids are essentially incompressible and do not count.
2. Adding an inert gas (e.g. argon) at constant volume raises the total pressure but changes none of the partial pressures of the reacting gases: there is no shift. Only compression (change of volume) or changing the amount of a reacting gas matters.

2.5 Effect of Changing Temperature

Temperature is the most subtle constraint, because it is the only one of the three that changes the value of the equilibrium constant (as we shall see in Section 4). The rule, however, is easy to apply once the sign of the enthalpy change is known.

PROPERTY **Temperature changes**

For an exothermic forward reaction ($\Delta H < 0$), raising the temperature shifts the equilibrium to the left (the endothermic, heat-absorbing direction is favoured); lowering the temperature shifts it to the right. For an endothermic forward reaction ($\Delta H > 0$), the reverse holds: raising the temperature shifts the equilibrium to the right.

The logic follows directly from Le Chatelier: heat added to the system is opposed by the reaction direction that absorbs heat, i.e. the endothermic direction. A useful memory trick is to treat heat as a chemical species: for $N_2 + 3\,H_2 \rightleftharpoons 2\,NH_3, \Delta H = -92\,kJ\,mol^{-1}$, write "$N_2 + 3\,H_2 \rightleftharpoons 2\,NH_3$ + heat". Adding heat then obviously pushes the equilibrium left, exactly like adding a product.

EXEMPLE **Heating the brown gas**

A sealed tube containing the equilibrium $N_2O_4(g) \rightleftharpoons 2\,NO_2(g)$, $\Delta H = +57\,kJ\,mol^{-1}$ (endothermic to the right), darkens when warmed and pales when cooled: heating favours the endothermic production of brown NO_2; cooling favours the exothermic reverse reaction.

2.6 Effect of a Catalyst

PROPERTY **A catalyst does not shift the equilibrium**

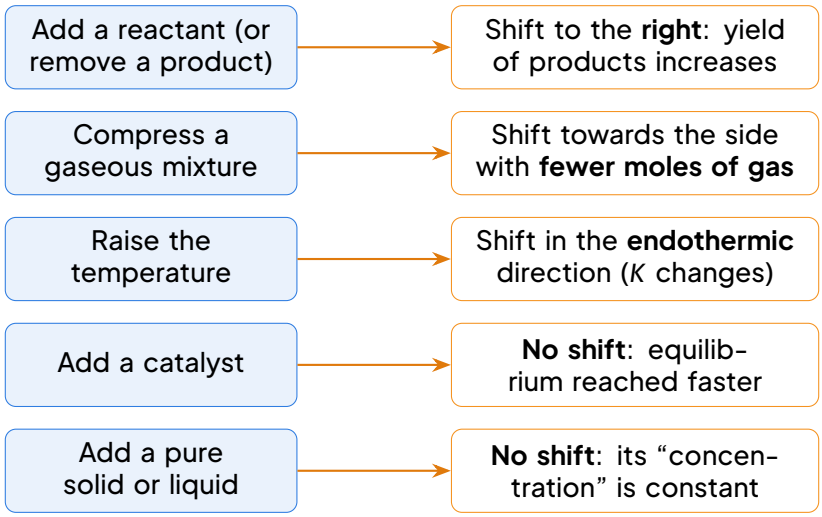
A catalyst speeds up the forward and reverse reactions equally (by providing an alternative pathway of lower activation energy for both). Consequently:

- the equilibrium position is unchanged — the same equilibrium mixture is obtained, with or without catalyst;
- the only effect is that equilibrium is **reached faster**.

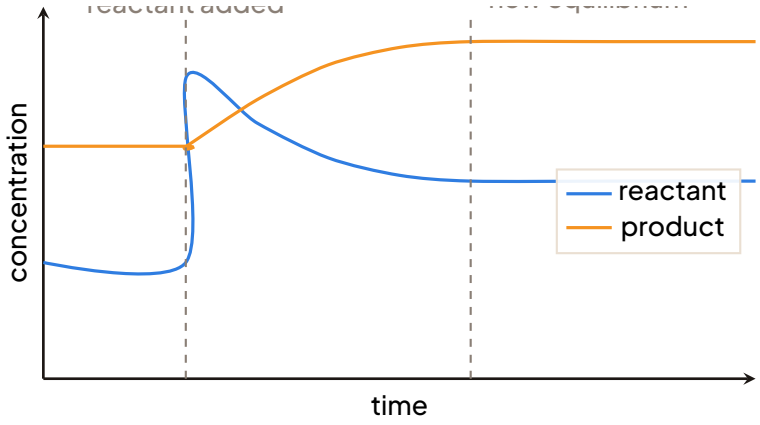
(The proof is not examinable; the statement and its application are.)

This is why catalysts are industrially precious even though they cannot improve the equilibrium yield: a process that reaches 30% conversion in one hour instead of one month is the difference between profit and bankruptcy.

2.7 Summary Diagram and Graphical View



How a system at equilibrium responds to each type of constraint (Le Chatelier's principle).



Concentration–time graph: at the dashed line a reactant is suddenly added (instantaneous jump of its concentration); the system then shifts right until a new equilibrium is established, with more product and slightly more reactant than originally.

Note carefully the shape of this graph: the added reactant's concentration jumps vertically at the moment of addition, then falls back — but it settles above its original value. This is the graphical signature of Le Chatelier's "partial opposition".

2.8 Industrial Applications: the Haber and Contact Processes

DID YOU KNOW? **Feeding the world from thin air**

Ammonia, NH_3, is the starting material for nitrogen fertilisers without which modern agriculture — including the maize and cocoa farms of Cameroon — could not feed today's population. It is manufactured from nitrogen (from air) and hydrogen (from natural gas) by the Haber process, developed by Fritz Haber and Carl Bosch in the early twentieth century. Choosing the operating conditions is a masterclass in applied Le Chatelier.

The equilibrium is

$$N_2(g) + 3\,H_2(g) \rightleftharpoons 2\,NH_3(g) \qquad \Delta H^{\ominus} = -92\ \text{kJ mol}^{-1}\ \text{(exothermic)}.$$

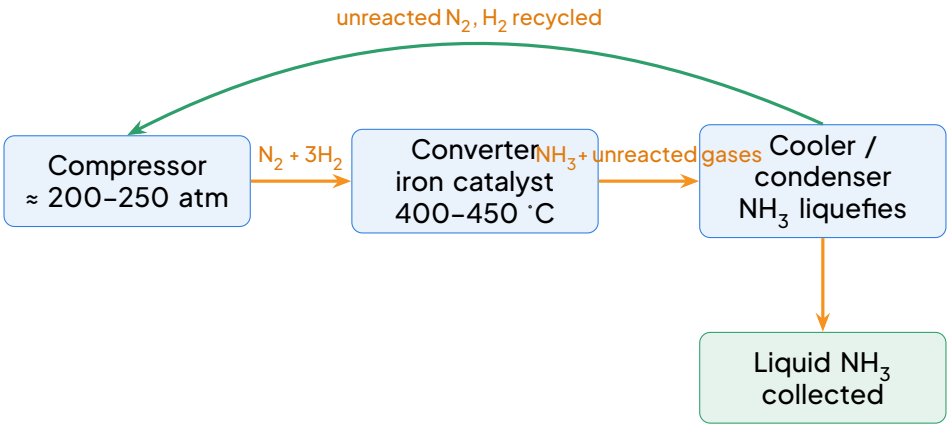
Let us apply our method to each condition:

Pressure. There are 4 moles of gas on the left and 2 on the right, so high pressure shifts the equilibrium to the right and increases the yield of ammonia. High pressure also speeds up the reaction. Industry uses about 200–250 atm. Why not 1000 atm? Because compressing gases and building vessels that withstand enormous pressures is extremely expensive and dangerous: the extra yield does not repay the extra cost.

Temperature. The forward reaction is exothermic, so low temperature favours a high yield of ammonia. But at low temperature the reaction is hopelessly slow — a high yield is useless if it takes years to obtain. Industry therefore uses a compromise temperature of about 400–450 °C: the yield per pass is modest, but it is obtained quickly.

Catalyst. Finely divided iron (with promoters) is used. It cannot improve the equilibrium yield, but it allows the compromise temperature to deliver that yield at an acceptable rate.

Removing the product. The gas mixture leaving the converter is cooled; ammonia liquefies and is removed. By Le Chatelier's principle, removing NH_3 pulls the equilibrium to the right, and the unreacted N_2 and H_2 are recycled through the converter, pushing the overall conversion towards 98 %.



Schematic of the Haber process. High pressure and product removal favour the yield; the moderate temperature and iron catalyst favour the rate; recycling pushes the overall conversion close to completion.

EXEMPLE **The Contact process for sulphuric acid**

The key step in manufacturing sulphuric acid (used in fertiliser and battery production) is

$$2\,SO_2(g) + O_2(g) \rightleftharpoons 2\,SO_3(g) \qquad \Delta H^{\ominus} = -196\ \text{kJ mol}^{-1}.$$

By the same reasoning: 3 moles of gas become 2, so high pressure favours SO_3 — but the yield is already so good that industry uses only about 2 atm, saving the cost of compression. A compromise temperature of about 450 °C and a vanadium(V) oxide catalyst, V_2O_5, give a fast reaction with an excellent yield.

KEY POINTS **Key points**

- The equilibrium position describes whether reactants or products predominate at equilibrium ("left" or "right").
- Le Chatelier's principle: a system at equilibrium subjected to a change adjusts so as to oppose the change — but only partially.
- Adding a reactant (or removing a product) shifts the equilibrium right; the reverse shifts it left. Pure solids and liquids have no effect.
- Compressing a gaseous equilibrium shifts it towards the side with fewer moles of gas; equal moles of gas means no shift; inert gas at constant volume means no shift.
- Raising the temperature favours the endothermic direction — and is the only change that alters the value of K.

- A catalyst speeds up both directions equally: equilibrium is reached faster but the position (and the yield) is unchanged.
- Industry chooses compromise conditions: the Haber process (≈ 450 °C, ≈ 200 atm, iron catalyst, product removed and recycled) balances yield against rate and cost.

EXERCISE 2 **Predicting shifts**

For each equilibrium below, state the effect on the yield of the indicated product of (i) increasing the pressure by compression, (ii) raising the temperature, (iii) adding a catalyst. Justify each answer.

1. $N_2(g) + 3\,H_2(g) \rightleftharpoons 2\,NH_3(g)$, $\Delta H = -92\ kJ\,mol^{-1}$ (consider NH_3 as the product).
2. $CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$, $\Delta H = +178\ kJ\,mol^{-1}$ (consider CO_2 as the product). What is the effect of adding more $CaCO_3(s)$?
3. $H_2(g) + I_2(g) \rightleftharpoons 2\,HI(g)$, $\Delta H = +53\ kJ\,mol^{-1}$ (consider HI as the product).

SOLUTION **Exercise 1**

1. (i) 4 moles of gas → 2 moles: compression shifts right, yield of NH_3 increases. (ii) Forward reaction exothermic: raising T shifts left (endothermic direction), yield decreases. (iii) No change in yield; equilibrium reached faster.
2. (i) 0 moles of gas on the left, 1 on the right: compression shifts left, yield of CO_2 decreases. (ii) Forward reaction endothermic: raising T shifts right, yield increases (this is why lime kilns are very hot). (iii) No change in yield. Adding more $CaCO_3(s)$: no effect — it is a pure solid, whose "concentration" is constant.
3. (i) 2 moles of gas on each side: compression causes no shift, yield unchanged. (ii) Forward reaction endothermic: raising T shifts right, yield of HI increases. (iii) No change in yield; equilibrium reached faster.

3 THE EQUILIBRIUM LAW: KC AND KP

Le Chatelier's principle, which we studied in the previous section, is a wonderful qualitative tool: it tells us the direction in which an equilibrium shifts when we disturb it. But an examiner, an industrial chemist at a fertiliser plant, or a laboratory technician in Douala needs more than a direction — they need numbers. How much product will actually be present when equilibrium is reached? Is a reaction worth running industrially? To answer such questions we need a quantitative description of equilibrium. That description is the **equilibrium law**, and the number at its heart is the **equilibrium constant**.

3.1 From Observation to Law: The Equilibrium Constant K_c

EXPERIMENT **The observation behind the law**

Consider the reversible reaction between hydrogen and iodine at a fixed temperature:

$$H_2(g) + I_2(g) \rightleftharpoons 2\,HI(g)$$

If we start several sealed tubes with different initial amounts of H_2, I_2 and HI, allow each to reach equilibrium at the same temperature, and then analyse the mixtures, we find that the equilibrium concentrations differ from tube to tube. Yet the quantity

$$\frac{[HI]^2}{[H_2][I_2]}$$

comes out to the same numerical value in every tube. This remarkable regularity, established experimentally in the nineteenth century by Guldberg and Waage (the "law of mass action"), is the basis of the equilibrium law: at a given temperature, the equilibrium composition is not arbitrary — it is fixed by a constant.

DEFINITION **The equilibrium constant K_c**

For a reversible reaction at equilibrium at a given temperature,

$$aA + bB \rightleftharpoons cC + dD$$

the **equilibrium constant** in terms of concentration is defined by

$$K_c = \frac{[C]^c\,[D]^d}{[A]^a\,[B]^b}$$

where [X] denotes the equilibrium concentration of species X in mol dm^{-3}, and the powers a, b, c, d are the stoichiometric coefficients of the balanced equation. The value of K_c is constant at a fixed temperature.

KEY POINTS **Writing a K_c expression correctly**

- Products go on top, reactants on the bottom.
- Each concentration is raised to the power of its coefficient in the balanced equation.
- Only species whose concentration can vary appear: gases and species in solution. (Pure solids and pure liquids are omitted — see the section on heterogeneous equilibria.)

- The expression depends on the way the equation is written: for $2\,SO_2 + O_2 \rightleftharpoons 2\,SO_3$, $K_c = \dfrac{[SO_3]^2}{[SO_2]^2[O_2]}$; for the reverse reaction the constant is $1/K_c$, and halving all coefficients gives $\sqrt{K_c}$.

WARNING **Equilibrium concentrations, not initial concentrations!**

The single most common error at GCE is to substitute initial concentrations (or initial amounts) into the K_c expression. The law holds only at equilibrium. If you are given initial amounts, you must first work out the equilibrium composition (the ICE method below) before applying the expression.

3.2 Units of K_c

K_c is not automatically dimensionless. Its units are obtained by substituting mol dm^{-3} for every concentration and simplifying. For a general reaction, if $\Delta n = (c + d) - (a + b)$ is the change in the number of moles of (concentration-dependent) species, then

$$\text{units of } K_c = \left(\text{mol dm}^{-3}\right)^{\Delta n}.$$

EXEMPLE **Deducing units of K_c**

- $H_2(g) + I_2(g) \rightleftharpoons 2\,HI(g)$: $K_c = \dfrac{[HI]^2}{[H_2][I_2]}$ has units $\dfrac{(\text{mol dm}^{-3})^2}{(\text{mol dm}^{-3})(\text{mol dm}^{-3})}$ = no units ($\Delta n = 0$).
- $N_2(g) + 3\,H_2(g) \rightleftharpoons 2\,NH_3(g)$: units = $\dfrac{(\text{mol dm}^{-3})^2}{(\text{mol dm}^{-3})(\text{mol dm}^{-3})^3} = (\text{mol dm}^{-3})^{-2} =$ **mol^{-2} dm^6**.
- $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$: units = mol dm^{-3} ($\Delta n = +1$).

WARNING **K_c and K_p generally have units**

Never write "K_c has no units" as a reflex. Deduce the units each time from the expression and the balanced equation. Examiners award a mark for correct units, and a wrong or missing unit loses it.

3.3 Calculating K_c: The ICE Table Method

Most GCE problems give initial amounts and one piece of equilibrium information (an amount, a concentration, or a percentage dissociation). The systematic tool is the **ICE table**: Initial, Change, Equilibrium.

METHOD **The ICE table procedure**

1. Write the balanced equation and the expression for K_c.
2. Initial row: enter the initial moles (or concentrations) of each species. Products are usually zero initially.
3. Change row: let the extent of reaction be x. Reactants change by $-ax$, $-bx$, …; products by $+cx$, $+dx$, …, according to the coefficients.
4. Equilibrium row: add the two rows algebraically.
5. Use the given equilibrium information to solve for x.
6. Convert equilibrium moles to equilibrium concentrations by dividing by the volume V (in dm^3).
7. Substitute into the K_c expression and state the units.

Stage	A	B	C
Initial (mol)	n_0(A)	n_0(B)	0
Change (mol)	$-ax$	$-bx$	$+cx$
Equilibrium	n_0(A) − ax	n_0(B) − bx	cx

then divide by V to get $[\,]_{eq}$

ICE table template for aA + bB ⇌ cC: the equilibrium row is initial + change; concentrations follow by dividing by the volume.

WORKED EXAMPLE 3 **Calculating K_c for the esterification of ethanol**

Statement. 1.0 mol of ethanoic acid and 1.0 mol of ethanol are mixed with a catalyst and allowed to reach equilibrium at 298 K:

$$CH_3COOH(l) + C_2H_5OH(l) \rightleftharpoons CH_3COOC_2H_5(l) + H_2O(l)$$

At equilibrium, 0.33 mol of ethanoic acid remains. Calculate K_c.

SOLUTION

Solution. Build the ICE table in moles (all species are in the same liquid mixture of volume V):

	CH_3COOH	C_2H_5OH	$CH_3COOC_2H_5$	H_2O
Initial / mol	1.0	1.0	0	0
Change / mol	$-x$	$-x$	$+x$	$+x$
Equilibrium / mol	$1.0 - x$	$1.0 - x$	x	x

Given $1.0 - x = 0.33$, so $x = 0.67$ mol. Since all coefficients are 1, every concentration is (moles)/V and V cancels:

$$K_c = \frac{[CH_3COOC_2H_5][H_2O]}{[CH_3COOH][C_2H_5OH]} = \frac{(0.67/V)(0.67/V)}{(0.33/V)(0.33/V)} = \frac{0.67^2}{0.33^2} \approx 4.1.$$

Here $\Delta n = 0$, so K_c has no units. Note how the unknown volume cancelled — a useful check.

WORKED EXAMPLE 4 **Calculating K_c when the volume does not cancel**

Statement. 2.00 mol of PCl_5 vapour is introduced into a 1.00 dm^3 vessel and heated. At equilibrium, 0.80 mol of Cl_2 is present:

$$PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$$

Calculate K_c at this temperature.

SOLUTION

Solution. ICE table in moles: start $(2.00, 0, 0)$; change $(-x, +x, +x)$; equilibrium $(2.00 - x, x, x)$. Given $x = 0.80$ mol, the equilibrium amounts are $n_{PCl_5} = 1.20$, $n_{PCl_3} = n_{Cl_2} = 0.80$ mol. Since $V = 1.00$ dm^3, the concentrations are numerically equal to the moles:

$$K_c = \frac{[PCl_3][Cl_2]}{[PCl_5]} = \frac{0.80 \times 0.80}{1.20} = 0.53 \text{ mol dm}^{-3}.$$

Units: $\dfrac{(\text{mol dm}^{-3})^2}{\text{mol dm}^{-3}} = \text{mol dm}^{-3}$ ($\Delta n = +1$). Here $\Delta n \neq 0$, so the volume does not cancel — always check.

WORKED EXAMPLE 5 **Finding an equilibrium concentration from K_c**

Statement. For $N_2(g) + 3\,H_2(g) \rightleftharpoons 2\,NH_3(g)$ at a certain temperature, $K_c = 0.50 \text{ mol}^{-2}\text{ dm}^6$.

At equilibrium in a 2.0 dm^3 vessel there are 0.10 mol of N_2 and 0.30 mol of H_2. Calculate the equilibrium concentration of NH_3.

SOLUTION

Solution. Equilibrium concentrations: $[N_2] = 0.10/2.0 = 0.050\ mol\ dm^{-3}$; $[H_2] = 0.30/2.0 = 0.15\ mol\ dm^{-3}$. Rearranging the equilibrium law:

$$[NH_3]^2 = K_c\,[N_2][H_2]^3 = 0.50 \times 0.050 \times (0.15)^3 = 8.4 \times 10^{-5}\ mol^2\ dm^{-6}$$

$$[NH_3] = \sqrt{8.4 \times 10^{-5}} = 9.2 \times 10^{-3}\ mol\ dm^{-3}.$$

3.4 Gaseous Equilibria: Partial Pressure and K_p

For reactions in the gas phase it is often more convenient to measure pressures than concentrations.

DEFINITION **Partial pressure**

In a mixture of gases, the **partial pressure** p_A of gas A is the pressure that A would exert if it alone occupied the whole volume at the same temperature. It is given by

$$p_A = x_A \times P_{total}$$

where $x_A = \dfrac{n_A}{n_{total}}$ is the **mole fraction** of A. The sum of all partial pressures equals the total pressure (Dalton's law). Pressures are measured in pascal (Pa, with kPa and atm also used).

DEFINITION **The equilibrium constant K_p**

For the gaseous equilibrium $aA(g) + bB(g) \rightleftharpoons cC(g) + dD(g)$,

$$K_p = \frac{p_C^c\, p_D^d}{p_A^a\, p_B^b}$$

where each p is the equilibrium partial pressure. The units of K_p are (pressure unit)$^{\Delta n}$, e.g. Pa^{-2} for the Haber equilibrium, deduced from the expression each time.

WORKED EXAMPLE 6 **Calculating K_p for the dissociation of PCl_5**

Statement. 1.00 mol of PCl_5 is placed in a vessel and heated. At equilibrium, 60% has dissociated and the total pressure is 200 kPa:

$$PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$$

Calculate K_p.

SOLUTION

Solution. ICE in moles: start $(1.00, 0, 0)$; change $(-0.60, +0.60, +0.60)$; equilibrium $(0.40, 0.60, 0.60)$, so $n_{total} = 1.60$ mol.

$$p_{PCl_5} = \frac{0.40}{1.60} \times 200 = 50\ \text{kPa}, \quad p_{PCl_3} = p_{Cl_2} = \frac{0.60}{1.60} \times 200 = 75\ \text{kPa}.$$

$$K_p = \frac{p_{PCl_3}\, p_{Cl_2}}{p_{PCl_5}} = \frac{75 \times 75}{50} = 112.5\ \text{kPa} \approx 1.1 \times 10^2\ \text{kPa}.$$

Units: $\frac{\text{kPa} \times \text{kPa}}{\text{kPa}} = \text{kPa}$ ($\Delta n = +1$).

WARNING **Mole fractions need the total number of moles**

When dissociation occurs, the total number of moles of gas changes (here from 1.00 to 1.60 mol). A frequent error is to divide by the initial number of moles instead of the equilibrium total. Always compute n_{total} from the equilibrium row of the ICE table.

PROPERTY **Relationship between K_p and K_c (GCE-useful extra)**

Using $p_i = \frac{n_i RT}{V} = [i]\,RT$ for each gas in the K_p expression gives

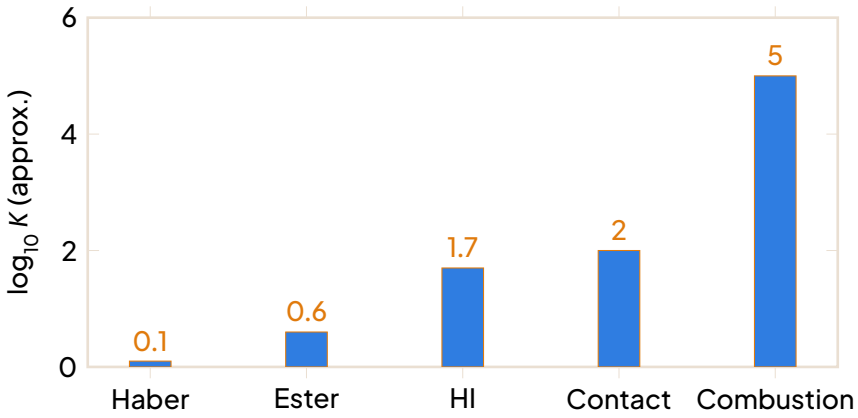
$$K_p = K_c\,(RT)^{\Delta n}$$

where Δn is the change in the number of moles of gas and $R = 8.314\ \text{J K}^{-1}\,\text{mol}^{-1}$ with T in kelvin (pressures then in Pa, concentrations in mol m^{-3} — take care with units). When $\Delta n = 0$, $K_p = K_c$. This relation is not required for routine GCE calculations but explains why both constants describe the same equilibrium. (Proof not examinable.)

3.5 What the Magnitude of K Tells Us

The size of the equilibrium constant is a direct measure of where the equilibrium position lies:

- $K \gg 1$ (e.g. 10^{10}): products dominate at equilibrium; the reaction goes essentially to **completion**.
- $K \ll 1$ (e.g. 10^{-10}): reactants dominate; the reaction **hardly proceeds**.
- $K \approx 1$: appreciable amounts of **both** reactants and products are present.



Illustrative magnitudes of K (log scale) for familiar reactions at typical working temperatures: combustion of fuels ($K \gg 1$, complete) versus the Haber process (small K, poor yield without optimisation). Values are indicative only — K varies with temperature.

WARNING **K says nothing about rate**

A huge K does not mean a fast reaction. The combustion of hydrogen has an enormous K, yet a hydrogen–oxygen mixture at room temperature shows no reaction without a spark, because the activation energy is high. K describes the equilibrium position; kinetics describes how quickly it is reached.

3.6 Factors Affecting the Equilibrium Constant

In the previous section we saw that changing concentration, pressure or adding a catalyst shifts the equilibrium position. Does the constant itself change?

PROPERTY **Only temperature changes K**

The value of an equilibrium constant for a given reaction depends only on temperature:

- Changing concentration: the system shifts to restore the ratio, but K is unchanged.
- Changing pressure: the position may shift (if $\Delta n \neq 0$), but K is unchanged.
- Adding a catalyst: equilibrium is reached faster, but the position and K are unchanged.
- Changing temperature: K itself changes — it increases with temperature for an en-

dothermic forward reaction and decreases for an exothermic one.

KEY POINTS **Key points of this section**

- $K_c = \dfrac{[C]^c[D]^d}{[A]^a[B]^b}$ and $K_p = \dfrac{p_C^c p_D^d}{p_A^a p_B^b}$, using equilibrium values only.
- Deduce the units of K_c and K_p from the expression every time: $(\text{mol dm}^{-3})^{\Delta n}$ and $(\text{pressure})^{\Delta n}$.
- Use the ICE table to convert initial amounts into equilibrium amounts before substituting.
- $p_A = x_A P_{\text{total}}$; $K_p = K_c(RT)^{\Delta n}$.
- $K \gg 1$: products favoured; $K \ll 1$: reactants favoured.
- Only a change in temperature alters the value of K.

EXERCISE 3 **Practice**

1. At equilibrium, a 1.0 dm^3 flask contains 0.20 mol SO_3, 0.40 mol SO_2 and 0.10 mol O_2 for $2\,SO_2(g) + O_2(g) \rightleftharpoons 2\,SO_3(g)$. Calculate K_c, stating its units.
2. For $N_2O_4(g) \rightleftharpoons 2\,NO_2(g)$, starting from 1.0 mol of pure N_2O_4, the mixture at equilibrium contains 0.40 mol of NO_2 at a total pressure of 150 kPa. Calculate K_p.
3. State, with reasons, the effect (if any) on the value of K_c for the Haber equilibrium of: (a) doubling the pressure; (b) adding iron catalyst; (c) raising the temperature (forward reaction exothermic).

SOLUTION **Exercise 1**

1. $[SO_3] = 0.20$, $[SO_2] = 0.40$, $[O_2] = 0.10\ \text{mol dm}^{-3}$. $K_c = \dfrac{(0.20)^2}{(0.40)^2(0.10)} = \dfrac{0.040}{0.016} = 2.5\ \text{mol}^{-1}\,\text{dm}^3$ ($\Delta n = 2 - 3 = -1$).
2. Change: +0.40 mol NO_2 means −0.20 mol N_2O_4. Equilibrium: $n_{N_2O_4} = 0.80$, $n_{NO_2} = 0.40$, total 1.20 mol. $p_{N_2O_4} = \frac{0.80}{1.20} \times 150 = 100$ kPa; $p_{NO_2} = 50$ kPa. $K_p = \dfrac{50^2}{100} = 25$ kPa.

3. (a) No change in K_c (position shifts, constant does not). (b) No change (catalyst speeds attainment only). (c) K_c decreases, because raising temperature favours the endothermic (reverse) direction.

4 HETEROGENEOUS EQUILIBRIA AND THE EFFECT OF TEMPERATURE ON K

In the previous section we wrote the equilibrium law for reactions in which every species is a gas or every species is in aqueous solution. But many equilibria that matter in industry and in everyday life involve more than one phase: think of limestone being roasted in a lime kiln, where a solid ($CaCO_3$) produces another solid (CaO) and a gas (CO_2). How do we write K_c and K_p when solids or pure liquids are present? And we have used Le Chatelier's principle to predict how temperature shifts an equilibrium; in this section we go further and ask: what does temperature do to the value of K itself?

4.1 Homogeneous and Heterogeneous Equilibria

DEFINITION **Homogeneous and heterogeneous equilibria**

A **homogeneous equilibrium** is an equilibrium in which all reactants and products are in the same phase (all gaseous, or all in the same solution). A **heterogeneous equilibrium** is an equilibrium in which the reactants and products are present in two or more different phases (for example solids and gases, or a pure liquid and gases).

The equilibrium $N_2(g) + 3\,H_2(g) \rightleftharpoons 2\,NH_3(g)$ is homogeneous: everything is gaseous. By contrast, the decomposition of calcium carbonate is heterogeneous:

$$CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$$

because two solids and one gas coexist.

EXPERIMENT **Observing a heterogeneous equilibrium**

Place a few grams of crushed limestone, $CaCO_3$, in a strong, sealed container fitted with a pressure gauge, and heat it strongly. The solid begins to decompose: $CaCO_3(s) \longrightarrow CaO(s) + CO_2(g)$. The gauge shows the pressure of CO_2 rising, then becoming constant even though plenty of $CaCO_3$ and CaO remain. The system has reached equilibrium: the forward and reverse reactions continue, but the pressure of CO_2 no longer changes. Remarkably, the equilibrium pressure of CO_2 is the same whether we start with one lump or ten lumps of limestone.

This last observation is the key to the whole section: the amount of solid present does not affect the equilibrium at all.

4.2 Why Pure Solids and Pure Liquids Are Omitted from K_c and K_p

PROPERTY **Rule for pure solids and pure liquids**

The concentrations of pure solids and pure liquids are constant (they equal the density divided by the molar mass, which does not change as more of the substance is added). They are therefore absorbed into the value of K and do not appear in the expressions for K_c or K_p. Only gases and species in solution are written.

Justification. Concentration means "amount per unit volume of the phase". For a pure solid such as $CaCO_3$, $[CaCO_3] = \dfrac{\text{density}}{M} = \dfrac{2.7\ \text{g cm}^{-3}}{100\ \text{g mol}^{-1}}$, a fixed number at a given temperature. Doubling the mass of solid doubles both the amount and the volume of the solid phase, so its concentration is unchanged. Since it never varies, we fold it into the constant. The same argument holds for a pure liquid such as water when it is the solvent or a pure liquid reactant.

WARNING **Do not write solids or pure liquids in K**

A very common GCE error is to write $K_c = \dfrac{[CaO][CO_2]}{[CaCO_3]}$. This is wrong. The correct expression is $K_c = [CO_2]$. Remember: **solids and pure liquids are invisible** in equilibrium expressions.

4.3 Writing K_c and K_p for Heterogeneous Systems

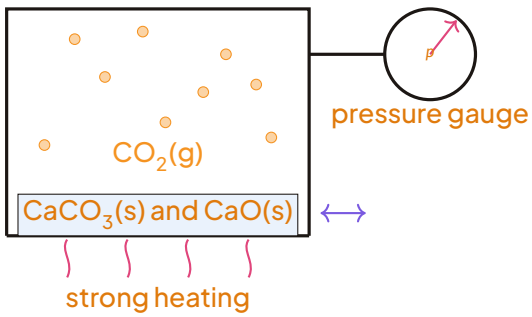
EXEMPLE **The three classic examples**

- $CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$: $K_c = [CO_2]$ and $K_p = p(CO_2)$.
- $NH_4Cl(s) \rightleftharpoons NH_3(g) + HCl(g)$: $K_c = [NH_3][HCl]$ and $K_p = p(NH_3)\,p(HCl)$.
- $C(s) + H_2O(g) \rightleftharpoons CO(g) + H_2(g)$: $K_c = \dfrac{[CO][H_2]}{[H_2O]}$ and $K_p = \dfrac{p(CO)\,p(H_2)}{p(H_2O)}$.

In each case the solid is simply left out.

METHOD **Writing K_p for a decomposition producing a single gas**

1. Write the balanced equation with state symbols.
2. Identify the only gaseous species; ignore all solids and pure liquids.
3. Write K_p as the partial pressure of that gas raised to its stoichiometric coefficient.
4. If the gas is the only species in the gas phase, its partial pressure equals the total measured pressure of the system at equilibrium: $K_p = p_{\text{total}}$.
5. Give units: for $K_p = p(CO_2)$ the units are those of pressure (e.g. atm or Pa).



Thermal decomposition of $CaCO_3$ in a closed container. At equilibrium the gauge reads a constant pressure: $K_p = p(CO_2)$, whatever the amounts of the two solids.

WORKED EXAMPLE 7 **Decomposition of ammonium chloride**

Solid ammonium chloride is heated in an evacuated 5.0 dm^3 vessel at 600 K:

$$NH_4Cl(s) \rightleftharpoons NH_3(g) + HCl(g)$$

At equilibrium the total pressure is 1.0 atm. Calculate K_p at this temperature, stating its units.

SOLUTION

Step 1 — Write K_p. The solid is omitted:

$$K_p = p(NH_3)\,p(HCl)$$

Step 2 — Find the partial pressures. The two gases are produced in a 1 : 1 ratio and are the only gases present, so

$$p(NH_3) = p(HCl) = \tfrac{1}{2} \times 1.0 = 0.50 \text{ atm}.$$

Step 3 — Compute.

$$K_p = 0.50 \times 0.50 = 0.25\ \text{atm}^2.$$

The units are $(\text{atm})^2$ because two pressures are multiplied. Note that the volume of the vessel and the mass of NH_4Cl are irrelevant, provided some solid remains at equilibrium.

WORKED EXAMPLE 8 **From K_p to K_c for a heterogeneous system**

For the equilibrium $CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$ at 1100 K, $K_p = 0.40$ atm. Calculate K_c at the same temperature, in mol dm^{-3}. (Take $R = 0.082\ \text{dm}^3\ \text{atm K}^{-1}\ \text{mol}^{-1}$.)

SOLUTION

Step 1 — Relate concentration and pressure. For the only gaseous species,

$$[CO_2] = \frac{n}{V} = \frac{p(CO_2)}{RT}.$$

Step 2 — Substitute.

$$K_c = [CO_2] = \frac{0.40}{0.082 \times 1100} = \frac{0.40}{90.2} = 4.4 \times 10^{-3}\ \text{mol dm}^{-3}.$$

This illustrates the general relation $K_p = K_c(RT)^{\Delta n_g}$, where here $\Delta n_g = 1$ (one mole of gas produced, none consumed).

DID YOU KNOW? **Lime production in Cameroon**

Quicklime, CaO, is produced by roasting limestone in kilns: $CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$, an endothermic process. Because $K_p = p(CO_2)$, the decomposition only proceeds significantly when the temperature is high enough for the equilibrium pressure of CO_2 to exceed the pressure of CO_2 in the kiln atmosphere. Kiln operators also remove the CO_2 continuously (it is swept away by the draught), so the reverse reaction cannot occur and the limestone decomposes completely. This is Le Chatelier's principle applied on an industrial scale — and the same chemistry underlies cement works such as those at Figuil in the North Region.

4.4 The Effect of Temperature on the Value of K

We already know that changing concentration or pressure shifts the equilibrium position but leaves K unchanged. Temperature is different: it is the **only factor that changes the value of the equilibrium constant**.

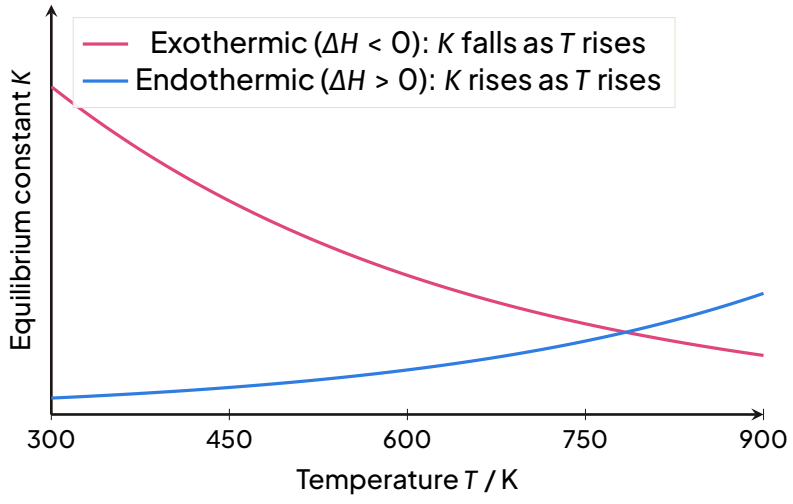
PROPERTY **Temperature and K**

Raising the temperature increases K for an endothermic reaction ($\Delta H > 0$) and decreases K for an exothermic reaction ($\Delta H < 0$). Lowering the temperature has the opposite effect in each case. (The qualitative reasoning via Le Chatelier is examinable; the van 't Hoff equation is not.)

Qualitative justification (Le Chatelier). Treat heat as if it were a "species" in the equation. For an endothermic forward reaction, heat is effectively a reactant: adding heat (raising T) pushes the equilibrium to the right, so more products form and $K = \dfrac{\text{products}}{\text{reactants}}$ grows. For an exothermic forward reaction, heat is a product: raising T pushes the equilibrium to the left, and K falls.

EXEMPLE **Two contrasting cases**

- Endothermic: $CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$, $\Delta H > 0$. Heating the kiln increases $K_p = p(CO_2)$: decomposition is favoured at high temperature.
- Exothermic: $N_2(g) + 3\,H_2(g) \rightleftharpoons 2\,NH_3(g)$, $\Delta H = -92\ \text{kJ mol}^{-1}$. Raising the temperature decreases K_p: the yield of ammonia falls. This is why the Haber process uses a compromise temperature of about 450 °C — high enough for a reasonable rate, low enough for a reasonable yield.



Variation of K with temperature. For an exothermic reaction K decreases as T increases; for an endothermic reaction K increases as T increases.

WARNING **Position versus value: a subtle but crucial distinction**

Do not confuse the effect of temperature on the equilibrium position with its effect on the value of K — the two are perfectly consistent. When K increases, the position shifts towards the products; when K decreases, it shifts towards the reactants. Temperature is special: it changes K itself. Concentration, pressure and catalysts never change K; they only change the position (or, for a catalyst, merely the speed at which equilibrium is reached). Saying "the catalyst increases K" is a serious error that costs marks.

4.5 The van 't Hoff Relationship (Extension)

DID YOU KNOW? **A quantitative link between T, ΔH and K**

Beyond the GCE syllabus, the quantitative link is the van 't Hoff equation:

$$\ln K = -\frac{\Delta H}{R} \cdot \frac{1}{T} + \text{constant}$$

A graph of $\ln K$ against $\frac{1}{T}$ is a straight line whose gradient is $-\frac{\Delta H}{R}$. For an endothermic reaction ($\Delta H > 0$) the gradient is negative; for an exothermic reaction it is positive. Chemists use this plot to measure enthalpy changes experimentally. You only need to know it qualitatively: it confirms that K depends exponentially on temperature, which is why even modest heating can transform the yield of a kiln or a reactor.

WORKED EXAMPLE 9 **Predicting the effect of heating**

For the water-gas reaction $C(s) + H_2O(g) \rightleftharpoons CO(g) + H_2(g)$, $\Delta H = +131\ \text{kJ mol}^{-1}$.

1. Write expressions for K_c and K_p.
2. State and explain the effect of raising the temperature on (i) the value of K_p and (ii) the yield of hydrogen.

SOLUTION

1. The carbon is a pure solid, so it is omitted:

$$K_c = \frac{[CO][H_2]}{[H_2O]}, \qquad K_p = \frac{p(CO)\,p(H_2)}{p(H_2O)}.$$

2. The forward reaction is endothermic ($\Delta H > 0$). By Le Chatelier's principle, raising the temperature shifts the equilibrium in the direction that absorbs heat, i.e. to the right. Consequently (i) K_p increases (more products relative to reactants at equilibrium) and (ii) the equilibrium yield of hydrogen increases. This is why industrial hydrogen production

by this route is carried out at high temperature.

EXERCISE 4 **Limestone in a sealed vessel**

A sample of $CaCO_3$ is placed in a sealed, evacuated vessel and heated to 1100 K. At equilibrium the pressure in the vessel is 0.40 atm.

1. Write the expression for K_p and calculate its value, with units.
2. A second, larger lump of $CaCO_3$ is used under identical conditions. State, with a reason, the new equilibrium pressure.
3. The decomposition is endothermic. State and explain the effect on K_p of heating the vessel to 1200 K.

SOLUTION **Exercise 1**

1. $CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$; solids are omitted, so $K_p = p(CO_2)$. Since CO_2 is the only gas, $p(CO_2) = 0.40$ atm, hence $K_p = 0.40$ atm.
2. The pressure is still 0.40 atm. The concentration (and hence the contribution) of a pure solid is constant, so the equilibrium pressure of CO_2 depends only on temperature, not on the amount of solid — provided at least some of each solid remains.
3. K_p increases. The forward reaction is endothermic, so by Le Chatelier's principle raising the temperature shifts the equilibrium to the right, increasing $p(CO_2)$ and therefore K_p.

EXERCISE 5 **Steam over hot carbon**

Steam is passed over red-hot carbon in a closed reactor at 1000 K:

$$C(s) + H_2O(g) \rightleftharpoons CO(g) + H_2(g) \qquad \Delta H = +131\ \text{kJ mol}^{-1}$$

1. Write the expression for K_p.
2. At equilibrium the partial pressures are $p(H_2O) = 0.20$ atm, $p(CO) = 0.80$ atm and $p(H_2) = 0.80$ atm. Calculate K_p, with units.
3. State two changes that would increase the equilibrium yield of hydrogen, justifying each.

SOLUTION **Exercise 2**

1. Carbon is a pure solid and is omitted: $K_p = \dfrac{p(\mathrm{CO})\,p(\mathrm{H_2})}{p(\mathrm{H_2O})}$.

2. $K_p = \dfrac{0.80 \times 0.80}{0.20} = 3.2$ atm. The units are atm because there are two pressure terms in the numerator and one in the denominator ($\Delta n_g = +1$).

3. (i) Raise the temperature: the forward reaction is endothermic, so by Le Chatelier's principle the equilibrium shifts to the right and K_p increases. (ii) Decrease the total pressure (or increase the volume): there are two moles of gas on the right and one on the left, so the equilibrium shifts to the side with more moles of gas, i.e. towards the products. (Removing CO or H_2 as they form would also work.)

KEY POINTS **Key points of this section**

- A heterogeneous equilibrium involves species in more than one phase.
- Pure solids and pure liquids never appear in K_c or K_p; their concentrations are constant.
- For $\mathrm{CaCO_3(s)} \rightleftharpoons \mathrm{CaO(s)} + \mathrm{CO_2(g)}$: $K_p = p(\mathrm{CO_2})$ — the basis of lime manufacture.
- K_p and K_c remain linked by $K_p = K_c(RT)^{\Delta n_g}$, counting only gaseous species in Δn_g.
- Temperature is the only factor that changes the value of K: raising T increases K for an endothermic reaction and decreases it for an exothermic one.
- The van 't Hoff equation, $\ln K = -\dfrac{\Delta H}{R} \cdot \dfrac{1}{T} + \text{constant}$, makes this quantitative (extension, qualitative use only).

5 REDOX SYSTEMS AND BALANCING REDOX EQUATIONS

In the previous section we saw that the position of a heterogeneous equilibrium, and indeed the value of any equilibrium constant, depends on temperature through the enthalpy change of the reaction. We treated equilibrium as a competition between forward and backward processes. Many of the most important reactions in chemistry — the rusting of iron roofing sheets in Douala, the working of a car battery, the extraction of metals from their ores — are equilibria of a special kind: they involve a **transfer of electrons** from one species to another. Such reactions are called **redox reactions**, and this section builds the language and the techniques needed to describe, balance and (in the next sections) quantify them.

5.1 Oxidation, Reduction and Redox Agents

Intuition from a simple experiment. When a strip of zinc metal is dipped into a blue solution of

copper(II) sulphate, the blue colour fades and a reddish-brown deposit of copper forms on the zinc. Analysis shows that zinc has gone into solution as Zn^{2+} ions while Cu^{2+} ions have become copper metal:

$$Zn(s) + Cu^{2+}(aq) \longrightarrow Zn^{2+}(aq) + Cu(s)$$

Each zinc atom has lost two electrons; each copper(II) ion has gained two electrons. The two processes cannot occur separately: the electrons lost by zinc are exactly those gained by copper. This coupling of electron loss and electron gain is the essence of every redox reaction.

DEFINITION **Oxidation and Reduction**

Oxidation is the loss of electrons by a species; equivalently, it is an increase in oxidation number. **Reduction** is the gain of electrons by a species; equivalently, it is a decrease in oxidation number. The mnemonic OILRIG summarises this: Oxidation Is Loss, Reduction Is Gain (of electrons).

DEFINITION **Oxidising and Reducing Agents**

An **oxidising agent** (oxidant) is a species that accepts electrons and is therefore itself reduced during the reaction. A **reducing agent** (reductant) is a species that donates electrons and is therefore itself oxidised.

In the zinc–copper reaction above, Zn is the reducing agent (it is oxidised to Zn^{2+}) and Cu^{2+} is the oxidising agent (it is reduced to Cu).

WARNING **The agent is named after what it does to the other species**

Students frequently confuse the labels. The oxidising agent causes oxidation but is itself reduced. Always ask: "Which species gains electrons?" — that species is reduced and is the oxidising agent.

5.2 Oxidation Numbers: the Book-keeping Tool

To decide whether oxidation or reduction has occurred, we assign an **oxidation number** (oxidation state) to each atom, using the conventional rules:

- an uncombined element has oxidation number 0;
- a monatomic ion has an oxidation number equal to its charge;
- oxygen is usually −2 (except peroxides, −1, and OF_2, +2);
- hydrogen is usually +1 (except metal hydrides such as NaH, −1);
- the sum of oxidation numbers equals the overall charge of the species.

EXEMPLE **Oxidation number of Mn in MnO_4^-**

Let the oxidation number of Mn be x. Then $x + 4(-2) = -1$, so $x = +7$. Manganese is in the +7 state in the permanganate ion. When MnO_4^- is converted to Mn^{2+} (oxidation number +2), manganese has been reduced — its oxidation number has decreased by 5, corresponding to a gain of 5 electrons.

5.3 Redox Systems: Half-Cells, Cells, Anode and Cathode

Separating the two processes. In the zinc–copper reaction, electrons pass directly from zinc to Cu^{2+}. But suppose we physically separate the two processes: place a zinc rod in a solution of Zn^{2+} ions in one beaker, and a copper rod in a solution of Cu^{2+} ions in another, then connect the rods by a wire and the solutions by a salt bridge. Electrons now flow through the wire from zinc to copper, and we can make them do useful work. Each beaker is a **half-cell**; the combination is an **electrochemical cell**.

DEFINITION **Half-cell and electrochemical cell**

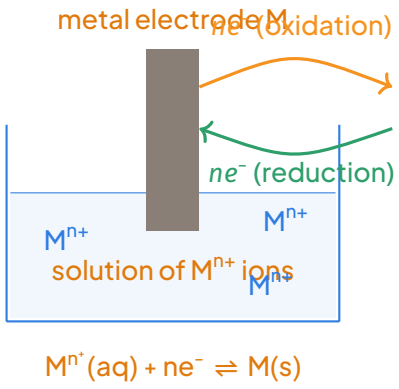
A **half-cell** consists of an **electrode** (an electronic conductor, often a metal or inert graphite/platinum) in contact with an electrolyte containing the oxidised and reduced forms of a redox couple. An **electrochemical cell** is formed by coupling two half-cells so that electrons can flow through an external circuit while ions complete the circuit internally (through a salt bridge or porous partition).

DEFINITION **Anode and cathode**

The **anode** is the electrode at which oxidation occurs; the **cathode** is the electrode at which reduction occurs. Mnemonic: AN OX, RED CAT — ANode = OXidation; REDuction at the CAThode.

WARNING **The anode is always the site of oxidation**

This definition is universal: it holds for galvanic (voltaic) cells and for electrolytic cells. In a galvanic cell the anode is the negative terminal, whereas in electrolysis the anode is connected to the positive terminal of the supply — but in both cases oxidation occurs at the anode. Never define the anode as "the positive electrode".



A generic metal/metal-ion half-cell: the electrode and the ions in solution are in dynamic redox equilibrium, $M^{n+} + ne^- \rightleftharpoons M$.

5.4 Redox Potential and Redox Equilibria

When a metal electrode is dipped into a solution of its ions, an equilibrium is rapidly established:

$$M^{n+}(aq) + n\,e^- \rightleftharpoons M(s)$$

Some metals (like zinc) "push" electrons onto the electrode strongly — the equilibrium lies towards electron release; others (like copper) hold their electrons more tightly. A **potential difference** therefore develops between the electrode and the solution. This leads to the key concept:

DEFINITION **Redox (electrode) potential**

The **redox potential** (electrode potential) of a half-cell is a measure of the tendency of the oxidised form of a redox couple to gain electrons (be reduced). The more positive the redox potential, the greater the tendency of the couple $Ox + ne^- \rightleftharpoons Red$ to proceed to the right, i.e. the stronger the oxidising agent Ox and the weaker the reducing agent Red.

Every redox reaction can be regarded as the **combination of two half-equations**, one for each couple. For the zinc–copper reaction:

$$\text{oxidation:}\quad Zn(s) \longrightarrow Zn^{2+}(aq) + 2\,e^-$$

$$\text{reduction:}\quad Cu^{2+}(aq) + 2\,e^- \longrightarrow Cu(s)$$

$$\text{overall:}\quad Zn(s) + Cu^{2+}(aq) \longrightarrow Zn^{2+}(aq) + Cu(s)$$

The electrons cancel because the number lost equals the number gained — a point we shall exploit systematically when balancing equations.

5.5 Writing Half-Equations for Simple Systems

A **half-equation** shows the oxidised and reduced forms of one couple, balanced for atoms and

for charge using electrons. Common examples you must be able to write quickly:

$$Zn^{2+}(aq) + 2\,e^- \rightleftharpoons Zn(s) \qquad Fe^{3+}(aq) + e^- \rightleftharpoons Fe^{2+}(aq)$$
$$Cl_2(g) + 2\,e^- \rightleftharpoons 2\,Cl^-(aq) \qquad I_2(aq) + 2\,e^- \rightleftharpoons 2\,I^-(aq)$$
$$2\,H^+(aq) + 2\,e^- \rightleftharpoons H_2(g) \qquad Cu^{2+}(aq) + 2\,e^- \rightleftharpoons Cu(s)$$

For oxoanions in acidic solution, water and H^+ must be included, e.g.

$$MnO_4^-(aq) + 8\,H^+(aq) + 5\,e^- \rightleftharpoons Mn^{2+}(aq) + 4\,H_2O(l)$$

$$Cr_2O_7^{2-}(aq) + 14\,H^+(aq) + 6\,e^- \rightleftharpoons 2\,Cr^{3+}(aq) + 7\,H_2O(l)$$

5.6 Balancing Redox Equations: the Ion-Electron (Half-Equation) Method

Many redox equations are far too complicated to balance by inspection. The **ion-electron method** (valid in acidic medium) is systematic and always works.

METHOD **Balancing a redox equation in acidic medium**

1. Identify the species oxidised and reduced (use oxidation numbers) and write the two skeleton half-equations.
2. Balance the atoms other than O and H in each half-equation.
3. Balance oxygen by adding H_2O molecules to the side deficient in O.
4. Balance hydrogen by adding H^+ ions to the side deficient in H.
5. Balance charge by adding electrons, e^-, to the more positive side.
6. Multiply each half-equation so that the numbers of electrons are equal, then add the half-equations.
7. Cancel electrons, H^+ and H_2O appearing on both sides; check atoms and charges balance.

WORKED EXAMPLE 10 **Balancing MnO_4^- + Fe^{2+} ⟶ Mn^{2+} + Fe^{3+} in acidic solution**

Statement: Acidified potassium permanganate oxidises iron(II) ions to iron(III) ions, the permanganate being reduced to manganese(II). Construct the balanced overall ionic equation.

SOLUTION

Step 1–5, reduction half-equation. Mn goes from +7 in MnO_4^- to +2 in Mn^{2+} (gain of $5e^-$).

Balance O with H_2O, H with H^+:

$$MnO_4^-(aq) + 8\,H^+(aq) + 5\,e^- \longrightarrow Mn^{2+}(aq) + 4\,H_2O(l)$$

Check charge: left $-1 + 8 - 5 = +2$; right $+2$. ✓
Step 1–5, oxidation half-equation. Fe goes from +2 to +3 (loss of $1e^-$):

$$Fe^{2+}(aq) \longrightarrow Fe^{3+}(aq) + e^-$$

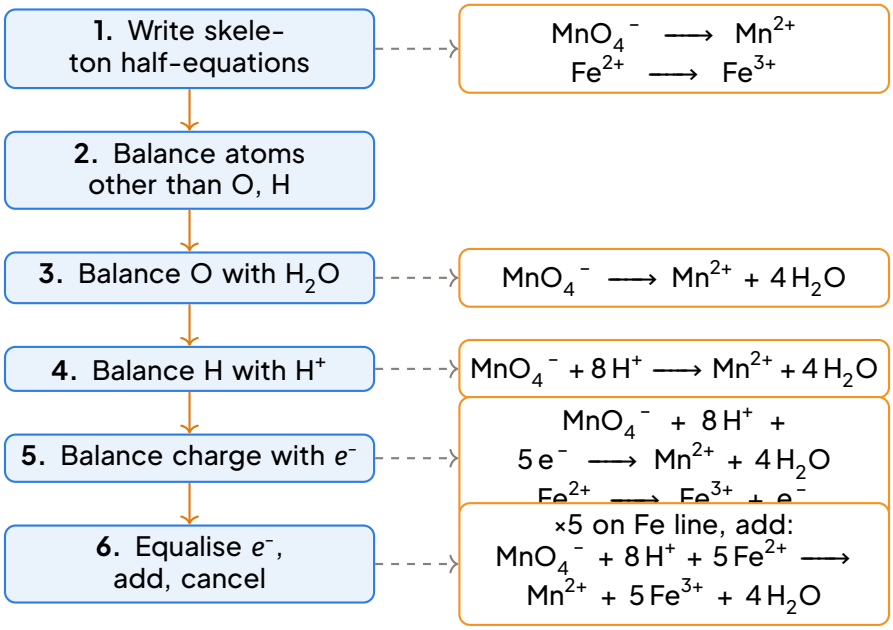
Step 6. Multiply the iron half-equation by 5 and add:

$$MnO_4^- + 8\,H^+ + 5\,e^- + 5\,Fe^{2+} \longrightarrow Mn^{2+} + 4\,H_2O + 5\,Fe^{3+} + 5\,e^-$$

Step 7. Cancel the $5e^-$:

$$\boxed{MnO_4^-(aq) + 8\,H^+(aq) + 5\,Fe^{2+}(aq) \longrightarrow Mn^{2+}(aq) + 5\,Fe^{3+}(aq) + 4\,H_2O(l)}$$

Final check — atoms: Mn 1/1, O 4/4, H 8/8, Fe 5/5; charge: left $-1 + 8 + 10 = +17$, right $+2 + 15 = +17$. ✓



Flowchart of the ion-electron method, applied to the MnO_4^-/Fe^{2+} reaction in acidic medium.

WARNING **No electrons in the final equation**

Electrons are intermediates of the method, not reactants or products. If e^- appears in

your final overall equation, you have forgotten to multiply and cancel — a very common GCE error. Also check that the total charge is the same on both sides, not only the atoms.

A note on alkaline medium. If a reaction occurs in alkaline solution, first balance it exactly as for acidic medium, then add to both sides as many OH^- ions as there are H^+ ions; each $H^+ + OH^-$ pair becomes H_2O. Cancel any water molecules that then appear on both sides. For example, the oxidation of Cr^{3+} to CrO_4^{2-} by hydrogen peroxide in alkaline solution gives

$$2\,Cr^{3+}(aq) + 3\,H_2O_2(aq) + 10\,OH^-(aq) \longrightarrow 2\,CrO_4^{2-}(aq) + 8\,H_2O(l)$$

At GCE Advanced Level, acidic-medium balancing is by far the most frequently examined; the alkaline conversion is a useful extension.

5.7 Balancing Using Oxidation Numbers

An alternative, equally acceptable, method uses oxidation numbers directly: the **total increase in oxidation number must equal the total decrease**.

WORKED EXAMPLE 11 **Balancing $Cu + HNO_3 \longrightarrow Cu(NO_3)_2 + NO + H_2O$**

Statement: Dilute nitric acid oxidises copper metal, being itself reduced to nitrogen monoxide. Balance the equation.

SOLUTION

Oxidation number changes: Cu: 0 → +2, an increase of 2. N (in the HNO_3 that is reduced): +5 → +2 in NO, a decrease of 3. To equalise, take 3 Cu (total increase $3 \times 2 = 6$) and 2 N reduced (total decrease $2 \times 3 = 6$):

$$3\,Cu + 2\,HNO_3 \longrightarrow 3\,Cu(NO_3)_2 + 2\,NO + H_2O$$

Now balance the remaining nitrate: $3\,Cu(NO_3)_2$ needs 6 more NO_3^-, i.e. 6 further HNO_3, giving 8 HNO_3 in total; then balance H and O with water:

$$3\,Cu(s) + 8\,HNO_3(aq) \longrightarrow 3\,Cu(NO_3)_2(aq) + 2\,NO(g) + 4\,H_2O(l)$$

Check: Cu 3/3, H 8/8, N 8/8, O 24/24. ✓

5.8 Disproportionation: a Useful Extra

DEFINITION **Disproportionation**

Disproportionation is a redox reaction in which the same element, in the same species, is simultaneously oxidised and reduced — its oxidation number both rises and falls.

Classic examples include the reaction of chlorine with cold dilute alkali,

$$Cl_2(aq) + 2\,OH^-(aq) \longrightarrow Cl^-(aq) + ClO^-(aq) + H_2O(l)$$

in which chlorine (oxidation number 0) is reduced to Cl^- (−1) and oxidised to ClO^- (+1), and the decomposition of hydrogen peroxide,

$$2\,H_2O_2(aq) \longrightarrow 2\,H_2O(l) + O_2(g)$$

in which oxygen (−1) goes to −2 (in water) and 0 (in O_2). Disproportionation questions appear regularly at GCE: the key skill is recognising that one reactant plays the roles of both oxidising and reducing agent.

DID YOU KNOW? **Redox in everyday Cameroonian life**

The lead–acid battery that starts taxis and motorbikes across Cameroon works on the redox couples $PbO_2/PbSO_4$ and $Pb/PbSO_4$; the rusting of corrugated iron roofs in the humid coastal climate is the slow oxidation of iron by atmospheric oxygen; and the iodine value of palm oil, important in the agro-food industry, is measured by a redox titration with iodine. Redox chemistry is never far away.

KEY POINTS **Key points of this section**

- Oxidation = loss of electrons / increase in oxidation number; reduction = gain of electrons / decrease in oxidation number (OILRIG).
- The oxidising agent is reduced; the reducing agent is oxidised.
- A half-cell is an electrode in contact with a redox couple, $Ox + ne^- \rightleftharpoons Red$; two half-cells make an electrochemical cell.
- Oxidation always occurs at the anode, reduction at the cathode — in every type of cell.
- The redox potential measures the tendency of a couple to gain electrons.
- Balance redox equations by half-equations (atoms, then O with H_2O, then H with H^+, then charge with e^-) or by oxidation numbers; electrons must cancel in the final equation.

EXERCISE 6 **Practice**

1. Assign oxidation numbers to S in SO_2, SO_3^{2-} and SO_4^{2-}, and state whether the conversion $SO_2 \longrightarrow SO_4^{2-}$ is an oxidation or a reduction.

2. Write balanced half-equations for: (a) Br_2 reduced to Br^-; (b) Sn^{2+} oxidised to Sn^{4+}.
3. Balance, in acidic solution: $Cr_2O_7^{2-}(aq) + Fe^{2+}(aq) \longrightarrow Cr^{3+}(aq) + Fe^{3+}(aq)$.
4. Show that the reaction $2\,Cu^+(aq) \longrightarrow Cu(s) + Cu^{2+}(aq)$ is a disproportionation.

SOLUTION **Exercise 1**

1. S in SO_2: $x+2(-2) = 0 \Rightarrow x = +4$. In SO_3^{2-}: $x-6 = -2 \Rightarrow x = +4$. In SO_4^{2-}: $x-8 = -2 \Rightarrow x = +6$. The change $+4 \rightarrow +6$ is an increase, so SO_2 is oxidised.
2. (a) $Br_2(aq) + 2\,e^- \longrightarrow 2\,Br^-(aq)$; (b) $Sn^{2+}(aq) \longrightarrow Sn^{4+}(aq) + 2\,e^-$.
3. Reduction: $Cr_2O_7^{2-} + 14\,H^+ + 6\,e^- \longrightarrow 2\,Cr^{3+} + 7\,H_2O$. Oxidation: $Fe^{2+} \longrightarrow Fe^{3+} + e^-$, multiplied by 6. Overall:

$$Cr_2O_7^{2-}(aq) + 14\,H^+(aq) + 6\,Fe^{2+}(aq) \longrightarrow 2\,Cr^{3+}(aq) + 6\,Fe^{3+}(aq) + 7\,H_2O(l)$$

Check charge: left $-2 + 14 + 12 = +24$; right $+6 + 18 = +24$. ✓
4. In Cu^+, copper has oxidation number +1. One Cu^+ is reduced to Cu (0) while the other is oxidised to Cu^{2+} (+2): the same element in the same species is both oxidised and reduced — a disproportionation.

6 TYPES OF HALF-CELLS AND STANDARD ELECTRODE POTENTIALS

In the previous section we learnt to recognise redox systems, to write half-equations and to balance full redox equations. We saw, for example, that when a strip of zinc is dipped into a solution of copper(II) sulphate, electrons flow spontaneously from zinc atoms to copper(II) ions. A natural question now arises: how strongly does each redox couple "pull" or "push" electrons? Can we attach a number to each couple that tells us how powerful an oxidising or reducing agent it is? The answer is yes, and that number is the **standard electrode potential**, E^θ. To measure it, we must first understand how a redox couple is turned into a physical device called a **half-cell**, and we must agree on a universal reference against which all half-cells are compared. That is the business of this section.

6.1 From a Redox Couple to a Half-Cell

DEFINITION **Half-cell**

A **half-cell** is an assembly consisting of an **electrode** (an electronic conductor) in contact

with an **electrolyte** containing the two species of a redox couple (the oxidised form and the reduced form). A half-cell is the physical realisation of one half-equation, for example $Cu^{2+}(aq) + 2\,e^- \rightleftharpoons Cu(s)$.

A half-cell alone cannot drive a sustained current: electrons must have somewhere to go. Only when two half-cells are connected (through a wire and a salt bridge) does a complete electrochemical cell exist, as we shall study in the next section. Here we focus on the structure of the half-cell itself. Depending on the physical nature of the two members of the redox couple, three types of half-cell are distinguished, exactly as listed in the syllabus.

6.2 The Three Types of Half-Cells

Type 1: Metal / metal-ion systems.

This is the simplest case: the reduced form of the couple is a solid metal, which is itself an excellent conductor of electricity. The metal strip serves both as the reduced species and as the electrode. It is simply dipped into a solution of its own ions. Examples:

- $Zn^{2+}(aq) + 2\,e^- \rightleftharpoons Zn(s)$: a zinc rod in zinc sulphate solution;
- $Cu^{2+}(aq) + 2\,e^- \rightleftharpoons Cu(s)$: a copper strip in copper(II) sulphate solution;
- $Ag^+(aq) + e^- \rightleftharpoons Ag(s)$: a silver wire in silver nitrate solution.

Type 2: Non-metal / non-metal-ion systems.

Here neither member of the couple is a metallic conductor. A common example is the chlorine / chloride couple $Cl_2(g) + 2\,e^- \rightleftharpoons 2\,Cl^-(aq)$: chlorine is a gas and chloride ions are in solution; there is no solid metal to carry electrons into or out of the system. The solution is to bubble the gas over an **inert electrode**, usually platinum, which dips into the solution containing the ions. The hydrogen half-cell $2\,H^+(aq) + 2\,e^- \rightleftharpoons H_2(g)$ (often written $H^+(aq) + e^- \rightleftharpoons \frac{1}{2}\,H_2(g)$) is built in exactly the same way, with hydrogen gas bubbled over platinum. Other examples include the oxygen / hydroxide couple $O_2(g) + 2\,H_2O(l) + 4\,e^- \rightleftharpoons 4\,OH^-(aq)$.

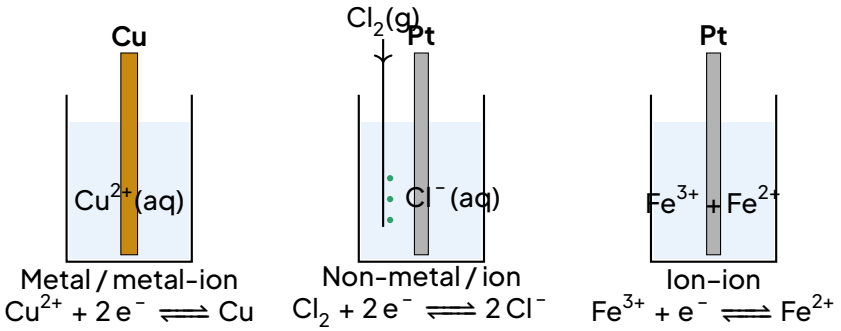
WARNING **Why platinum?**

Platinum is used as an **inert electrode** whenever the half-cell contains no metallic conductor. Platinum is chosen because it conducts electricity, is chemically unreactive (it does not take part in the redox reaction), and its surface catalyses the electron transfer between the dissolved/gaseous species and the external circuit. A very common exam mistake is to say that "platinum is one of the reactants" — it is not; it appears in no half-equation.

Type 3: Ion–ion systems.

Here both members of the couple are ions in the same solution, for example $Fe^{3+}(aq) + e^- \rightleftharpoons Fe^{2+}(aq)$ or $MnO_4^-(aq) + 8\,H^+(aq) + 5\,e^- \rightleftharpoons Mn^{2+}(aq) + 4\,H_2O(l)$. Again there is no metal to act as electrode, so a platinum wire is dipped into a solution containing both ions of the couple.

Electrons are exchanged between the ions and the external circuit through the inert platinum surface.



The three types of half-cells. Platinum (Pt) is used as an inert electrode whenever the couple contains no metallic conductor.

KEY POINTS **Choosing the electrode**

If the reduced form of the couple is a metal, that metal is the electrode. If not (gas, or two dissolved ions), use an inert platinum electrode. For ion–ion half-cells, the solution must contain both ions of the couple.

6.3 The Need for a Reference: the Standard Hydrogen Electrode (SHE)

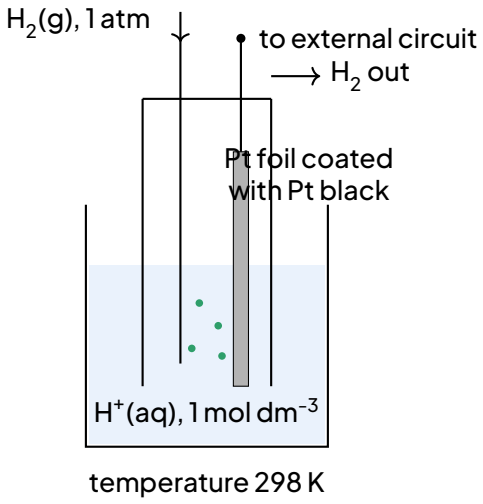
There is no instrument that measures the "absolute" tendency of a single half-cell to gain electrons. What a voltmeter can measure is the difference of potential between two half-cells. Scientists therefore agreed to compare every half-cell with one universal reference, exactly as altitudes on Earth are measured relative to sea level. The chosen "sea level" of electrochemistry is the **standard hydrogen electrode** (SHE).

DEFINITION **Standard hydrogen electrode**

The **standard hydrogen electrode** consists of a platinum electrode coated with finely divided platinum ("platinum black"), over which pure hydrogen gas at a pressure of 1 atm (101 kPa) is bubbled, immersed in a solution of hydrogen ions (e.g. dilute HCl) of concentration 1 mol dm^{-3}, at 298 K. Its half-equation is

$$2\,H^+(aq) + 2\,e^- \rightleftharpoons H_2(g) \qquad E^\theta = 0.00\ V$$

By international agreement, its standard electrode potential is assigned the value exactly zero at all temperatures.



The standard hydrogen electrode (SHE): hydrogen at 1 atm bubbled over platinised platinum in $H^+(aq)$ at 1 mol dm^{-3} and 298 K.

PROPERTY **Standard conditions**

Standard electrode potentials are measured under **standard conditions**:

- temperature: 298 K (25 °C);
- concentration of every aqueous species: 1 mol dm^{-3};
- pressure of any gas: 1 atm (101 kPa);
- any solid present must be pure.

(These conditions must be known and quoted precisely; the proof is not examinable.)

6.4 Standard Electrode Potential E^θ

DEFINITION **Standard electrode potential**

The **standard electrode potential** E^θ of a half-cell is the e.m.f. of a cell formed by coupling that half-cell, under standard conditions, to a standard hydrogen electrode. It is measured with a high-resistance voltmeter so that (almost) no current flows. By convention, E^θ is always quoted for the reduction half-equation written as oxidised form + ne^- → reduced form.

WARNING **E^θ is a reduction potential**

E^θ values are always quoted for the reduction half-equation, with electrons on the left: $Zn^{2+} + 2\,e^- \longrightarrow Zn$, $E^\theta = -0.76$ V. Never write the oxidation equation ($Zn \longrightarrow Zn^{2+} + 2\,e^-$) and attach the same value to it. Also, E^θ is not multiplied when the half-equation is multiplied: for $2\,Zn^{2+} + 4\,e^- \longrightarrow 2\,Zn$ the potential is still -0.76 V, because E^θ is an intensive quantity (a voltage, not an amount of energy).

METHOD **Measuring E^θ of a half-cell against the SHE**

1. Set up the half-cell under standard conditions (1 mol dm^{-3} solutions, 298 K, 1 atm for gases).
2. Connect it to a standard hydrogen electrode with a salt bridge (e.g. filter paper soaked in saturated KNO_3 solution).
3. Join the two electrodes through a high-resistance voltmeter.
4. Read the e.m.f.: its magnitude is $|E^\theta|$ of the half-cell.
5. Note the polarity: if the electrode under test is the negative terminal (electrons flow from it to the SHE), its E^θ is negative; if it is the positive terminal, its E^θ is positive.

EXEMPLE **Sign convention in practice**

Coupling a standard Zn^{2+}/Zn half-cell to the SHE gives a voltmeter reading of 0.76 V, and the zinc electrode is found to be the negative terminal (electrons flow from zinc to hydrogen electrode, where $2\,H^+ + 2\,e^- \longrightarrow H_2$ occurs). Hence $E^\theta(Zn^{2+}/Zn) = -0.76$ V. With a Cu^{2+}/Cu half-cell, the reading is 0.34 V but copper is the positive terminal, so $E^\theta(Cu^{2+}/Cu) = +0.34$ V. A negative E^θ means the couple is a better reducing system than H^+/H_2; a positive E^θ means it is a better oxidising system.

6.5 Other Reference Electrodes

The SHE is the defining reference, but it is awkward in daily laboratory work: it needs a supply of pure hydrogen gas, a precise pressure control, and the platinum surface is easily poisoned by impurities. In practice, chemists use robust secondary reference electrodes whose potentials against the SHE are accurately known.

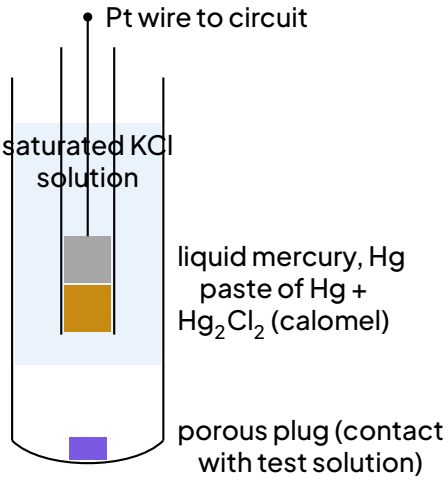
The calomel electrode.

"Calomel" is the old name for mercury(I) chloride, Hg_2Cl_2. The electrode consists of mercury in contact with a paste of mercury and solid Hg_2Cl_2, immersed in a potassium chloride solution of

known concentration. The half-equation is

$$Hg_2Cl_2(s) + 2\,e^- \rightleftharpoons 2\,Hg(l) + 2\,Cl^-(aq)$$

With saturated KCl (the "saturated calomel electrode", SCE) its potential is +0.24 V versus the SHE at 298 K. It is compact, stable and easy to use, which is why it is the reference built into many pH meters.



The saturated calomel electrode (SCE), a practical secondary reference electrode: E = +0.24 V versus the SHE at 298 K.

The silver / silver chloride electrode.

A silver wire coated with a layer of solid silver chloride is dipped into a solution of chloride ions (usually KCl):

$$AgCl(s) + e^- \rightleftharpoons Ag(s) + Cl^-(aq) \qquad E \approx +0.22\ V \text{ (saturated KCl, vs SHE)}$$

It is even smaller and more convenient than the calomel electrode and is the reference most commonly found inside modern pH and ion-selective electrodes. To use any secondary reference, one simply subtracts its known potential: if a half-cell shows +0.52 V against the SCE, then E^θ = 0.52 + 0.24 = +0.76 V against the SHE.

6.6 The Electrochemical Series

Once every half-cell has been measured against the SHE, the couples can be ranked in a single table ordered by E^θ: this is the **electrochemical series**.

Reduction half-equation		E^θ / V
$Li^+ + e^- \rightleftharpoons Li$	strongest reducing agent on the right	-3.04
$Zn^{2+} + 2\,e^- \rightleftharpoons Zn$		-0.76
$Fe^{2+} + 2\,e^- \rightleftharpoons Fe$		-0.44
$2\,H^+ + 2\,e^- \rightleftharpoons H_2$	reference	0.00
$Cu^{2+} + 2\,e^- \rightleftharpoons Cu$		+0.34
$Fe^{3+} + e^- \rightleftharpoons Fe^{2+}$		+0.77
$Cl_2 + 2\,e^- \rightleftharpoons 2\,Cl^-$	strongest oxidising agent on the left	+1.36

The more negative E^θ, the more readily the reduced form (right-hand side) gives away electrons: lithium and zinc are powerful reducing agents. The more positive E^θ, the more readily the oxidised form (left-hand side) accepts electrons: chlorine is a powerful oxidising agent. This single table will allow us, in the next section, to predict the direction of redox reactions and to calculate the e.m.f. of any cell.

EXPERIMENT **Setting up a half-cell in the laboratory**

Objective: Construct a standard Cu^{2+}/Cu half-cell and a Fe^{3+}/Fe^{2+} half-cell. Apparatus: Two beakers (100 cm^3), copper strip, platinum wire, $CuSO_4$ solution (1 mol dm^{-3}), $Fe_2(SO_4)_3$ and $FeSO_4$ solutions (mixed to give 1 mol dm^{-3} each), connecting wires, voltmeter, salt bridge (filter paper soaked in KNO_3), SHE apparatus (or a secondary reference). Procedure:

1. Polish the copper strip and platinum wire with emery paper; rinse with distilled water.
2. Place the copper strip in the $CuSO_4$ solution (Type 1 half-cell).
3. Place the platinum wire in the mixed Fe^{3+}/Fe^{2+} solution (Type 3 half-cell).
4. Connect each half-cell to the reference electrode via a salt bridge.
5. Measure the e.m.f. with a high-resistance voltmeter. Note the polarity.

Observation: The copper electrode is positive relative to SHE (E^θ = +0.34 V). The platinum electrode in the iron solution is positive relative to SHE (E^θ = +0.77 V). Conclusion: The measured potentials confirm the positions of these couples in the electrochemical series.

DID YOU KNOW? **The Daniell Cell and the Telegraph**

The first practical application of electrochemical cells was the telegraph, revolutionising communication in the 19th century. The Daniell cell ($Zn|Zn^{2+}||Cu^{2+}|Cu$) provided a steadier voltage than the earlier Voltaic pile because the porous pot (early salt bridge) prevented polarisation. In Cameroon, the principles of these cells underpin the lead-acid batteries that start our vehicles and the dry cells powering torches in rural areas. Understanding E^θ

allows engineers to choose the best couples for high-energy batteries (like Lithium-ion, $E^\theta \approx -3.04$ V for Li) or for corrosion protection (sacrificial anodes using Zn or Mg).

WORKED EXAMPLE 12 **Identifying half-cell types and predicting polarity**

For each of the following redox couples, state the type of half-cell, the electrode material, and whether its E^θ is positive or negative relative to the SHE: (a) Ag^+/Ag ($E^\theta = +0.80$ V); (b) MnO_4^-/Mn^{2+} ($E^\theta = +1.51$ V); (c) Fe^{2+}/Fe ($E^\theta = -0.44$ V).

SOLUTION

(a) Ag^+/Ag: the reduced form is a metal, so this is a metal/metal-ion half-cell: a silver wire dipped in $AgNO_3$ solution (1 mol dm^{-3}). $E^\theta = +0.80$ V is positive: silver is the positive terminal against the SHE (reduction $Ag^+ + e^- \longrightarrow Ag$ occurs there).

(b) MnO_4^-/Mn^{2+}: both species are ions in solution, so this is an ion–ion half-cell requiring an inert platinum electrode dipped in a solution containing MnO_4^-, Mn^{2+} and H^+, each at 1 mol dm^{-3}. $E^\theta = +1.51$ V is positive: acidified permanganate is a very strong oxidising agent.

(c) Fe^{2+}/Fe: metal/metal-ion half-cell: an iron nail in $FeSO_4$ solution. $E^\theta = -0.44$ V is negative: iron is the negative terminal against the SHE; electrons flow from iron to the hydrogen electrode, so iron is a reducing agent stronger than H_2.

EXERCISE 7 **Half-cells and the SHE**

1. Draw a fully labelled diagram of the half-cell you would use to measure E^θ for the couple Cl_2/Cl^-, stating all standard conditions.
2. A standard Ni^{2+}/Ni half-cell is connected to the SHE. The voltmeter reads 0.25 V and nickel is the negative terminal. Deduce $E^\theta(Ni^{2+}/Ni)$ and write the half-equation that occurs at each electrode.
3. A half-cell gives an e.m.f. of +0.10 V when measured against a saturated calomel electrode ($E = +0.24$ V vs SHE), the unknown electrode being positive. Calculate its standard electrode potential.

SOLUTION **Exercise 1**

1. Beaker containing $Cl^-(aq)$ at 1 mol dm^{-3} (e.g. NaCl solution); chlorine gas bubbled at 1 atm over a platinum electrode; temperature 298 K. The diagram is the middle half-cell

of the figure of the three types, labelled with these conditions.
2. Nickel is the negative terminal, so $E^{\theta}(Ni^{2+}/Ni) = -0.25\,V$. At the nickel electrode (oxidation): $Ni(s) \longrightarrow Ni^{2+}(aq) + 2\,e^{-}$. At the SHE (reduction): $2\,H^{+}(aq) + 2\,e^{-} \longrightarrow H_2(g)$.
3. $E^{\theta} = E_{measured} + E_{ref} = 0.10 + 0.24 = +0.34\,V$ versus the SHE.

KEY POINTS **Key points**

- Three types of half-cells: metal/metal-ion (the metal is the electrode), non-metal/ion and ion–ion (both need an inert platinum electrode).
- The SHE ($2\,H^{+} + 2\,e^{-} \rightleftharpoons H_2$, $1\,mol\,dm^{-3}\,H^{+}$, H_2 at 1 atm, 298 K, platinised Pt) is assigned $E^{\theta} = 0.00\,V$.
- E^{θ} of a half-cell = e.m.f. of the cell it forms with the SHE under standard conditions; it is always quoted for the reduction half-equation and is never multiplied by stoichiometric coefficients.
- Negative E^{θ}: the electrode is the negative terminal versus the SHE (strong reducing system); positive E^{θ}: strong oxidising system.
- Practical reference electrodes: calomel (+0.24 V, saturated KCl) and silver/silver chloride (≈ +0.22 V).
- Ranking couples by E^{θ} gives the electrochemical series, the tool for predicting redox reactions and cell e.m.f.s.

7 ELECTROCHEMICAL CELLS, CELL EMF AND USES OF E^{θ}

In the previous section we learnt how to measure the **standard electrode potential** E^{θ} of any half-cell by coupling it to the standard hydrogen electrode. But why couple a half-cell to the SHE at all? Because a single half-cell can do nothing on its own: a zinc rod dipped in zinc sulphate solution simply sits there. The moment we connect two half-cells together, something remarkable happens: electrons begin to flow through a wire, and we have built a battery. This section is about that idea: how two half-cells combine to make an **electrochemical cell**, how we write it down compactly, how we calculate its voltage, and how the table of E^{θ} values becomes a powerful predictive tool.

7.1 Coupling Two Half-Cells: the Electrochemical Cell

Consider two half-cells, for example $Zn^{2+}(aq)/Zn(s)$ and $Cu^{2+}(aq)/Cu(s)$. Each on its own is at equilibrium:

$$Zn^{2+}(aq) + 2\,e^{-} \rightleftharpoons Zn(s) \qquad E^{\theta} = -0.76\,V$$

$$Cu^{2+}(aq) + 2\,e^- \rightleftharpoons Cu(s) \qquad E^\theta = +0.34\,V$$

Zinc has the more negative E^θ: it is the stronger **reducing agent**, so zinc atoms release electrons more readily than copper atoms. If we now connect the two metal rods by a wire, electrons flow from zinc to copper. Zinc is oxidised ($Zn \longrightarrow Zn^{2+} + 2\,e^-$) and copper(II) ions are reduced ($Cu^{2+} + 2\,e^- \longrightarrow Cu$).

Two practical problems arise immediately. First, the solutions must not mix directly, otherwise Cu^{2+} ions would simply react on the zinc surface and no current would flow through the wire. Second, as the reaction proceeds, the zinc beaker would accumulate positive charge (extra Zn^{2+}) and the copper beaker negative charge (loss of Cu^{2+}); this charge build-up would stop the electron flow almost instantly. Both problems are solved by a **salt bridge**.

DEFINITION **Electrochemical (galvanic) cell**

An **electrochemical cell** (or galvanic/voltaic cell) is a device made by coupling two half-cells so that a spontaneous redox reaction drives a flow of electrons through an external circuit. The electrode at which oxidation occurs is the **anode** (negative pole); the electrode at which reduction occurs is the **cathode** (positive pole).

DEFINITION **Cell emf**

The **electromotive force** (emf) of a cell, E_{cell}, is the potential difference between its two electrodes measured when no current is drawn (e.g. with a high-resistance voltmeter). Under standard conditions it is the **standard cell emf** E^θ_{cell}, measured in volts.

7.2 The Daniell Cell

The classic example is the **Daniell cell**, built from a Zn/Zn^{2+} half-cell and a Cu/Cu^{2+} half-cell.

EXPERIMENT **The Daniell cell**

Set up a zinc rod in 1.0 mol dm^{-3} $ZnSO_4(aq)$ and a copper rod in 1.0 mol dm^{-3} $CuSO_4(aq)$. Connect the solutions with a salt bridge (a U-tube containing a gel saturated with KNO_3 or KCl, plugged with cotton wool) and connect the rods through a voltmeter. Observations: the voltmeter reads about 1.10 V with copper as the positive terminal; the zinc rod slowly dissolves; the blue colour of the copper sulphate fades and pink copper deposits on the copper rod. Interpretation:

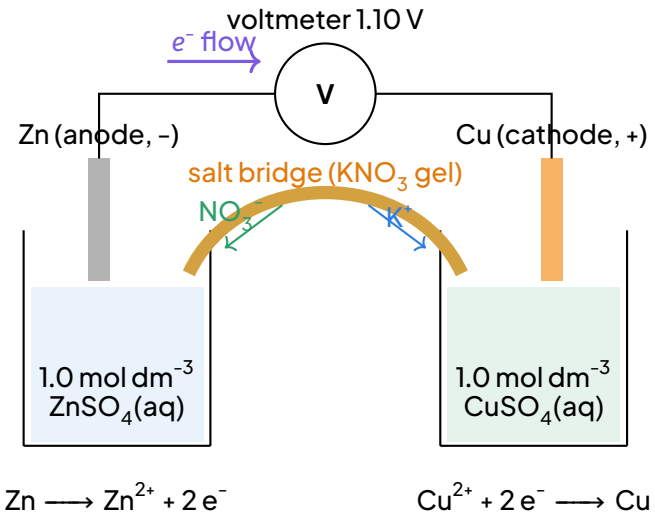
$$\text{Anode (oxidation):} \quad Zn(s) \longrightarrow Zn^{2+}(aq) + 2\,e^-$$

$$\text{Cathode (reduction):} \quad Cu^{2+}(aq) + 2\,e^- \longrightarrow Cu(s)$$

$$\text{Overall:} \quad Zn(s) + Cu^{2+}(aq) \longrightarrow Zn^{2+}(aq) + Cu(s)$$

Electrons flow through the external circuit from zinc to copper; in the salt bridge, cations

(K^+) migrate towards the copper half-cell and anions (NO_3^-) towards the zinc half-cell, keeping both solutions electrically neutral.



The Daniell cell: electrons flow through the external circuit from the zinc anode to the copper cathode; the salt bridge completes the circuit and maintains electrical neutrality.

WARNING **The salt bridge is not optional**

Without a salt bridge (or porous pot), charge builds up in each beaker and the cell stops working within moments. In examinations you must state its two functions: (i) it completes the circuit, (ii) it maintains electrical neutrality by allowing ion migration, without the two solutions mixing. Use KNO_3 or KCl because K^+ and NO_3^- (or Cl^-) have similar mobilities and do not react with the half-cell species.

7.3 Cell Diagrams (Cell Notation)

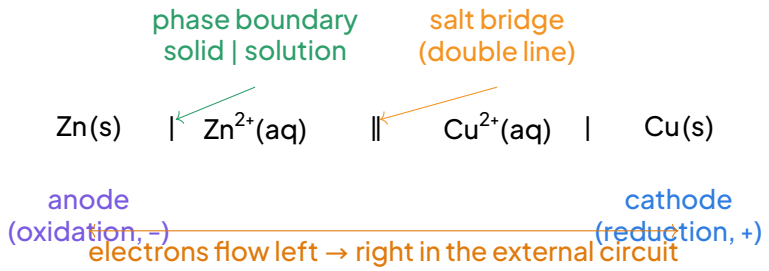
Drawing beakers every time is tedious, so chemists use a shorthand called the **cell diagram** or **cell notation**. The rules are strict and examinable:

METHOD **Writing a cell diagram and computing E^θ_{cell}**

1. Identify the two half-cells and look up their E^θ values.
2. The half-cell with the more negative E^θ is the anode (oxidation): write it on the left.

3. The half-cell with the more positive E^θ is the cathode (reduction): write it on the right.
4. Use a single vertical line | for a phase boundary (e.g. solid|solution) and a double vertical line ‖ for the salt bridge.
5. Write species in the order: electrode | ions ‖ ions | electrode. For half-cells without a solid conductor (gas electrodes, ion–ion systems), use an inert electrode (usually Pt) on the outside: $Pt|H_2(g)|H^+(aq)$, $Pt|Fe^{2+}(aq), Fe^{3+}(aq)$.
6. Compute the standard emf:

$$E^\theta_{cell} = E^\theta_{right} - E^\theta_{left} = E^\theta_{cathode} - E^\theta_{anode}.$$



Anatomy of a cell diagram: anode on the left, cathode on the right; single lines mark phase boundaries, the double line the salt bridge.

WORKED EXAMPLE 13 **Emf of the Daniell cell**

Given $E^\theta(Zn^{2+}/Zn) = -0.76$ V and $E^\theta(Cu^{2+}/Cu) = +0.34$ V, write the cell diagram and calculate E^θ_{cell}.

SOLUTION

Solution. Zinc has the more negative E^θ, so it is the anode (left); copper is the cathode (right):

$$Zn(s)|Zn^{2+}(aq)||Cu^{2+}(aq)|Cu(s)$$

$$E^\theta_{cell} = E^\theta_{right} - E^\theta_{left} = (+0.34) - (-0.76) = +1.10\ \text{V}.$$

The positive value confirms that the reaction $Zn(s) + Cu^{2+}(aq) \longrightarrow Zn^{2+}(aq) + Cu(s)$ is spontaneous under standard conditions.

WORKED EXAMPLE 14 **Cell diagram for a hydrogen electrode and a metal/metal-ion electrode**

Write the cell diagram and calculate E^{θ}_{cell} for a cell made from the standard hydrogen electrode and a Cu^{2+}/Cu half-cell. $E^{\theta}(Cu^{2+}/Cu) = +0.34\,V$.

SOLUTION

Solution. SHE has $E^{\theta} = 0.00\,V$; copper has $+0.34\,V$. SHE is more negative, so it is the anode (left). The SHE uses an inert Pt electrode:

$$Pt(s)|H_2(g, 1\,atm)|H^+(aq, 1\,M)||Cu^{2+}(aq, 1\,M)|Cu(s)$$

$$E^{\theta}_{cell} = 0.34 - 0.00 = +0.34\,V.$$

WORKED EXAMPLE 15 **Cell diagram for an ion-ion half-cell**

Write the cell diagram and calculate E^{θ}_{cell} for a cell made from Fe^{3+}/Fe^{2+} ($E^{\theta} = +0.77\,V$) and Cu^{2+}/Cu ($E^{\theta} = +0.34\,V$).

SOLUTION

Solution. Copper has the more negative E^{θ}, so it is the anode (left). The Fe^{3+}/Fe^{2+} half-cell requires an inert Pt electrode:

$$Cu(s)|Cu^{2+}(aq)||Fe^{3+}(aq), Fe^{2+}(aq)|Pt(s)$$

$$E^{\theta}_{cell} = 0.77 - 0.34 = +0.43\,V.$$

WARNING **Sign mistakes in E^{θ}_{cell}**

The most common GCE error is to subtract the wrong way round, or to change the sign of the anode potential because "oxidation is the reverse". Never reverse the sign of a tabulated E^{θ}. All E^{θ} values are written as reduction potentials; simply apply $E^{\theta}_{cell} = E^{\theta}_{cathode} - E^{\theta}_{anode}$ with the tabulated values as they stand.

7.4 Uses of Standard Electrode Potentials

The table of E^{θ} values is one of the most useful predictive tools in A-Level chemistry.

1. Predicting the direction of electron flow and electrode polarity. In any cell, electrons flow through the external circuit from the half-cell with the more negative E^{θ} (which becomes the negative electrode, the anode) to the one with the more positive E^{θ} (the positive electrode, the cathode).

2. Predicting whether a redox reaction is feasible.

PROPERTY **Feasibility criterion**

A redox reaction is **feasible** (spontaneous) under standard conditions if the emf of the corresponding cell is positive:

$$E^{\theta}_{\text{cell}} = E^{\theta}_{\text{reduced species}} - E^{\theta}_{\text{oxidised species}} > 0.$$

Equivalently: a species will oxidise another if its couple has the more positive E^{θ}. (Proof not examinable; the criterion follows from $\Delta G^{\theta} = -nFE^{\theta}_{\text{cell}}$.)

EXEMPLE **Will zinc displace copper?**

Does Zn(s) react with $Cu^{2+}(aq)$? Here Cu^{2+} is reduced (E^{θ} = +0.34 V) and Zn is oxidised (E^{θ} = −0.76 V):

$$E^{\theta}_{\text{cell}} = 0.34 - (-0.76) = +1.10\ \text{V} > 0 \quad \Rightarrow \quad \text{feasible.}$$

Conversely, Cu(s) cannot displace Zn^{2+}: $E^{\theta}_{\text{cell}} = -0.76 - 0.34 = -1.10\ \text{V} < 0$.

3. Comparing oxidising and reducing strengths.

METHOD **Ranking oxidising and reducing agents from E^{θ}**

1. The more positive the E^{θ} of a couple, the stronger the oxidising agent is the oxidised form (left-hand side of the reduction half-equation).
2. The more negative the E^{θ}, the stronger the reducing agent is the reduced form (right-hand side).
3. Example: since $E^{\theta}(Cl_2/Cl^-) = +1.36\ \text{V} > E^{\theta}(Br_2/Br^-) = +1.07\ \text{V}$, Cl_2 is a stronger oxidising agent than Br_2, and Br^- is a stronger reducing agent than Cl^-.

WARNING **Feasible does not mean fast**

A positive E^{θ}_{cell} tells us a reaction is energetically feasible, but says nothing about its rate. Some feasible reactions have a very high activation energy and are extremely slow in practice (e.g. many oxidations by MnO_4^- need warming). E^{θ} predicts possibility, not speed.

7.5 Factors Affecting Electrode Potential

The tabulated values are standard potentials, measured at 298 K, with all solutions at 1.0 mol dm^{-3} and gases at 1 atm (101 kPa). Change these conditions and the actual electrode potential changes. Each half-cell is an equilibrium, so **Le Chatelier's principle** predicts the direction of the change.

- **Concentration:** For $Cu^{2+}(aq) + 2\,e^- \rightleftharpoons Cu(s)$, increasing $[Cu^{2+}]$ shifts the equilibrium to the right (Le Chatelier), so the electrode becomes a better electron acceptor: E becomes more positive. Diluting the solution makes E more negative.
- **Temperature:** Changing temperature shifts the position of the half-cell equilibrium (and changes K), so E changes; this is why standard potentials are quoted at 298 K.
- **Pressure (gaseous electrodes):** For a half-cell involving a gas, e.g. $Cl_2(g)+2\,e^- \rightleftharpoons 2\,Cl^-(aq)$, increasing the pressure of Cl_2 shifts the equilibrium right (Le Chatelier) and makes E more positive.

The quantitative relationship is given by the Nernst equation, which is beyond the GCE syllabus but explains exactly how E varies with concentration.

WARNING **Non-standard conditions can reverse predictions**

Because concentration changes the actual electrode potential, a reaction predicted unfeasible from E^θ values may occur under non-standard conditions. For example, concentrated hydrochloric acid reduces MnO_2 to Mn^{2+} (the laboratory preparation of chlorine) although standard potentials suggest otherwise: the very high $[Cl^-]$ and $[H^+]$ shift the half-cell potentials. Always state "under standard conditions" when using E^θ to predict feasibility.

DID YOU KNOW? **Batteries around us**

Every battery is an electrochemical cell: the dry cell used in torches and radios across Cameroon is based on a Zn anode and a MnO_2 cathode; car batteries couple lead and lead(IV) oxide plates in sulphuric acid. The Daniell cell itself, invented in 1836, was one of the first reliable sources of electric current and powered early telegraph networks.

EXERCISE 8 **Using E^θ values**

Use the following standard electrode potentials:

$E^\theta(Fe^{2+}/Fe) = -0.44\,V$, $E^\theta(Cu^{2+}/Cu) = +0.34\,V$, $E^\theta(Ag^+/Ag) = +0.80\,V$, $E^\theta(Cl_2/Cl^-) = +1.36\,V$

1. Write the cell diagram for the cell made from the Fe^{2+}/Fe and Ag^+/Ag half-cells, and calculate its standard emf.

2. Predict, with a calculation, whether iron metal will displace copper(II) ions from solution.
3. Arrange Fe, Cu and Ag in order of increasing reducing power.
4. Write the cell diagram for a cell constructed from a Cl_2/Cl^- half-cell and a Cu^{2+}/Cu half-cell. Calculate E^θ_{cell} and state which electrode is the anode.
5. Will chlorine gas oxidise copper metal to Cu^{2+} under standard conditions? Justify with a calculation.

SOLUTION **Exercise 1**

1. Fe has the more negative E^θ, so it is the anode (left): $Fe(s)|Fe^{2+}(aq)||Ag^+(aq)|Ag(s)$; $E^\theta_{cell} = 0.80 - (-0.44) = +1.24\,V$.
2. Fe oxidised, Cu^{2+} reduced: $E^\theta_{cell} = 0.34 - (-0.44) = +0.78\,V > 0$, so the reaction $Fe(s) + Cu^{2+}(aq) \longrightarrow Fe^{2+}(aq) + Cu(s)$ is feasible.
3. Reducing power increases as E^θ becomes more negative: Ag < Cu < Fe.
4. Cu has the more negative E^θ (+0.34 V vs +1.36 V), so it is the anode (left). The chlorine half-cell requires an inert Pt electrode: $Cu(s)|Cu^{2+}(aq)||Cl\text{-}(aq)|Cl_2(g)|Pt(s)$; $E^\theta_{cell} = 1.36 - 0.34 = +1.02\,V$.
5. For Cl_2 oxidising Cu: Cl_2 reduced ($E^\theta = +1.36$ V), Cu oxidised ($E^\theta = +0.34$ V). $E^\theta_{cell} = 1.36 - 0.34 = +1.02\,V > 0$. Yes, the reaction is feasible.

KEY POINTS **Key points**

- An electrochemical cell couples two half-cells; oxidation occurs at the **anode** (negative), reduction at the **cathode** (positive); electrons flow anode → cathode through the external circuit.
- The **salt bridge** completes the circuit and keeps the half-cells electrically neutral (use KNO_3 or KCl).
- Cell diagram: anode on the left, cathode on the right; | = phase boundary, ‖ = salt bridge. Use Pt for half-cells without a solid conductor.
- $E^\theta_{cell} = E^\theta_{right} - E^\theta_{left} = E^\theta_{cathode} - E^\theta_{anode}$; never change the sign of a tabulated E^θ.
- $E^\theta_{cell} > 0 \Rightarrow$ reaction feasible under standard conditions; but feasibility says nothing about rate, and non-standard concentrations can alter the actual potentials.
- More positive $E^\theta \Rightarrow$ stronger oxidising agent; more negative $E^\theta \Rightarrow$ stronger reducing agent.

- Electrode potential changes with concentration (Le Chatelier), temperature, and gas pressure.

8 ACID-BASE THEORIES, PH AND THE IONIC PRODUCT OF WATER

In the previous section we studied redox equilibria, in which the particle transferred between chemical species is the electron. We now turn to another enormous family of equilibria in which the transferred particle is the proton, H^+: the **acid–base equilibria**. You will see that the same ideas of dynamic equilibrium, equilibrium constants and Le Chatelier's principle apply here too. Acids and bases are part of everyday Cameroonian life: the sourness of lime juice and palm wine, the slippery feel of soap, the sting of an ant bite — all of these are acid-base chemistry.

8.1 What is an acid? Three theories

The Arrhenius theory

The oldest useful definition is due to the Swedish chemist Svante Arrhenius. It is based on what happens when a substance dissolves in water.

DEFINITION **Arrhenius acid and base**

An **Arrhenius acid** is a substance which releases hydrogen ions, H^+, when dissolved in water. An **Arrhenius base** is a substance which releases hydroxide ions, OH^-, when dissolved in water.

$$HCl(g) \xrightarrow{H_2O} H^+(aq) + Cl^-(aq) \qquad NaOH(s) \xrightarrow{H_2O} Na^+(aq) + OH^-(aq)$$

This theory explains the behaviour of common laboratory acids (HCl, H_2SO_4, HNO_3) and alkalis (NaOH, KOH), but it has serious **limitations**:

- It only works for solutions in water, yet acid-base reactions occur in other solvents and even in the gas phase.
- It cannot explain why NH_3 is a base: ammonia contains no OH^- group to release.
- The free proton H^+ does not exist alone in water; it attaches to a water molecule to give the **oxonium ion**, H_3O^+.

The Brønsted–Lowry theory

In 1923, Brønsted and Lowry independently proposed a more general theory centred on proton transfer.

DEFINITION **Brønsted–Lowry acid and base**

A **Brønsted–Lowry acid** is a proton donor; a **Brønsted–Lowry base** is a proton acceptor. An acid–base reaction is a transfer of a proton, H^+, from the acid to the base.

Consider hydrogen chloride dissolving in water:

$$HCl(aq) + H_2O(l) \longrightarrow H_3O^+(aq) + Cl^-(aq)$$

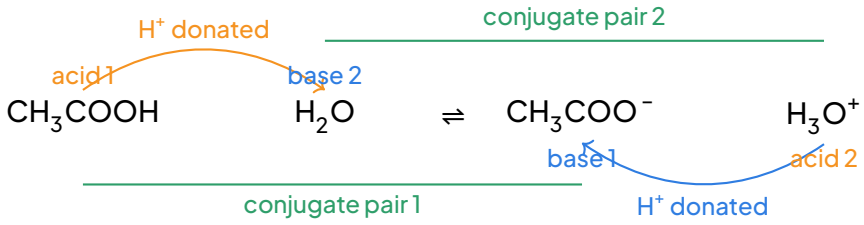
HCl donates a proton (acid); H_2O accepts it (base). Notice that water, which Arrhenius ignored, is here an active participant. Ammonia is now easily explained as a base:

$$NH_3(aq) + H_2O(l) \rightleftharpoons NH_4^+(aq) + OH^-(aq)$$

DEFINITION **Conjugate acid–base pairs**

A **conjugate acid–base pair** consists of two species that differ by exactly one proton. When an acid donates a proton, the species left behind is its **conjugate base**; when a base accepts a proton, the species formed is its **conjugate acid**.

For the reaction $HCl + H_2O \rightarrow H_3O^+ + Cl^-$, the two conjugate pairs are HCl/Cl^- and H_3O^+/H_2O. The stronger the acid, the weaker its conjugate base: Cl^- is an extremely weak base because HCl is a very strong acid.



Proton transfer in the equilibrium $CH_3COOH + H_2O \rightleftharpoons CH_3COO^- + H_3O^+$, showing the two conjugate acid–base pairs.

The Lewis theory

G. N. Lewis generalised the idea still further, to reactions where no proton is transferred at all.

DEFINITION **Lewis acid and base**

A **Lewis acid** is an electron-pair acceptor; a **Lewis base** is an electron-pair donor. A Lewis acid–base reaction forms a dative (coordinate) covalent bond.

EXEMPLE **A Lewis acid–base reaction**

Boron trifluoride, BF_3, has only six electrons around boron — an incomplete octet — so it can accept an electron pair. Ammonia, NH_3, has a lone pair on nitrogen:

$$BF_3 + NH_3 \rightarrow F_3B \leftarrow NH_3$$

BF_3 is the Lewis acid, NH_3 the Lewis base, and the product contains a dative bond B ← N. Metal ions such as Cu^{2+} accepting lone pairs from water ligands are also Lewis acids.

KEY POINTS **Three theories compared**

Theory	Acid	Base
Arrhenius	releases H^+ in water	releases OH^- in water
Brønsted–Lowry	proton (H^+) donor	proton acceptor
Lewis	electron-pair acceptor	electron-pair donor

Each theory includes the previous one as a special case: every Brønsted–Lowry base is a Lewis base, but BF_3 is a Lewis acid without being a Brønsted–Lowry acid.

8.2 Amphoteric (amphiprotic) species

Some species can behave as an acid or as a base depending on the partner. They are called **amphoteric** (or **amphiprotic** when proton transfer is involved).

EXEMPLE **Water and the hydrogencarbonate ion**

Water accepts a proton from HCl (acting as a base) but donates a proton to NH_3 (acting as an acid). Similarly:

$$HCO_3^- + H_2O \rightleftharpoons H_2CO_3 + OH^- \qquad (HCO_3^- \text{ as a base})$$
$$HCO_3^- + H_2O \rightleftharpoons CO_3^{2-} + H_3O^+ \qquad (HCO_3^- \text{ as an acid})$$

8.3 The auto-ionisation of water and the ionic product K_w

Pure water seems unreactive, yet a sensitive conductivity meter shows that it conducts electricity very slightly. This proves that a few ions exist in pure water. The explanation is the **auto-ionisation** (self-ionisation) of water, a Brønsted–Lowry proton transfer between two water molecules:

$$2\,H_2O(l) \rightleftharpoons H_3O^+(aq) + OH^-(aq)$$

This equilibrium lies very far to the left, but it is vitally important. Applying the equilibrium law, and treating the concentration of liquid water as constant, we absorb $[H_2O]^2$ into the constant.

DEFINITION **Ionic product of water**

The **ionic product of water** is

$$K_w = [H_3O^+][OH^-]$$

At 25 °C, $K_w = 1.0 \times 10^{-14}\ \text{mol}^2\ \text{dm}^{-6}$.

PROPERTY **A universal relation**

In any aqueous solution — acidic, neutral or alkaline — at a given temperature,

$$[H_3O^+][OH^-] = K_w$$

The two ion concentrations are linked: if one increases, the other must decrease so that the product stays constant. (Proof not examinable; it follows directly from the equilibrium law.)

In pure water, the equation produces equal numbers of the two ions, so $[H_3O^+] = [OH^-] = \sqrt{K_w} = 1.0 \times 10^{-7}\ \text{mol dm}^{-3}$ at 25 °C.

8.4 The pH scale

Concentrations of H_3O^+ range over many powers of ten, which is clumsy. The Danish chemist Sørensen introduced a logarithmic scale to compress this range.

DEFINITION **pH**

The **pH** of a solution is defined by

$$\text{pH} = -\log_{10}[H_3O^+] \quad \Longleftrightarrow \quad [H_3O^+] = 10^{-\text{pH}}$$

with $[H_3O^+]$ in mol dm^{-3}. Similarly, $\text{pOH} = -\log_{10}[OH^-]$.

Taking $-\log_{10}$ of $K_w = [H_3O^+][OH^-]$ gives the very useful relation, valid at 25 °C:

$$\text{pH} + \text{pOH} = 14$$

METHOD **Converting between $[H_3O^+]$, $[OH^-]$, pH and pOH at 25 °C**

1. From $[H_3O^+]$ to pH: compute $pH = -\log_{10}[H_3O^+]$.
2. From pH to $[H_3O^+]$: compute $[H_3O^+] = 10^{-pH}$.
3. From $[H_3O^+]$ to $[OH^-]$: use $[OH^-] = K_w/[H_3O^+] = 10^{-14}/[H_3O^+]$.
4. From pH to pOH: use $pOH = 14 - pH$, then $[OH^-] = 10^{-pOH}$.

WORKED EXAMPLE 16 **pH of a lime juice sample**

The oxonium ion concentration of a sample of lime juice is $[H_3O^+] = 5.0 \times 10^{-3}$ mol dm^{-3} at 25 °C. Calculate (a) the pH, (b) $[OH^-]$ in the juice.

SOLUTION

(a) Apply the definition:

$$pH = -\log_{10}(5.0 \times 10^{-3}) = -(\log_{10} 5.0 - 3) = 3 - 0.70 = 2.30$$

(b) Use the ionic product:

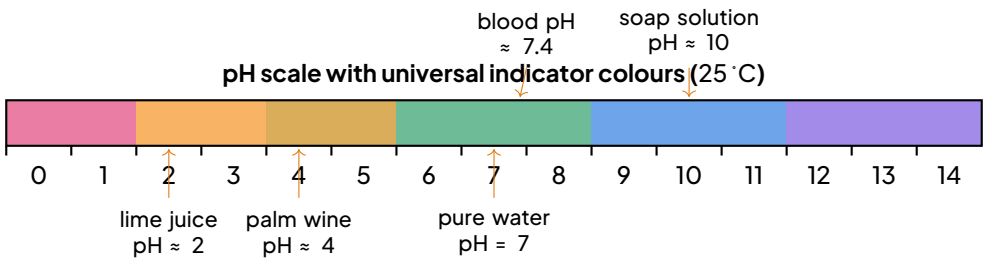
$$[OH^-] = \frac{K_w}{[H_3O^+]} = \frac{1.0 \times 10^{-14}}{5.0 \times 10^{-3}} = 2.0 \times 10^{-12} \text{ mol dm}^{-3}$$

Check: $pOH = -\log_{10}(2.0 \times 10^{-12}) = 11.70$, and indeed $2.30 + 11.70 = 14$. The juice is acidic, as expected since $[H_3O^+] > [OH^-]$.

8.5 Acidic, neutral and alkaline solutions

KEY POINTS **Classifying solutions at** 25 °C

Solution	$[H_3O^+]$ vs $[OH^-]$	$[H_3O^+]$ (mol dm^{-3})	pH
Acidic	$[H_3O^+] > [OH^-]$	$> 1.0 \times 10^{-7}$	< 7
Neutral	$[H_3O^+] = [OH^-]$	$= 1.0 \times 10^{-7}$	= 7
Alkaline	$[H_3O^+] < [OH^-]$	$< 1.0 \times 10^{-7}$	> 7



The pH scale with universal indicator colours and familiar examples.

8.6 Measuring pH

- **Universal indicator paper or solution (pH paper).** A mixture of indicators whose colour changes gradually from red (strongly acidic) through green (neutral) to dark purple (strongly alkaline). The colour is matched against a printed chart. It is quick and cheap but only approximate (to about ±1 pH unit).
- **The pH meter.** A glass electrode sensitive to H_3O^+ connected to a voltmeter calibrated directly in pH units. It must first be **calibrated** using standard buffer solutions of known pH. It gives accurate readings (to 0.01 pH unit) and is essential for following pH changes during a titration.

8.7 Effect of temperature on K_w and on the pH of pure water

The auto-ionisation of water is endothermic:

$$2\,H_2O(l) \rightleftharpoons H_3O^+(aq) + OH^-(aq) \qquad \Delta H > 0$$

By Le Chatelier's principle, raising the temperature shifts the equilibrium to the right: more ions form, so K_w increases. For example, $K_w \approx 5.5 \times 10^{-14}$ mol^2 dm^{-6} near 40 °C. In pure water we still have $[H_3O^+] = [OH^-]$, so the water is still neutral — but $[H_3O^+] > 10^{-7}$, so the pH is less than 7.

WARNING **Neutral does not always mean pH 7**

Pure water is always neutral because $[H_3O^+] = [OH^-]$ at every temperature. But its pH equals 7 only at 25 °C. At 40 °C, neutral water has pH ≈ 6.8. Neutrality is defined by the equality of the two ion concentrations, not by a fixed pH value.

WARNING **pH values outside 0–14**

The familiar 0–14 range applies only to moderately dilute solutions at 25 °C. For a concentrated strong acid such as 10 mol dm^{-3} HCl, pH = $-\log_{10}(10) = -1$: a negative pH. Likewise, a very concentrated strong base can have pH greater than 14. The scale does not stop at

0 and 14.

EXERCISE 9 **Practising pH calculations**

At 25 °C, a sample of soapy water from a laundry in Bamenda has pH = 10.5.

1. Calculate $[H_3O^+]$ and $[OH^-]$ in the solution.
2. State, with a reason, whether the solution is acidic, neutral or alkaline.
3. At 60 °C, $K_w = 9.3 \times 10^{-14}$ mol^2 dm^{-6}. Calculate the pH of pure water at this temperature and explain why the water is still neutral.

SOLUTION **Exercise 1**

1. $[H_3O^+] = 10^{-pH} = 10^{-10.5} = 3.2 \times 10^{-11}$ mol dm^{-3}.

$$[OH^-] = \frac{K_w}{[H_3O^+]} = \frac{1.0 \times 10^{-14}}{3.2 \times 10^{-11}} = 3.2 \times 10^{-4} \text{ mol dm}^{-3}$$

2. Since $[OH^-] > [H_3O^+]$ (and pH > 7), the solution is alkaline, as expected for soap. 3. In pure water $[H_3O^+] = \sqrt{K_w} = \sqrt{9.3 \times 10^{-14}} = 3.0 \times 10^{-7}$ mol dm^{-3}, so pH = $-\log_{10}(3.0 \times 10^{-7}) = 6.5$. The water is still neutral because auto-ionisation produces equal concentrations of H_3O^+ and OH^-; neutrality depends on the equality $[H_3O^+] = [OH^-]$, not on pH being 7.

DID YOU KNOW? **Why blood pH matters**

Human blood is kept within the very narrow pH range 7.35–7.45 by natural buffer systems (mainly the H_2CO_3/HCO_3^- pair you will study in Section 11). A shift of only 0.2 pH unit can be life-threatening. The same buffering principle is used in fermented foods and in the preparation of traditional medicines, where controlling acidity prevents the growth of harmful microorganisms.

KEY POINTS **Key points of this section**

- Arrhenius: acids give H^+, bases give OH^- in water (limited to aqueous solutions).
- Brønsted–Lowry: acid = proton donor, base = proton acceptor; species differing by one H^+ form a conjugate pair.

- Lewis: acid = electron-pair acceptor, base = electron-pair donor (e.g. $BF_3 + NH_3$).
- Water auto-ionises: $2\,H_2O \rightleftharpoons H_3O^+ + OH^-$, with $K_w = [H_3O^+][OH^-] = 1.0 \times 10^{-14}\ mol^2\ dm^{-6}$ at 25 °C.
- $pH = -\log_{10}[H_3O^+]$; $pH + pOH = 14$ at 25 °C.
- Neutral means $[H_3O^+] = [OH^-]$; $pH = 7$ only at 25 °C since K_w increases with temperature.

9 pH of Strong and Weak Acids and Bases, and of Mixtures

In the previous section we defined pH, established the ionic product of water $K_w = [H_3O^+][OH^-] = 1.0 \times 10^{-14}$ at 25 °C, and learnt how to measure pH. We now put these tools to work. The central question of this section is simple: if I dissolve a given amount of an acid or a base in water, what is the pH of the solution? The answer depends crucially on whether the acid or base is **strong** or **weak** — that is, on how completely it ionises in water.

9.1 Strong Acids and Strong Bases: Complete Ionisation

When hydrogen chloride gas is dissolved in water, essentially every HCl molecule donates its proton to a water molecule:

$$HCl(aq) + H_2O(l) \longrightarrow H_3O^+(aq) + Cl^-(aq)$$

The reaction goes to completion: no detectable HCl molecules remain in solution. Such an acid is called **strong**. Similarly, sodium hydroxide is an ionic solid that dissociates completely:

$$NaOH(s) \xrightarrow{H_2O} Na^+(aq) + OH^-(aq)$$

DEFINITION **Strong and weak acids and bases**

An acid is **strong** if it ionises completely in aqueous solution (degree of ionisation $\alpha \approx 1$); it is **weak** if it ionises only partially ($\alpha \ll 1$), so that an equilibrium is set up between the molecules and the ions. The same distinction applies to bases: a **strong base** (e.g. NaOH, KOH) dissociates completely to give OH^- ions, while a **weak base** (e.g. NH_3) reacts only partially with water.

WARNING **'Strong' is not 'concentrated'**

Strength describes the extent of ionisation; **concentration** describes the amount of solute per unit volume. A 10 mol L^{-1} solution of ethanoic acid is concentrated but weak; a 10^{-4} mol L^{-1} solution of HCl is dilute but strong. Never use the two words interchangeably in the examination.

Common strong acids: HCl, HNO_3, H_2SO_4 (first proton fully ionised), HBr, $HClO_4$. Common strong bases: NaOH, KOH, $Ba(OH)_2$ (moderately soluble, but what dissolves dissociates completely).

pH of a strong monoprotic acid. Since ionisation is complete, each mole of acid gives one mole of H_3O^+:

$$[H_3O^+] = c_a \quad \Longrightarrow \quad \boxed{\text{pH} = -\log c_a}$$

pH of a strong monobasic base (e.g. NaOH, KOH). Complete dissociation gives $[OH^-] = c_b$. We then pass through K_w:

$$[H_3O^+] = \frac{K_w}{[OH^-]} = \frac{K_w}{c_b} \quad \Longrightarrow \quad \text{pH} = 14 + \log c_b \quad (\text{at } 25\,^\circ\text{C})$$

For a strong diacidic base like $Ba(OH)_2$, $[OH^-] = 2c_b$ (solubility permitting), so pH = 14 + log($2c_b$).

EXEMPLE **pH of strong acid and strong base solutions**

(a) 0.05 mol L^{-1} HCl: pH = − log(0.05) = 1.30.

(b) 0.02 mol L^{-1} NaOH: $[OH^-] = 0.02$ mol L^{-1}, so $[H_3O^+] = \frac{1.0 \times 10^{-14}}{0.02} = 5.0 \times 10^{-13}$ mol L^{-1} and pH = 12.30.

(c) 0.01 mol L^{-1} H_2SO_4: the first proton is fully released; the second proton has $K_{a2} = 1.2 \times 10^{-2}$, so at this concentration it is not fully dissociated. A rigorous treatment gives $[H_3O^+] \approx 0.011$ mol L^{-1} and pH ≈ 1.96. (The simplified assumption "both protons fully released" would give $[H_3O^+] = 0.02$ mol L^{-1}, pH = 1.70; this overestimates the acidity.)

9.2 Weak Acids and Weak Bases: Partial Ionisation, K_a and K_b

Ethanoic acid, the acid in vinegar and in the palm-oil kitchens of Cameroon, behaves very differently from HCl. Dissolved in water, only a small fraction of its molecules ionise; an equilibrium is established:

$$CH_3COOH(aq) + H_2O(l) \rightleftharpoons CH_3COO^-(aq) + H_3O^+(aq)$$

Likewise ammonia, a weak base, reacts only partially:

$$NH_3(aq) + H_2O(l) \rightleftharpoons NH_4^+(aq) + OH^-(aq)$$



HCl (strong)
only H_3O^+
and Cl^- ions

CH_3COOH (weak)
mostly molecules,
few ions

Particle picture of ionisation. Strong acid: complete ionisation, only ions present. Weak acid: partial ionisation, mostly un-ionised molecules in equilibrium with a few ions.

Because the ionisation of a weak acid is an equilibrium, we can apply the equilibrium law from earlier in this chapter.

DEFINITION K_a, pK_a, K_b, pK_b

For a weak acid $HA(aq) + H_2O(l) \rightleftharpoons A^-(aq) + H_3O^+(aq)$, the **acid dissociation constant** is

$$K_a = \frac{[A^-][H_3O^+]}{[HA]} \quad \text{and} \quad pK_a = -\log K_a.$$

For a weak base $B(aq) + H_2O(l) \rightleftharpoons BH^+(aq) + OH^-(aq)$, the **base dissociation constant** is

$$K_b = \frac{[BH^+][OH^-]}{[B]} \quad \text{and} \quad pK_b = -\log K_b.$$

The larger K_a (the smaller pK_a), the stronger the weak acid. For a conjugate acid–base pair, $K_a \times K_b = K_w$, i.e. $pK_a + pK_b = 14$ at 25 °C.

EXEMPLE **Typical values at** 25 °C

Ethanoic acid: $K_a = 1.8 \times 10^{-5}$ mol L^{-1}, $pK_a = 4.74$. Ammonia: $K_b = 1.8 \times 10^{-5}$ mol L^{-1}, so for NH_4^+, $pK_a = 14 - 4.74 = 9.26$.

9.3 Calculating the pH of a Weak Acid

Consider a weak acid HA of analytical concentration c. Let x be the concentration that ionises. The equilibrium table (in mol L^{-1}) is:

	HA	A^-	H_3O^+
Initial	c	0	0
Equilibrium	$c - x$	x	x

Hence $K_a = \dfrac{x^2}{c - x}$. For a weak acid, $x \ll c$, so $c - x \approx c$, giving

$$K_a \approx \frac{x^2}{c} \quad \Longrightarrow \quad [H_3O^+] \approx \sqrt{K_a\, c} \quad \Longrightarrow \quad \text{pH} = \tfrac{1}{2}(pK_a - \log c).$$

METHOD **pH of a weak acid using K_a**

1. Write the ionisation equation $HA + H_2O \rightleftharpoons A^- + H_3O^+$ and the expression for K_a.
2. Use the approximation $[H_3O^+] = \sqrt{K_a \cdot c}$.
3. Compute $\text{pH} = -\log[H_3O^+]$.
4. Check the approximation: the degree of ionisation $\alpha = [H_3O^+]/c$ must be small (less than about 5%). If not, solve the quadratic $x^2 + K_a x - K_a c = 0$ (the positive root is $x = \frac{-K_a + \sqrt{K_a^2 + 4K_a c}}{2}$).

WARNING $[H_3O^+] \neq c$ **for a weak acid**

Because ionisation is partial, $[H_3O^+]$ is much smaller than the acid concentration. Writing $\text{pH} = -\log c$ for a weak acid is a classic examination error: it is only valid for strong monoprotic acids.

WORKED EXAMPLE 17 **pH of ethanoic acid**

Calculate the pH of a 0.10 mol L^{-1} solution of ethanoic acid, given $K_a(CH_3COOH) = 1.8 \times 10^{-5}$ mol L^{-1} at 25 °C.

SOLUTION

Solution. The equilibrium is $CH_3COOH + H_2O \rightleftharpoons CH_3COO^- + H_3O^+$.
Applying the approximation:

$$[H_3O^+] = \sqrt{K_a\, c} = \sqrt{1.8 \times 10^{-5} \times 0.10} = \sqrt{1.8 \times 10^{-6}} = 1.34 \times 10^{-3} \text{ mol L}^{-1}.$$

$$\text{pH} = -\log(1.34 \times 10^{-3}) = 2.87.$$

Check: $\alpha = \dfrac{1.34 \times 10^{-3}}{0.10} = 0.0134 = 1.3\% < 5\%$. The approximation is valid. (Compare: a strong acid of the same concentration would have pH = 1.00.)

pH of a weak base. For $B + H_2O \rightleftharpoons BH^+ + OH^-$ with concentration c_b, the same reasoning gives $[OH^-] \approx \sqrt{K_b\, c_b}$; we then use K_w to obtain $[H_3O^+]$ and hence the pH.

EXEMPLE **pH of ammonia solution**

For 0.10 mol L^{-1} NH_3 ($K_b = 1.8 \times 10^{-5}$ mol L^{-1}):

$$[OH^-] = \sqrt{1.8 \times 10^{-5} \times 0.10} = 1.34 \times 10^{-3} \text{ mol L}^{-1},$$

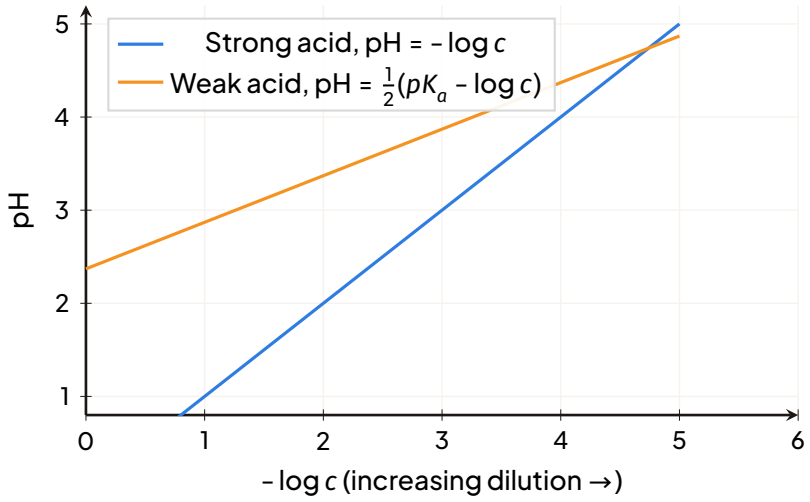
$$[H_3O^+] = \frac{1.0 \times 10^{-14}}{1.34 \times 10^{-3}} = 7.5 \times 10^{-12} \text{ mol L}^{-1} \quad \Rightarrow \quad \text{pH} = 11.13.$$

9.4 Degree of Ionisation: Dependence on Concentration and Temperature

The **degree of ionisation** is $\alpha = \dfrac{[H_3O^+]}{c} = \sqrt{\dfrac{K_a}{c}}$ (using the approximation).

PROPERTY **Effect of dilution and temperature on α**

- Dilution: as c decreases, $\alpha = \sqrt{K_a/c}$ increases: a weak acid ionises more when diluted (this is Ostwald's dilution law). Note, however, that $[H_3O^+] = \sqrt{K_a c}$ still decreases, so the pH rises towards 7.
- Temperature: K_a depends on temperature (ionisation of most weak acids is endothermic, so K_a increases slightly with temperature, by Le Chatelier's principle). K_w also increases strongly with temperature, so the pH of pure water falls below 7 on heating while remaining neutral ($[H_3O^+] = [OH^-]$).



pH versus dilution for a strong acid and a weak acid ($pK_a = 4.74$) of the same initial concentration. Both pH values rise on dilution; the weak acid is always less acidic (higher pH) and its pH changes by only 0.5 unit per tenfold dilution, against 1 unit for the strong acid.

9.5 pH of Mixtures of Acids and Bases

When an acid and a base are mixed, they neutralise each other. The strategy is always the same: treat the neutralisation as a reaction going to completion, find what remains, then compute the pH.

METHOD **pH of an acid–base mixture**

1. Write the neutralisation equation (ionic form: $H_3O^+ + OH^- \longrightarrow 2\,H_2O$; with a weak partner, write the molecular equation, e.g. $CH_3COOH + OH^- \longrightarrow CH_3COO^- + H_2O$).
2. Calculate the moles of acid and base mixed: $n = c\,V$ (volumes in L).
3. Subtract to find the excess moles of the reagent in excess.
4. Divide by the total volume to get the excess concentration, then compute the pH (using K_w if the excess is OH^-).
5. If neither is in excess (equivalence), the pH is decided by the salt formed (see hydrolysis, later section).

Case 1: strong acid + strong base. The reaction $H_3O^+ + OH^- \longrightarrow 2\,H_2O$ is complete. The pH is fixed by whichever ion is in excess; at exact equivalence, pH = 7 (at 25 °C).

WORKED EXAMPLE 18 **Strong acid in excess**

20.0 cm^3 of 0.10 mol L^{-1} HCl is mixed with 15.0 cm^3 of 0.10 mol L^{-1} NaOH. Calculate the pH of the mixture.

SOLUTION

Solution. Moles mixed:

$$n(H_3O^+) = 0.10 \times 0.0200 = 2.0 \times 10^{-3}\ \text{mol}, \qquad n(OH^-) = 0.10 \times 0.0150 = 1.5 \times 10^{-3}\ \text{mol}.$$

Neutralisation $H_3O^+ + OH^- \longrightarrow 2\,H_2O$ consumes all the OH^-; excess:

$$n(H_3O^+)_{\text{excess}} = (2.0 - 1.5) \times 10^{-3} = 5.0 \times 10^{-4}\ \text{mol}.$$

Total volume = 35.0 cm^3 = 0.0350 L, so

$$[H_3O^+] = \frac{5.0 \times 10^{-4}}{0.0350} = 1.43 \times 10^{-2}\ \text{mol L}^{-1} \quad \Rightarrow \quad \text{pH} = 1.85.$$

Case 2: strong acid + weak base. Example: $HCl + NH_3 \longrightarrow NH_4Cl$. The reaction is effectively complete. If the acid is in excess, the pH comes from the excess H_3O^+ as above. At equivalence, the solution contains NH_4^+, a weak acid ($K_a = K_w/K_b$), so the pH is less than 7 — this is salt hydrolysis, treated in a later section.

Case 3: weak acid + strong base. Example: $CH_3COOH + NaOH \longrightarrow CH_3COONa + H_2O$. If the base is in excess, the excess OH^- fixes the pH (the weak acid is entirely consumed). At equivalence, the solution contains CH_3COO^-, a weak base, so the pH is greater than 7. If the acid is in excess, the mixture contains a weak acid and its salt — a **buffer**, whose pH is calculated by the methods of the next section.

EXEMPLE **Base in excess over a weak acid**

Mixing 25.0 cm^3 of 0.10 mol L^{-1} CH_3COOH with 30.0 cm^3 of 0.10 mol L^{-1} NaOH: excess $n(OH^-) = (3.0 - 2.5) \times 10^{-3} = 5.0 \times 10^{-4}$ mol in 55.0 cm^3, giving $[OH^-] = 9.1 \times 10^{-3}$ mol L^{-1}, $[H_3O^+] = 1.1 \times 10^{-12}$ mol L^{-1}, pH = 11.96.

WARNING **Do not forget the total volume**

After mixing, the excess reagent is diluted in the combined volume. Dividing the excess moles by the volume of one solution only is a very common source of error.

KEY POINTS **Key points**

- Strong acids/bases ionise completely: pH = $-\log c_a$ (monoprotic acid) and pH = $14 + \log c_b$ (monobasic base) at 25 °C. For $Ba(OH)_2$, $[OH^-] = 2c_b$.
- Weak acids/bases ionise partially: $[H_3O^+] = \sqrt{K_a c}$ (acid), $[OH^-] = \sqrt{K_b c}$ (base); always check $\alpha < 5\%$.
- $pK_a = -\log K_a$; for a conjugate pair $pK_a + pK_b = 14$ at 25 °C.
- Dilution increases the degree of ionisation of a weak acid (Ostwald's law) but raises its pH towards 7.
- For mixtures: write the neutralisation equation, find the excess moles, divide by the total volume, then compute the pH. At equivalence the pH depends on the salt (hydrolysis).

EXERCISE 10 **Practice**

1. Calculate the pH of: (a) 0.025 mol L^{-1} HNO_3; (b) 0.050 mol L^{-1} KOH.
2. Calculate the pH of 0.050 mol L^{-1} methanoic acid HCOOH ($K_a = 1.8 \times 10^{-4}$ mol L^{-1}) and its degree of ionisation.
3. 30.0 cm^3 of 0.10 mol L^{-1} NaOH is added to 25.0 cm^3 of 0.10 mol L^{-1} HCl. Find the pH of the mixture.
4. Explain, with an equation, why the pH at equivalence of a CH_3COOH–NaOH titration is greater than 7.

SOLUTION **Exercise 1**

1. (a) Strong acid: pH = $-\log(0.025) = 1.60$. (b) $[H_3O^+] = 10^{-14}/0.050 = 2.0 \times 10^{-13}$, pH = 12.70.
2. $[H_3O^+] = \sqrt{1.8 \times 10^{-4} \times 0.050} = 3.0 \times 10^{-3}$ mol L^{-1}; pH = 2.52; $\alpha = 3.0 \times 10^{-3}/0.050 = 6\%$ (borderline; the quadratic gives pH ≈ 2.55).
3. Excess $n(OH^-) = (3.0 - 2.5) \times 10^{-3} = 5.0 \times 10^{-4}$ mol in 55.0 cm^3: $[OH^-] = 9.1 \times 10^{-3}$ mol L^{-1}, pH = 11.96.
4. At equivalence only CH_3COONa is present; CH_3COO^- is a weak base: $CH_3COO^- + H_2O \rightleftharpoons CH_3COOH + OH^-$, producing OH^-, so pH > 7.

10 ACID-BASE INDICATORS AND TITRATION CURVES

In the previous section we learnt how to calculate the pH of strong and weak acids and bases, and of their mixtures. We now put that knowledge to work in one of the most important practical activities of the Upper Sixth course: the **acid–base titration**. In a titration we add a base of known concentration progressively to an acid (or vice versa) and follow the pH as the reaction proceeds. Two questions immediately arise: how do we know when the acid has been exactly neutralised, and how does the pH vary during the addition? The answers are, respectively, acid-base indicators and titration curves — the two subjects of this section.

10.1 Acid–Base Indicators and How They Work

EXPERIMENT **Observing an indicator**

Place a few drops of methyl orange in a beaker of dilute hydrochloric acid: the solution is red. Add dilute sodium hydroxide solution drop by drop while stirring. At first nothing seems to happen; then, quite suddenly, one extra drop turns the whole solution yellow. The indicator has "signalled" that neutralisation is complete. How can a dye "know" the pH?

DEFINITION **Acid–base indicator**

An **acid–base indicator** is a weak acid (or weak base) whose acid form and conjugate base form have different colours. For an indicator that is a weak acid, we write the equilibrium:

$$\mathrm{HIn(aq)} \rightleftharpoons \mathrm{H^+(aq) + In^-(aq)} \qquad K_{In} = \frac{[\mathrm{H^+}][\mathrm{In^-}]}{[\mathrm{HIn}]}$$

where HIn and $\mathrm{In^-}$ have different colours. We define $pK_{In} = -\log K_{In}$.

Rearranging the equilibrium expression gives the key to the indicator's behaviour:

$$[\mathrm{H^+}] = K_{In}\frac{[\mathrm{HIn}]}{[\mathrm{In^-}]} \quad \Longrightarrow \quad \mathrm{pH} = pK_{In} + \log\frac{[\mathrm{In^-}]}{[\mathrm{HIn}]}$$

- In **acidic** solution (excess $\mathrm{H^+}$), Le Chatelier's principle pushes the equilibrium to the left: the colour of HIn dominates.
- In **alkaline** solution, $\mathrm{H^+}$ is removed, the equilibrium shifts right: the colour of $\mathrm{In^-}$ dominates.
- When $[\mathrm{HIn}] = [\mathrm{In^-}]$, $\mathrm{pH} = pK_{In}$: this is the **transition point**, where the intermediate (mixed) colour is seen.

The human eye can only detect that one form dominates when it is roughly **ten times** more concentrated than the other. Since $\log 10 = 1$, this leads to a very useful rule.

PROPERTY **Colour-change range of an indicator**

An indicator changes colour gradually over a pH range of approximately

$$\text{pH} = pK_{In} \pm 1$$

Below $pK_{In} - 1$ the acid colour is seen; above $pK_{In} + 1$ the base colour is seen; in between, the colour changes progressively. (Proof not examinable, but you must be able to justify it using the ratio $[\text{In}^-]/[\text{HIn}]$ varying from 1/10 to 10.)

Indicator	pK_{In}	pH range	Acid colour	Base colour
Methyl orange	3.7	3.1 – 4.4	red	yellow
Litmus	7.0	5.0 – 8.0	red	blue
Phenolphthalein	9.3	8.3 – 10.0	colourless	pink

WARNING **Litmus is a poor titration indicator**

Litmus changes over the very wide range pH 5–8, so its colour change is gradual and imprecise. It is fine for testing whether a solution is acidic or alkaline, but never choose it to detect the end point of a titration.

10.2 pH Changes During a Titration: Equivalence Point and End Point

Consider adding 0.100 mol dm^{-3} NaOH from a burette to 25.0 cm^3 of 0.100 mol dm^{-3} HCl in a conical flask. The reaction is:

$$\text{HCl(aq)} + \text{NaOH(aq)} \longrightarrow \text{NaCl(aq)} + \text{H}_2\text{O(l)} \qquad \text{i.e.} \qquad \text{H}^+\text{(aq)} + \text{OH}^-\text{(aq)} \longrightarrow \text{H}_2\text{O(l)}$$

Let us track the pH using the methods of the previous section.

- **Start (0 cm^3):** strong acid, $[\text{H}^+] = 0.100$ mol dm^{-3}, so pH = 1.00.
- **After 10.0 cm^3:** moles H^+ remaining $= 0.100 \times (25.0 - 10.0) \times 10^{-3} = 1.50 \times 10^{-3}$ mol in a total volume of 35.0 cm^3, giving $[\text{H}^+] = 0.0429$ mol dm^{-3} and pH = 1.37. The pH has barely moved!
- **After 24.9 cm^3:** $[\text{H}^+] = \dfrac{0.100 \times 0.1 \times 10^{-3}}{49.9 \times 10^{-3}} = 2.0 \times 10^{-4}$ mol dm^{-3}, so pH = 3.70.
- **After 25.0 cm^3:** exactly enough OH^- has been added to react with all the H^+. Only NaCl and water remain: pH = 7.00.
- **After 25.1 cm^3:** excess $\text{OH}^- = 0.100 \times 0.1 \times 10^{-3} = 1.0 \times 10^{-5}$ mol in 50.1 cm^3, giving $[\text{OH}^-] = 2.0 \times 10^{-4}$ mol dm^{-3}, pOH = 3.70, pH = 10.30.

Look at the last three values: a single drop (from 24.9 to 25.1 cm^3) makes the pH jump from 3.7 to 10.3 — a leap of nearly 7 pH units! This near-vertical jump is the most important feature of every titration curve.

DEFINITION **Equivalence point and end point**

The **equivalence point** of a titration is the point at which the amount of titrant added is exactly the stoichiometric amount needed to react completely with the analyte ($n_{acid} = n_{base}$, adjusted for stoichiometric coefficients). It is a fixed, theoretical point.
The **end point** is the point at which the indicator changes colour. It is an experimental observation. A good titration is one where the end point coincides with the equivalence point.

WARNING **Equivalence point ≠ end point**

These two words are not synonyms, and GCE examiners penalise candidates who confuse them. The equivalence point is stoichiometric; the end point depends on the indicator chosen. We choose the indicator precisely so that its end point falls at (or extremely close to) the equivalence point.

10.3 Titration Curves and the Choice of Indicator

A **titration curve** is a graph of pH (vertical axis) against volume of titrant added (horizontal axis). Its shape depends on the strengths of the acid and the base.

Strong acid titrated with strong base (SA–SB).

The curve starts at very low pH (≈ 1), rises very slowly, then shoots up almost vertically through pH 7 at the equivalence point, and finally levels off near pH 13. The vertical region is very long (roughly pH 3 to 11). **Any indicator** whose range lies inside this window is suitable: both methyl orange and phenolphthalein work.

Weak acid titrated with strong base (WA–SB).

The curve starts higher (pH ≈ 3 for a 0.1 mol dm^{-3} weak acid) and shows a gently rising "buffer region" before equivalence. At half-equivalence, [HA] = [A$^-$], so pH = pK_a — a very useful way of measuring pK_a experimentally. At the equivalence point the solution contains only the salt NaA, whose anion A$^-$ is a weak base (salt hydrolysis, next section): **the pH at equivalence is above 7** (typically 8–9). The vertical region is short (roughly pH 7 to 11), so only **phenolphthalein** is suitable; methyl orange would change colour far too early.

WARNING **The equivalence point of a WA–SB titration is NOT at pH 7**

A very common mistake is to write "pH = 7 at equivalence" for every titration. pH 7 at equivalence occurs only for a strong acid–strong base titration. For a weak acid–strong base titration the equivalence pH is above 7 (basic salt); for a strong acid–weak base titration it is below 7 (acidic salt).

Strong acid titrated with weak base (SA–WB).

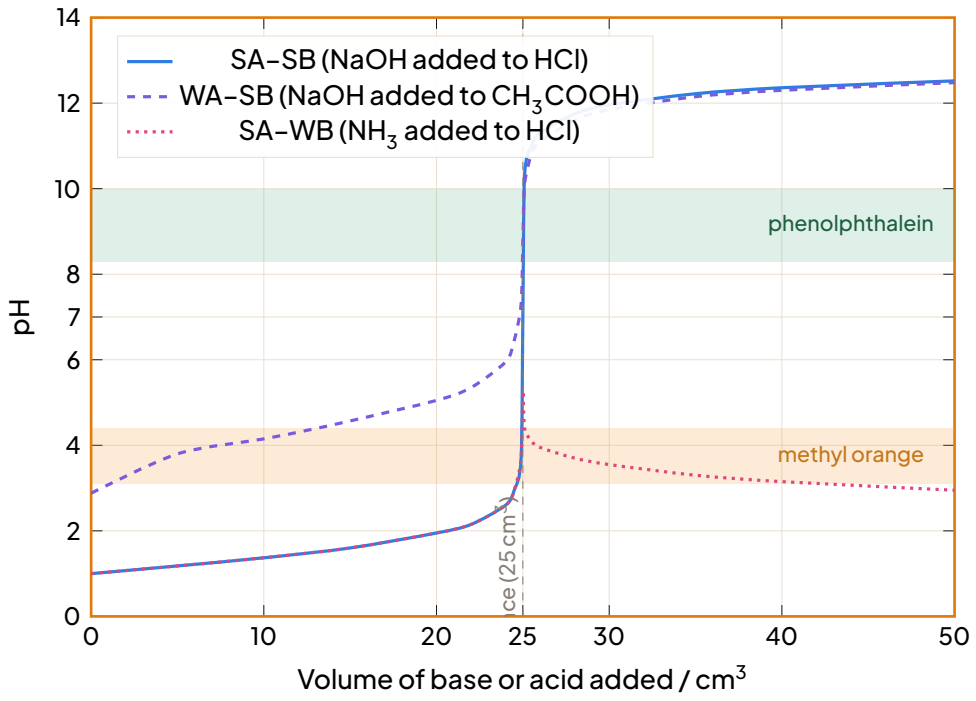
The curve starts at low pH; the vertical region lies roughly between pH 3 and 7, with the equivalence point **below 7** (the salt, e.g. NH_4Cl, is acidic). **Methyl orange** is the suitable indicator; phenolphthalein would change long after equivalence.

Weak acid titrated with weak base (WA–WB).

There is **no sharp vertical region**: the pH drifts gradually through the equivalence point. **No common indicator** changes sharply enough to locate the equivalence point, so this titration is never performed with an indicator (a pH meter must be used instead).

METHOD **Choosing the correct indicator**

1. Identify the strengths of the acid and the base (strong or weak).
2. Sketch (or recall) the titration curve and locate the vertical region and the equivalence point pH.
3. Choose the indicator whose range $pK_{In} \pm 1$ falls entirely within the vertical region.
4. Check: SA–SB → either indicator; WA–SB → phenolphthalein; SA–WB → methyl orange; WA–WB → no suitable indicator.



Titration curves for 0.1 mol dm^{-3} solutions: strong acid–strong base (solid blue), weak acid–strong base (dashed purple) and strong acid–weak base (dotted pink), with the colour-change ranges of methyl orange and phenolphthalein shaded. An indicator is suitable only if its band lies within the vertical part of the curve.

WORKED EXAMPLE 19 **Choosing and justifying an indicator**

25.0 cm^3 of 0.100 mol dm^{-3} ethanoic acid, CH_3COOH ($K_a = 1.8 \times 10^{-5}$ mol dm^{-3}), is titrated with 0.100 mol dm^{-3} NaOH. (a) Calculate the pH at the start and at half-equivalence. (b) State, with reasons, which of methyl orange or phenolphthalein is suitable.

SOLUTION

(a) At the start, for a weak acid $[H^+] \approx \sqrt{K_a c} = \sqrt{1.8 \times 10^{-5} \times 0.100} = 1.34 \times 10^{-3}$ mol dm^{-3}, so pH = 2.87.

At half-equivalence, half the acid has been converted to CH_3COO^-, so $[CH_3COOH] = [CH_3COO^-]$ and

$$\text{pH} = pK_a = -\log(1.8 \times 10^{-5}) = 4.74.$$

(b) This is a weak acid–strong base titration: the equivalence point lies above pH 7 (the ethanoate ion is basic) and the vertical region runs roughly from pH 7 to 11. Phenolphthalein (range 8.3–10.0) lies inside this region and is suitable. Methyl orange (3.1–4.4) would change colour well before equivalence and is unsuitable.

10.4 Titrations Involving Diprotic Species: Two Equivalence Points

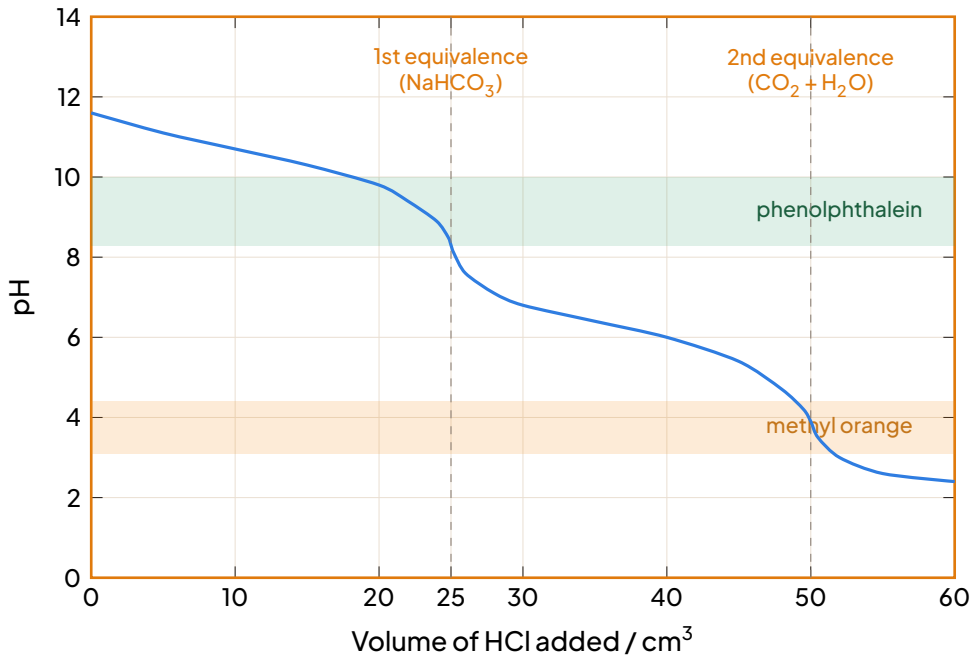
Some acids can donate **two** protons (diprotic acids, e.g. H_2SO_4, H_2CO_3) and some bases can accept two protons (e.g. the carbonate ion CO_3^{2-}). When such a species is titrated with a monoprotic partner, the reaction occurs in **two successive steps**, giving **two equivalence points** and **two vertical regions** on the curve.

The classic GCE example is the titration of sodium carbonate solution with dilute hydrochloric acid:

Step 1: $Na_2CO_3(aq) + HCl(aq) \longrightarrow NaHCO_3(aq) + NaCl(aq)$ i.e. $CO_3^{2-} + H^+ \longrightarrow HCO_3^-$

Step 2: $NaHCO_3(aq) + HCl(aq) \longrightarrow NaCl(aq) + H_2O(l) + CO_2(g)$ i.e. $HCO_3^- + H^+ \longrightarrow H_2CO_3$

The first equivalence point (near pH 8.3, where the solution contains $NaHCO_3$) is detected by **phenolphthalein** (pink → colourless). The second equivalence point (near pH 3.9, where all carbonate has become CO_2 and water) is detected by **methyl orange** (yellow → orange/red). Note that the volume of acid needed for step 2 equals the volume for step 1, because each mole of CO_3^{2-} produces exactly one mole of HCO_3^-, which then consumes a second mole of H^+.



Titration of 25.0 cm^3 of 0.10 mol dm^{-3} Na_2CO_3 with 0.10 mol dm^{-3} HCl. Two protons are accepted in succession, giving two equivalence points of equal spacing: the first within the phenolphthalein range, the second within the methyl orange range.

EXEMPLE **Reading the two end points**

In the titration above, phenolphthalein decolourises after 25.0 cm^3 of acid and methyl orange turns after a further 25.0 cm^3 (50.0 cm^3 in total). The equality of the two volumes

confirms that the base is CO_3^{2-} (diprotic): $n(\text{HCl})$ total = $2\,n(Na_2CO_3)$. This "double indicator" technique is a favourite GCE practical question.

WORKED EXAMPLE 20 **Double-indicator calculation**

25.0 cm^3 of a sodium carbonate solution is titrated with 0.100 mol dm^{-3} HCl. Phenolphthalein decolourises after 12.50 cm^3 of acid; methyl orange then changes after a further 12.50 cm^3. Calculate the concentration of the Na_2CO_3 solution.

SOLUTION

The first end point corresponds to step 1 only: $CO_3^{2-} + H^+ \longrightarrow HCO_3^-$, so

$$n(Na_2CO_3) = n(\text{HCl})_{\text{step 1}} = 0.100 \times 12.50 \times 10^{-3} = 1.25 \times 10^{-3}\ \text{mol}.$$

Hence

$$[Na_2CO_3] = \frac{1.25 \times 10^{-3}}{25.0 \times 10^{-3}} = 0.0500\ \text{mol dm}^{-3}.$$

Check with the total titre: $n(\text{HCl})_{\text{total}} = 0.100 \times 25.0 \times 10^{-3} = 2.50 \times 10^{-3}$ mol $= 2\,n(Na_2CO_3)$, as expected for a diprotic base.

10.5 Reading Titration Curves: Finding pK_a at Half-Equivalence

A titration curve is not only a way to detect equivalence — it is also an **experimental source of data**. For a weak acid titrated with a strong base, at the **half-equivalence point** exactly half of the acid HA has been converted into its conjugate base A^-. Then:

$$[\text{HA}] = [A^-] \quad \Longrightarrow \quad \text{pH} = pK_a + \log\frac{[A^-]}{[\text{HA}]} = pK_a$$

METHOD **Determining pK_a from a titration curve (GCE skill)**

1. Locate the equivalence point (midpoint of the vertical jump) and read its volume V_e.
2. Halve it: at $V_e/2$, read the pH from the curve.
3. This pH is the pK_a of the weak acid; then $K_a = 10^{-pK_a}$.

The same logic applies to a weak base titrated with a strong acid, giving pK_b (or pK_a of the conjugate acid) at half-equivalence.

WORKED EXAMPLE 21 **Reading pK_a from a curve**

The titration of 25.0 cm^3 of a weak acid HA with 0.100 mol dm^{-3} NaOH shows a vertical jump centred at 20.0 cm^3. The pH read at 10.0 cm^3 of added base is 4.80. Determine pK_a and K_a of the acid, and its initial concentration.

SOLUTION

Half-equivalence occurs at $V_e/2 = 10.0\ cm^3$, so $pK_a = \text{pH} = 4.80$ and

$$K_a = 10^{-4.80} = 1.6 \times 10^{-5}\ \text{mol dm}^{-3}.$$

At equivalence, $n(\text{HA}) = n(\text{NaOH}) = 0.100 \times 20.0 \times 10^{-3} = 2.00 \times 10^{-3}$ mol in 25.0 cm^3, so $c(\text{HA}) = \dfrac{2.00 \times 10^{-3}}{25.0 \times 10^{-3}} = 0.0800\ \text{mol dm}^{-3}$.

KEY POINTS **Key points of this section**

- An indicator is a weak acid/base whose two forms have different colours; it changes over pH = $pK_{In} \pm 1$.
- The equivalence point is stoichiometric; the end point is the indicator's colour change — choose the indicator so the two coincide.
- SA–SB: equivalence at pH 7, any common indicator. WA–SB: equivalence above 7, phenolphthalein. SA–WB: equivalence below 7, methyl orange. WA–WB: no suitable indicator.
- Diprotic systems (e.g. Na_2CO_3 with HCl) show two equivalence points of equal spacing; phenolphthalein detects the first, methyl orange the second.
- At half-equivalence, pH = pK_a: a powerful way to read K_a off a titration curve.

EXERCISE 11 **Indicator choice and curve reading**

1. 20.0 cm^3 of 0.0500 mol dm^{-3} ammonia solution is titrated with 0.100 mol dm^{-3} HCl. (a) Calculate the volume of acid needed to reach equivalence. (b) State whether the pH at equivalence is above, below or equal to 7, justifying your answer with an equation. (c) Choose a suitable indicator from methyl orange and phenolphthalein.
2. A student titrates ethanoic acid with sodium hydroxide using methyl orange. Explain why the titration result is unreliable, and name the indicator that should have been used.

3. Sketch the titration curve for 25 cm^3 of 0.1 mol dm^{-3} Na_2CO_3 titrated with 0.1 mol dm^{-3} HCl, marking both equivalence volumes and the two suitable indicators.

SOLUTION **Exercise 1**

1. (a) $n(NH_3) = 0.0500 \times 20.0 \times 10^{-3} = 1.00 \times 10^{-3}$ mol; reaction is 1:1, so $V(HCl) = \frac{1.00 \times 10^{-3}}{0.100} = 10.0\ cm^3$. (b) Below 7: the salt NH_4Cl formed contains NH_4^+, a weak acid: $NH_4^+(aq) + H_2O(l) \rightleftharpoons NH_3(aq) + H_3O^+(aq)$. (c) Strong acid–weak base: vertical region around pH 3–7, so methyl orange.
2. Ethanoic acid is weak and NaOH is strong: the equivalence point is above pH 7 and the vertical region is roughly pH 7–11. Methyl orange (3.1–4.4) changes colour well before equivalence, giving a falsely small titre. Phenolphthalein should be used.
3. Curve starts near pH 11.6; first vertical region (phenolphthalein range) at 25 cm^3 ($CO_3^{2-} \longrightarrow HCO_3^-$); buffer-like region around pH 6.3; second vertical region (methyl orange range) at 50 cm^3 ($HCO_3^- \longrightarrow H_2CO_3$); levels near pH 2–3 in excess acid.

11 BUFFER SOLUTIONS

In the previous section we followed the pH during a titration and saw how violently it can change near the equivalence point: a single drop of sodium hydroxide can send the pH of a strong acid solution from 3 to 11. In many real situations, however, such swings are unacceptable. Your blood must stay within a very narrow pH window (about 7.35 to 7.45) or enzymes stop working; the palm-wine tappers of the South-West Region rely on a stable pH for fermentation; a shampoo that is too acidic or too alkaline damages hair. Nature and industry solve this problem with **buffer solutions**, the subject of this section.

11.1 What is a Buffer Solution?

An observation to start with.

Take two beakers, each containing 100 cm^3 of liquid at pH 7. The first contains pure water, the second a mixture of ethanoic acid and sodium ethanoate. Add 1 cm^3 of 1.0 mol dm^{-3} hydrochloric acid to each. In the water, the pH crashes from 7 to about 2. In the second beaker, the pH barely moves, perhaps from 4.76 to 4.75. The second liquid has resisted the change. It is a buffer.

DEFINITION **Buffer solution**

A **buffer solution** is a solution that resists changes in pH when small amounts of acid (H^+) or base (OH^-) are added to it, or when it is moderately diluted.

WARNING **A buffer does not fix the pH**

A buffer does not keep the pH perfectly constant. It only minimises the change. Adding acid to a buffer always lowers the pH slightly; adding base always raises it slightly. If you keep adding acid, the buffer is eventually overwhelmed and the pH falls sharply. Examiners love the statement "a buffer keeps the pH constant" — it is false and costs marks.

11.2 Composition and Types of Buffers

How can a solution fight both added acid and added base? It must contain, in large amounts, **both** a species that can consume H^+ and a species that can consume OH^-. A weak acid HA and its conjugate base A^- do exactly this.

PROPERTY **Composition of a buffer**

An effective buffer contains large, comparable amounts of a weak acid and its conjugate base (or a weak base and its conjugate acid). The two species must be a conjugate pair, and neither may be strong.

There are two classic types.

1. Acidic buffers (pH < 7). Made from a **weak acid and one of its salts** (which supplies the conjugate base). Typical example: ethanoic acid and sodium ethanoate, CH_3COOH/CH_3COONa. Such a buffer can also be made by partially neutralising an excess of weak acid with sodium hydroxide: the unreacted acid and the ethanoate formed then coexist.

2. Alkaline buffers (pH > 7). Made from a **weak base and one of its salts** (which supplies the conjugate acid). Typical example: ammonia and ammonium chloride, $NH_{3/}NH_4Cl$.

WARNING **A strong acid and its salt do NOT form a buffer**

HCl and NaCl mixed together are not a buffer. HCl is fully ionised, so there is no reservoir of undissociated HA to absorb added OH^-; and Cl^- is such a weak base that it cannot absorb added H^+. Only a weak acid/base pair works, because the undissociated weak species and its conjugate partner must both exist in large amounts.

11.3 How a Buffer Works

The acidic buffer CH_3COOH/CH_3COO^-.

In the mixture, the weak acid is barely ionised, so we have a large reservoir of CH_3COOH molecules and (from the fully ionised salt) a large reservoir of CH_3COO^- ions.

- **Add acid (H^+):** the ethanoate ions mop it up:

$$CH_3COO^-(aq) + H^+(aq) \longrightarrow CH_3COOH(aq)$$

The added H^+ is converted into weakly ionised ethanoic acid, so the free H^+ concentration hardly changes.

- **Add base (OH^-):** the ethanoic acid neutralises it:

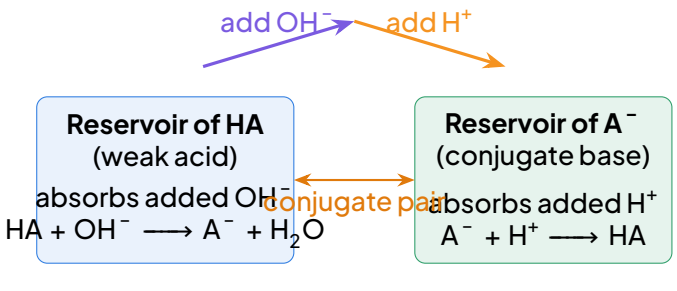
$$CH_3COOH(aq) + OH^-(aq) \longrightarrow CH_3COO^-(aq) + H_2O(l)$$

The added OH^- is removed as water.

In both cases the added strong reagent is "swallowed" by one reservoir and converted into the other partner of the pair; only the ratio $[CH_3COO^-]/[CH_3COOH]$ shifts slightly, and we shall see that the pH depends only on this ratio.

The alkaline buffer NH_3/NH_4^+.

- **Add acid:** $NH_3(aq) + H^+(aq) \longrightarrow NH_4^+(aq)$
- **Add base:** $NH_4^+(aq) + OH^-(aq) \longrightarrow NH_3(aq) + H_2O(l)$



An acidic buffer: two large reservoirs, HA and A^-, neutralise added base and added acid respectively.

11.4 Calculating the pH of an Acidic Buffer

For the weak acid equilibrium $HA(aq) + H_2O(l) \rightleftharpoons H_3O^+(aq) + A^-(aq)$:

$$K_a = \frac{[H_3O^+][A^-]}{[HA]}$$

Rearranging for $[H_3O^+]$:

$$[H_3O^+] = K_a \times \frac{[HA]}{[A^-]}$$

Because the acid is weak and the concentrations of acid and salt are reasonably large (typically $> 0.01\,mol\,dm^{-3}$), the equilibrium concentrations are essentially the analytical concentrations of acid and salt mixed: $[HA] \approx c_{acid}$ and $[A^-] \approx c_{salt}$. Taking $-\log$ of both sides gives the celebrated **Henderson–Hasselbalch equation**:

PROPERTY **Henderson–Hasselbalch equation**

For an acidic buffer made of a weak acid HA and its salt A^-:

$$\text{pH} = \text{p}K_a + \log\frac{[\text{salt}]}{[\text{acid}]} = \text{p}K_a + \log\frac{[A^-]}{[HA]}$$

Since the two species share the same volume, concentrations may be replaced by moles. The derivation from K_a is examinable and you should be able to reproduce it.

Note the immediate consequences: when [salt] = [acid], pH = pK_a; and because pH depends on a ratio, diluting the buffer with water changes both concentrations equally and leaves the pH (almost) unchanged — a point the GCE likes to test.

METHOD **Calculating the pH of an acidic buffer**

1. Identify the weak acid and its conjugate base (the salt).
2. Write K_a and rearrange: $[H_3O^+] = K_a\dfrac{[\text{acid}]}{[\text{salt}]}$.
3. Substitute concentrations (or moles) of acid and salt.
4. Compute $[H_3O^+]$, then pH = $-\log[H_3O^+]$ — or use Henderson–Hasselbalch directly.
5. To find the pH after adding strong acid or base, first adjust the moles of HA and A^- using the neutralisation equations, then reapply the formula.

WORKED EXAMPLE 22 **pH of an ethanoic acid / ethanoate buffer**

A buffer is prepared by dissolving 0.10 mol of ethanoic acid and 0.15 mol of sodium ethanoate in water and making up to 1.00 dm^3. Given $K_a(CH_3COOH) = 1.8 \times 10^{-5}$ mol dm^{-3}, calculate the pH of the buffer.

SOLUTION

Solution. Both species are in the same volume, so we may work with moles.

$$[H_3O^+] = K_a \times \frac{n(\text{acid})}{n(\text{salt})} = 1.8 \times 10^{-5} \times \frac{0.10}{0.15} = 1.2 \times 10^{-5}\ \text{mol dm}^{-3}$$

$$\text{pH} = -\log(1.2 \times 10^{-5}) = 4.92$$

Check with Henderson–Hasselbalch: $\text{p}K_a = -\log(1.8 \times 10^{-5}) = 4.74$, so pH = 4.74 + log(0.15/0.10) = 4.74 + 0.18 = 4.92. ✓

WORKED EXAMPLE 23 **Effect of adding a small amount of acid to the buffer**

To 1.00 dm^3 of the buffer above (0.10 mol CH_3COOH, 0.15 mol CH_3COO^-) we add 0.010 mol of HCl (negligible volume change). Calculate the new pH, and compare with adding the same acid to 1.00 dm^3 of pure water.

SOLUTION

Solution. The added H^+ converts ethanoate into ethanoic acid: $CH_3COO^- + H^+ \longrightarrow CH_3COOH$.

$$n(CH_3COO^-) = 0.15 - 0.010 = 0.14\ \text{mol}$$
$$n(CH_3COOH) = 0.10 + 0.010 = 0.11\ \text{mol}$$

$$\text{pH} = 4.74 + \log\frac{0.14}{0.11} = 4.74 + 0.10 = 4.84$$

The pH falls by only 0.08 unit. In pure water, $[H_3O^+] = 0.010\ \text{mol dm}^{-3}$, giving pH = 2.00: a fall of five pH units from 7. The buffer has reduced the change by a factor of more than 60.

11.5 Calculating the pH of an Alkaline Buffer

For an alkaline buffer such as NH_3/NH_4Cl, two equivalent routes exist.

Route 1: use K_b of the weak base. From $NH_3 + H_2O \rightleftharpoons NH_4^+ + OH^-$:

$$K_b = \frac{[NH_4^+][OH^-]}{[NH_3]} \quad\Longrightarrow\quad [OH^-] = K_b \times \frac{[\text{base}]}{[\text{salt}]}$$

then pOH = $-\log[OH^-]$ and pH = 14 − pOH (at 25 °C). Equivalently, $\text{pOH} = \text{p}K_b + \log\frac{[\text{salt}]}{[\text{base}]}$.

Route 2: use K_a of the conjugate acid NH_4^+ and apply Henderson–Hasselbalch to the pair NH_4^+/NH_3:

$$\text{pH} = \text{p}K_a(NH_4^+) + \log\frac{[NH_3]}{[NH_4^+]} \qquad \text{with } \text{p}K_a + \text{p}K_b = 14 \text{ at } 25\,^\circ\text{C}.$$

WORKED EXAMPLE 24 **pH of an ammonia / ammonium chloride buffer**

A solution contains 0.20 mol dm^{-3} NH_3 and 0.10 mol dm^{-3} NH_4Cl. Given $K_b(NH_3) = 1.8 \times 10^{-5}$ mol dm^{-3}, calculate its pH at 25 °C.

SOLUTION

Solution.

$$[OH^-] = K_b \times \frac{[NH_3]}{[NH_4^+]} = 1.8 \times 10^{-5} \times \frac{0.20}{0.10} = 3.6 \times 10^{-5}\ \text{mol dm}^{-3}$$

$$\text{pOH} = -\log(3.6 \times 10^{-5}) = 4.44 \quad\Longrightarrow\quad \text{pH} = 14.00 - 4.44 = 9.56.$$

Check via Route 2: $pK_b = 4.74$, so $pK_a(NH_4^+) = 14.00 - 4.74 = 9.26$. pH = 9.26 + log(0.20/0.10) = 9.26 + 0.30 = 9.56. ✓

11.6 Designing a Buffer of a Target pH

METHOD **Preparing a buffer of a given pH**

1. Choose a weak acid whose pK_a is close to the target pH (ideally within ±1 unit), because a buffer works best when [salt] ≈ [acid].
2. Use Henderson–Hasselbalch to find the required ratio: $\log \frac{[\text{salt}]}{[\text{acid}]} = \text{pH} - pK_a$.
3. Mix the acid and its salt in that mole ratio (or partially neutralise the weak acid with NaOH to generate the salt in situ).
4. For an alkaline buffer, choose a weak base whose conjugate acid has pK_a near the target pH.

EXEMPLE **Choosing the right acid**

To buffer at pH 5.0, ethanoic acid ($pK_a = 4.74$) is ideal. To buffer near pH 9.25, the NH_4^+/NH_3 system ($pK_a(NH_4^+) = 9.25$) is the natural choice. Using an acid with pK_a far from the target would require an extreme ratio of salt to acid, giving a poor buffer.

WORKED EXAMPLE 25 **Preparing a buffer by partial neutralisation**

Calculate the volume of 0.50 mol dm^{-3} NaOH that must be added to 500 cm^3 of 0.40 mol dm^{-3} ethanoic acid to produce a buffer of pH 5.00. ($K_a = 1.8 \times 10^{-5}$)

SOLUTION

Solution. Target pH = 5.00, $pK_a = 4.74$. Required ratio:

$$\log \frac{[\text{salt}]}{[\text{acid}]} = 5.00 - 4.74 = 0.26 \quad \Rightarrow \quad \frac{[\text{salt}]}{[\text{acid}]} = 10^{0.26} = 1.82$$

Let x = moles of NaOH added. Initial moles $CH_3COOH = 0.500 \times 0.40 = 0.20$ mol. After reaction: $n(CH_3COO^-) = x$, $n(CH_3COOH) = 0.20 - x$.

$$\frac{x}{0.20 - x} = 1.82 \quad \Rightarrow \quad x = 1.82(0.20 - x) \quad \Rightarrow \quad 2.82x = 0.364 \quad \Rightarrow \quad x = 0.129 \text{ mol}$$

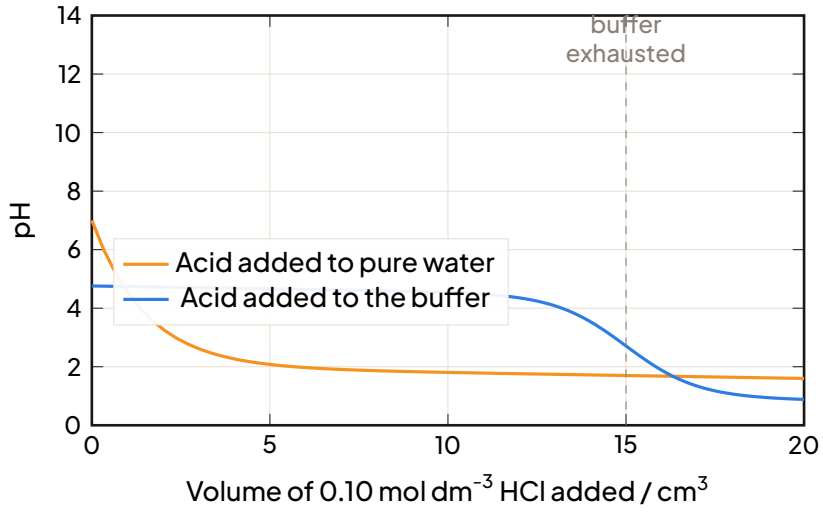
Volume of NaOH = 0.129/0.50 = 0.258 dm^3 = 258 cm^3.

11.7 Buffer Capacity and Dilution

DEFINITION **Buffer capacity**

The **buffer capacity** is the amount of strong acid or base a buffer can absorb before its pH changes significantly (often defined as the amount needed to change the pH of 1 dm^3 of buffer by 1 unit). It is greater when the total concentrations of the weak acid and its conjugate base are larger, and it is maximum when [salt] = [acid], i.e. when pH = pK_a.

Effect of dilution (GCE-useful extra). Diluting a buffer with pure water divides both [salt] and [acid] by the same factor, so their ratio — and hence the pH — is essentially unchanged. However, the capacity decreases: a diluted buffer resists smaller amounts of added acid or base. This is why concentrated buffer stock solutions are diluted just before use in laboratories.



Adding strong acid to pure water (orange) crashes the pH immediately; the buffer (blue) holds the pH nearly constant until its reservoirs are exhausted.

11.8 Uses of Buffer Solutions

DID YOU KNOW? **Buffers at work around us**

- Blood. The carbonic acid / hydrogencarbonate system, H_2CO_3/HCO_3^-, keeps blood pH near 7.4. CO_2 from respiration feeds the equilibrium $CO_2 + H_2O \rightleftharpoons H_2CO_3 \rightleftharpoons H^+ + HCO_3^-$, and the lungs and kidneys adjust the two reservoirs.
- Food and fermentation. Controlled pH is essential in fermenting cassava into garri, in brewing, and in yoghurt and cheese making; buffers keep micro-organisms in their op-

timal pH range.

- Pharmaceuticals. Injections and eye drops are buffered so they do not damage tissues.
- Cosmetics. Shampoos are buffered near pH 5.5 to protect hair keratin.
- Laboratory. Standard buffer solutions of known pH are used to calibrate pH meters.

KEY POINTS **Key points on buffers**

- A buffer resists (but does not prevent) pH changes when small amounts of acid or base are added.
- Acidic buffer: weak acid + its salt, e.g. CH_3COOH/CH_3COONa; alkaline buffer: weak base + its salt, e.g. $NH_{3/}NH_4Cl$.
- Added H^+ is consumed by the conjugate base; added OH^- by the weak acid.
- $pH = pK_a + \log \frac{[salt]}{[acid]}$; for alkaline buffers use K_b or the K_a of the conjugate acid.
- Choose the weak acid with pK_a close to the target pH; capacity is greatest when the two components are equal.
- Dilution leaves the pH almost unchanged but lowers the capacity.

EXERCISE 12 **Buffer calculations**

1. Calculate the pH of a buffer containing 0.25 mol dm^{-3} methanoic acid HCOOH (K_a = 1.6×10^{-4} mol dm^{-3}) and 0.10 mol dm^{-3} sodium methanoate.
2. 0.020 mol of NaOH is added to 1.00 dm^3 of a buffer containing 0.30 mol CH_3COOH and 0.30 mol CH_3COONa (pK_a = 4.74). Calculate the pH before and after the addition.
3. Given ethanoic acid ($K_a = 1.8 \times 10^{-5}$), what mole ratio of sodium ethanoate to ethanoic acid is required to prepare a buffer of pH 5.10?

SOLUTION **Exercise 1**

1. $[H_3O^+] = K_a \frac{[acid]}{[salt]} = 1.6 \times 10^{-4} \times \frac{0.25}{0.10} = 4.0 \times 10^{-4}$, so pH = 3.40.

2. Before: equal amounts, so pH = pK_a = 4.74. After adding OH^-: $n(CH_3COOH)$ = 0.30 − 0.020 = 0.28 mol; $n(CH_3COO^-)$ = 0.30 + 0.020 = 0.32 mol. pH = 4.74 + log(0.32/0.28) = 4.74 + 0.06 = 4.80. The pH rises by only 0.06 unit.

3. $pK_a = -\log(1.8 \times 10^{-5}) = 4.74$. $\log \frac{[salt]}{[acid]} = 5.10 - 4.74 = 0.36$, so $\frac{[salt]}{[acid]} = 10^{0.36} \approx 2.3$: mix salt and acid in the mole ratio 2.3 : 1.

12 SALT HYDROLYSIS, SOLUBILITY PRODUCT AND COMMON ION EFFECT

12.1 Introduction: From Buffers to the pH of Salt Solutions

In the previous section, we saw how to prepare a buffer solution. The buffer's pH depends on the relative strengths of the weak acid and the weak base. In the previous section, we saw how to calculate the pH of a solution of a weak acid and a

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12.2 Salt Hydrolysis

EXPERIMENT **A surprising observation in the school laboratory**

Dissolve a spatula of sodium chloride, NaCl, in distilled water: the pH paper shows pH = 7. Repeat with sodium ethanoate, CH_3COONa: the solution is alkaline (pH ≈ 9). Repeat again with ammonium chloride, NH_4Cl: the solution is acidic (pH ≈ 5). Yet none of these solids is an acid or a base in the Brønsted–Lowry sense! The explanation is that some ions of the salt react with water itself.

DEFINITION **Salt hydrolysis**

Salt hydrolysis is the reaction of the cation and/or the anion of a dissolved salt with water, producing H_3O^+ or OH^- ions and thereby altering the pH of the solution from 7.

The rule of thumb is simple: **the ion derived from the weak parent (weak acid or weak base) is the one that hydrolyses**. Ions derived from strong acids or strong bases are spectators, because the reverse reaction that would re-form the parent is negligible.

Case 1: Salt of a strong acid and a strong base.

Example: NaCl (from HCl and NaOH). In solution we have $Na^+(aq)$ and $Cl^-(aq)$. Neither ion reacts with water: Cl^- is the conjugate base of a strong acid, so it has no tendency to accept a proton; Na^+ is the conjugate acid of a strong base. There is **no hydrolysis** and the solution is **neutral**, pH = 7 at 25 °C.

Case 2: Salt of a weak acid and a strong base.

Example: CH_3COONa (from CH_3COOH and NaOH). The Na^+ ion is a spectator, but the ethanoate ion is the conjugate base of a weak acid, so it accepts a proton from water:

$$CH_3COO^-(aq) + H_2O(l) \rightleftharpoons CH_3COOH(aq) + OH^-(aq)$$

OH^- ions are produced, so the solution is **alkaline** (pH > 7).

Case 3: Salt of a strong acid and a weak base.

Example: NH_4Cl (from HCl and NH_3). The Cl^- ion is a spectator, but the ammonium ion is the conjugate acid of the weak base NH_3, so it donates a proton to water:

$$NH_4^+(aq) + H_2O(l) \rightleftharpoons NH_3(aq) + H_3O^+(aq)$$

H_3O^+ ions are produced, so the solution is **acidic** (pH < 7).

Case 4: Salt of a weak acid and a weak base.

Example: CH_3COONH_4. Both ions hydrolyse: CH_3COO^- produces OH^- while NH_4^+ produces H_3O^+. The resulting pH depends on the relative values of K_a (of the weak acid) and K_b (of the weak base):

- if $K_a = K_b$, the solution is neutral (this is approximately the case for CH_3COONH_4, pH ≈ 7);
- if $K_a > K_b$, the solution is acidic;
- if $K_a < K_b$, the solution is alkaline.

Classification of salts and the pH of their aqueous solutions

Strong acid + strong base	Weak acid + strong base	Strong acid + weak base	Weak acid + weak base
NaCl pH = 7 (neutral)	CH_3COONa pH > 7 (alkaline)	NH_4Cl pH < 7 (acidic)	CH_3COONH_4 pH set by K_a vs K_b

The pH of a salt solution is fixed by the strengths of the parent acid and base.

KEY POINTS **Key points on hydrolysis**

- Only the ion coming from the weak parent hydrolyses.
- Anion hydrolysis ⇒ OH^- ⇒ alkaline solution; cation hydrolysis ⇒ H_3O^+ ⇒ acidic solution.
- Salts of strong acid + strong base do not hydrolyse: pH = 7.

12.3 Solubility and the Solubility Product K_{sp}

When silver chloride is shaken with water, a tiny amount dissolves and a dynamic equilibrium is established between the solid and its ions:

$$AgCl(s) \rightleftharpoons Ag^+(aq) + Cl^-(aq)$$

Such a solution, in contact with undissolved solid, is called a **saturated solution**. The **solubility** s of the salt is the maximum amount (in mol dm^{-3}) that can dissolve at a given temperature.

DEFINITION **Solubility product**

For a sparingly soluble salt A_mB_n in equilibrium with its saturated solution,

$$A_mB_n(s) \rightleftharpoons m\,A^{n+}(aq) + n\,B^{m-}(aq), \qquad K_{sp} = [A^{n+}]^m\,[B^{m-}]^n$$

where the concentrations are the equilibrium concentrations in the saturated solution. The pure solid does not appear in the expression (its activity is 1). K_{sp} is constant at a given temperature.

The units of K_{sp} depend on the formula: for AgCl, $K_{sp} = [Ag^+][Cl^-]$ has units $mol^2\,dm^{-6}$; for $PbCl_2$, $K_{sp} = [Pb^{2+}][Cl^-]^2$ has units $mol^3\,dm^{-9}$.

WARNING **Stoichiometric powers**

In every K_{sp} expression, raise each ion concentration to its **stoichiometric coefficient**: for $CaF_2 \rightleftharpoons Ca^{2+} + 2\,F^-$, $K_{sp} = [Ca^{2+}][F^-]^2$, not $[Ca^{2+}][F^-]$. Forgetting the power is the most common error at the GCE.

METHOD **Relating solubility s to K_{sp}**

1. Write the dissolving equation and let the molar solubility be s mol dm^{-3}.
2. Express each ionic concentration in terms of s.
3. Substitute into K_{sp} and solve.

Salt AB (e.g. AgCl): $[A^+] = [B^-] = s$, so $\boxed{K_{sp} = s^2}$ and $s = \sqrt{K_{sp}}$.

Salt AB_2 (e.g. $PbCl_2$): $[A^{2+}] = s$, $[B^-] = 2s$, so $\boxed{K_{sp} = 4s^3}$ and $s = \sqrt[3]{K_{sp}/4}$.

Salt A_2B (e.g. Ag_2CrO_4): $[A^+] = 2s$, $[B^{2-}] = s$, so $K_{sp} = (2s)^2 s = 4s^3$.

WORKED EXAMPLE 26 **Solubility of silver chloride**

The solubility product of AgCl at 25 °C is $K_{sp} = 1.8 \times 10^{-10}\ mol^2\,dm^{-6}$. Calculate the solubility of AgCl in pure water in mol dm^{-3} and in g dm^{-3}.

SOLUTION

Solution. For $AgCl(s) \rightleftharpoons Ag^+(aq) + Cl^-(aq)$, let the solubility be s. Then $[Ag^+] = [Cl^-] = s$

and

$$K_{sp} = s^2 \Rightarrow s = \sqrt{1.8 \times 10^{-10}} = 1.34 \times 10^{-5}\ \text{mol dm}^{-3}.$$

With $M(\text{AgCl}) = 108 + 35.5 = 143.5\ \text{g mol}^{-1}$:

$$s = 1.34 \times 10^{-5} \times 143.5 = 1.9 \times 10^{-3}\ \text{g dm}^{-3}.$$

Silver chloride is indeed sparingly soluble: less than 2 mg per litre.

WORKED EXAMPLE 27 **K_{sp} from solubility: an AB_2 salt**

The solubility of lead(II) chloride, $PbCl_2$, in water at 25 °C is $1.6 \times 10^{-2}\ \text{mol dm}^{-3}$. Calculate $K_{sp}(PbCl_2)$.

SOLUTION

Solution. $PbCl_2(s) \rightleftharpoons Pb^{2+}(aq) + 2\,Cl^-(aq)$. With $s = 1.6 \times 10^{-2}$: $[Pb^{2+}] = s$ and $[Cl^-] = 2s = 3.2 \times 10^{-2}\ \text{mol dm}^{-3}$.

$$K_{sp} = [Pb^{2+}][Cl^-]^2 = s(2s)^2 = 4s^3 = 4(1.6 \times 10^{-2})^3 = 1.6 \times 10^{-5}\ \text{mol}^3\ \text{dm}^{-9}.$$

12.4 Predicting Precipitation: Ionic Product versus K_{sp}

When two solutions are mixed, the concentrations of the ions immediately after mixing define the **ionic product** Q, which has the same form as K_{sp} but uses the actual (not necessarily equilibrium) concentrations.

PROPERTY **Precipitation rule**

- $Q < K_{sp}$: the solution is unsaturated; no precipitate forms (more solid could dissolve).
- $Q = K_{sp}$: the solution is exactly saturated; equilibrium.
- $Q > K_{sp}$: the solution is supersaturated; precipitation occurs until $Q = K_{sp}$.

EXEMPLE **Will a precipitate form?**

$50\ \text{cm}^3$ of $2.0 \times 10^{-4}\ \text{mol dm}^{-3}$ $AgNO_3$ is mixed with $50\ \text{cm}^3$ of $2.0 \times 10^{-4}\ \text{mol dm}^{-3}$ NaCl. Does AgCl precipitate? ($K_{sp} = 1.8 \times 10^{-10}$.)
On mixing, each concentration is halved (volume doubles): $[Ag^+] = [Cl^-] = 1.0 \times$

10^{-4} mol dm^{-3}. Then

$$Q = (1.0 \times 10^{-4})^2 = 1.0 \times 10^{-8} > K_{sp} = 1.8 \times 10^{-10},$$

so a white precipitate of AgCl forms. (This is exactly the test for chloride ions in qualitative analysis.)

12.5 The Common Ion Effect

What happens if we try to dissolve AgCl not in pure water but in a solution that already contains chloride ions, say NaCl? By Le Chatelier's principle, the added Cl^- shifts the equilibrium $AgCl(s) \rightleftharpoons Ag^+ + Cl^-$ to the left: less solid dissolves.

DEFINITION **Common ion effect**

The **common ion effect** is the decrease in solubility of a sparingly soluble salt when a soluble compound providing an ion common to the salt is added to the solution.

WORKED EXAMPLE 28 **Solubility of AgCl in 0.10 mol dm^{-3} NaCl**

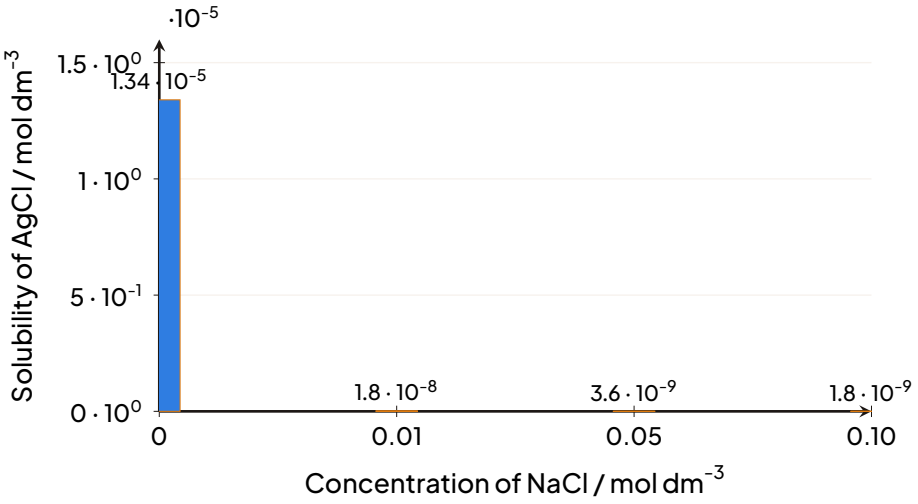
Calculate the solubility of AgCl in a 0.10 mol dm^{-3} NaCl solution and compare with its solubility in pure water. (K_{sp}(AgCl) = 1.8×10^{-10} mol^2 dm^{-6}.)

SOLUTION

Solution. Let s be the solubility of AgCl in the NaCl solution. Then $[Ag^+] = s$ and $[Cl^-] = 0.10 + s \approx 0.10$ (since $s \ll 0.10$).

$$K_{sp} = [Ag^+][Cl^-] = s(0.10) \Rightarrow s = \frac{1.8 \times 10^{-10}}{0.10} = 1.8 \times 10^{-9} \text{ mol dm}^{-3}.$$

In pure water $s = 1.34 \times 10^{-5}$ mol dm^{-3}: the solubility has been divided by about 7 400! The approximation $0.10 + s \approx 0.10$ is fully justified.



Solubility of AgCl in water and in NaCl solutions of increasing concentration at 25 °C: the common ion Cl^- drastically reduces the solubility.

WARNING **K_{sp} does not change!**

The common ion effect reduces the solubility s, but the value of K_{sp} **stays exactly the same** at the same temperature. Only a change of temperature changes K_{sp}. In the example above, $K_{sp} = 1.8 \times 10^{-10}$ in pure water and in the NaCl solution; what changed is how much solid must dissolve to satisfy it.

DID YOU KNOW? **Applications around us**

The common ion effect is exploited every day. In qualitative analysis at the GCE practical examination, group precipitations are controlled by adjusting a common ion. In water treatment, adding lime helps precipitate unwanted ions. Even the formation of kidney stones (CaC_2O_4, calcium oxalate) is a solubility-product problem: when the ionic product of Ca^{2+} and oxalate in urine exceeds K_{sp}, crystals grow. Drinking plenty of water dilutes the ions, keeping $Q < K_{sp}$.

EXERCISE 13 **Consolidation**

1. State, with a hydrolysis equation, whether an aqueous solution of each salt is acidic, neutral or alkaline: (a) KNO_3; (b) NH_4NO_3; (c) Na_2CO_3.

2. The solubility of silver chromate, Ag_2CrO_4, is 6.5×10^{-5} mol dm^{-3} at 25 °C. Calculate

$K_{sp}(Ag_2CrO_4)$.

3. Calculate the solubility of $PbCl_2$ ($K_{sp} = 1.6 \times 10^{-5}$ mol^3 dm^{-9}) in a 0.20 mol dm^{-3} solution of $Pb(NO_3)_2$.

SOLUTION **Exercise 1**

1. (a) KNO_3: strong acid + strong base, no hydrolysis, neutral. (b) $NH_4^+ + H_2O \rightleftharpoons NH_3 + H_3O^+$: acidic. (c) $CO_3^{2-} + H_2O \rightleftharpoons HCO_3^- + OH^-$: alkaline.
2. $Ag_2CrO_4 \rightleftharpoons 2\,Ag^+ + CrO_4^{2-}$: $K_{sp} = (2s)^2(s) = 4s^3 = 4(6.5 \times 10^{-5})^3 = 1.1 \times 10^{-12}$ mol^3 dm^{-9}.
3. Let s be the solubility: $[Pb^{2+}] = 0.20 + s \approx 0.20$, $[Cl^-] = 2s$. Then $K_{sp} = 0.20 \times (2s)^2 = 0.80\,s^2$, so $s = \sqrt{1.6 \times 10^{-5} / 0.80} = 4.5 \times 10^{-3}$ mol dm^{-3} (compare with 1.6×10^{-2} in pure water).

KEY POINTS **Chapter summary: the last pieces of the equilibria puzzle**

- Salt hydrolysis: the ion from the weak parent reacts with water and fixes the pH.
- $K_{sp} = [\text{cation}]^m[\text{anion}]^n$; for AB, $K_{sp} = s^2$; for AB_2 or A_2B, $K_{sp} = 4s^3$.
- Precipitation occurs when the ionic product exceeds K_{sp}.
- A common ion lowers solubility but never changes K_{sp}; only temperature does.

With this section, the chapter Equilibria is complete: from dynamic equilibrium and Le Chatelier, through K_c, K_p, redox potentials, pH, titrations and buffers, to hydrolysis and solubility, you now hold all the tools the GCE Advanced Level examiner can demand. Revise the worked examples, redo the exercises without looking, and go to the examination hall with confidence!

13 INTEGRATION ACTIVITY

14 INTEGRATION ACTIVITY: WATER TREATMENT AT BATCHAM

14.1 Aims of the activity

This integration activity brings together, in one realistic problem, the four great families of ideas studied in this chapter (Lessons 1–3, 19–24, 28, 32–33):

- **dynamic equilibrium** and **Le Chatelier's principle** applied to the hydrolysis of a metal cation (Lessons 1, 2, 3);
- the **ionic product of water** K_w and the definition of **pH** (Lessons 20, 21);
- **acid–base neutralisation** stoichiometry (strong base added to an acidic solution) (Lessons 22, 24);
- the **solubility product** K_{sp} and precipitation as a purification tool (Lessons 32, 33);
- a reasoned discussion of **buffer solutions** in a real treatment context (Lesson 28).

By the end of the activity you should be able to: extract quantitative information from a technical data sheet; perform pH, neutralisation and K_{sp} calculations with correct units and significant figures; justify a treatment decision using equilibrium arguments; and present a short, well-organised technical report, as a junior chemist of a water company (for example Camwater) would do.

14.2 Situation

EXPERIMENT **The Village Water Treatment Challenge**

The village of Batcham, near Bafoussam in the West Region, is supplied by a borehole drilled into iron-rich lateritic ground. Since the last rainy season, families complain that the water tastes sour and that washed clothes come out with reddish-brown stains. The village water committee sends a sample to the regional laboratory. You are the junior chemist who receives the analysis report:

Parameter	Measured value
pH (pH meter, 25 °C)	5.6
Total dissolved iron, as Fe^{3+}	$c_0 = 2.0 \times 10^{-4}$ mol L^{-1}
Temperature of all measurements	25 °C

Data provided by the laboratory (all at 25 °C):

- Ionic product of water: $K_w = 1.0 \times 10^{-14}\ \text{mol}^2\,\text{L}^{-2}$.
- Solubility product of iron(III) hydroxide: $K_{sp}(\text{Fe(OH)}_3) = 1.0 \times 10^{-38}\ \text{mol}^4\,\text{L}^{-4}$.
- Iron(III) ions hydrolyse in water; the first step is

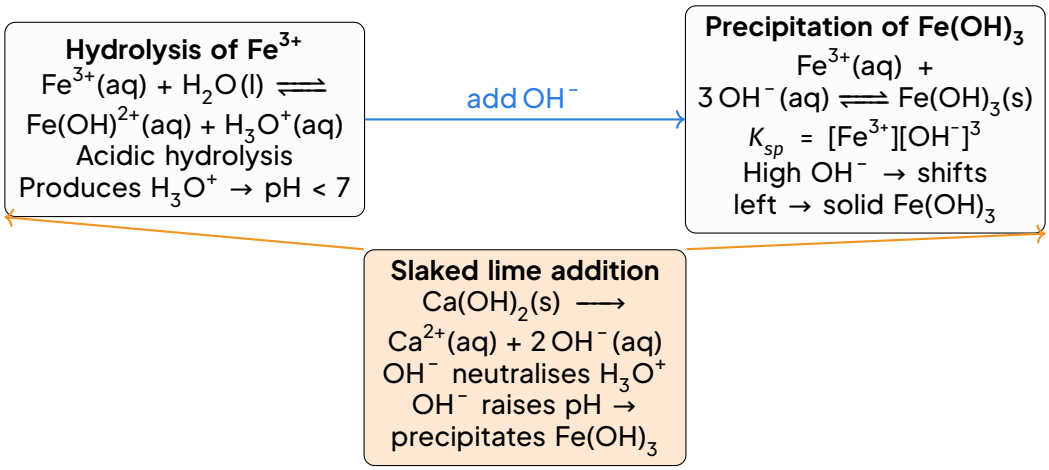
$$\text{Fe}^{3+}(\text{aq}) + \text{H}_2\text{O}(\text{l}) \rightleftharpoons \text{Fe(OH)}^{2+}(\text{aq}) + \text{H}_3\text{O}^+(\text{aq})$$

- The treatment reagent available locally is slaked lime, Ca(OH)_2 ($M = 74\ \text{g mol}^{-1}$), assumed to dissolve and dissociate completely:

$$\text{Ca(OH)}_2(\text{s}) \longrightarrow \text{Ca}^{2+}(\text{aq}) + 2\,\text{OH}^-(\text{aq})$$

- The borehole supplies a daily tank of volume $V = 10\,000\ \text{L}$ ($10\ \text{m}^3$).
- Health guideline: the iron concentration in drinking water should fall below $1.0 \times 10^{-6}\ \text{mol L}^{-1}$, and the pH should be brought close to 7.

Your team must design the treatment and write a one-page treatment report containing all calculations and justifications.



Equilibria involved in the Batcham water treatment: hydrolysis of Fe^{3+} (acidic), precipitation of Fe(OH)_3 (favoured by high pH), and the role of slaked lime.

14.3 Tasks

1. **Characterising the acidity.** Using the measured pH and K_w, calculate $[\text{H}_3\text{O}^+]$ and $[\text{OH}^-]$ in the borehole water. Is the water acidic, neutral or basic? Justify.

2. **Why is the water acidic?** The acidity is partly due to dissolved Fe^{3+} salts. Using the hydrolysis equation given and **Le Chatelier's principle**, explain why a solution of Fe^{3+} is acidic, and predict what happens to the hydrolysis equilibrium when OH^- ions are added. Name the

type of reaction by which a cation such as Fe^{3+} produces H_3O^+ in water.

3. **Dosing the lime – free acidity neutralisation.** The committee first considers neutralising only the free acidity. Calculate the amount (in moles) of OH^- needed to neutralise the free H_3O^+ in the daily tank, the corresponding amount of $Ca(OH)_2$, and the mass of slaked lime required. Write the neutralisation equation. Why, in practice, must much more lime than this be added (think of the hydrolysis equilibrium of Task 2 and the precipitation of iron)?

4. **Removing the iron.** Write the dissolution equilibrium of $Fe(OH)_3$ and the expression of its K_{sp}. Show that as lime raises the pH, $Fe(OH)_3$ precipitates. Calculate the residual concentration $[Fe^{3+}]$ when the treated water reaches pH = 7.0. Compare with the health guideline and conclude. What mass of iron (as Fe) is removed per day from the 10 000 L tank? (M(Fe) = 56 g mol^{-1}.)

5. **Total lime requirement.** Calculate the total amount of $Ca(OH)_2$ needed per day to (i) neutralise the free H_3O^+, (ii) precipitate essentially all Fe^{3+} as $Fe(OH)_3$, and (iii) provide a slight excess to reach pH 7. Give the total mass in kg.

6. **Would a buffer help?** A trainee suggests adding a buffer to "fix the pH at 7 permanently". Using what you know about the composition and functioning of buffer solutions, explain why installing a buffer is undesirable here (think of the quantities of reagents needed, the cost, and the purpose of a treatment).

7. **Report.** Assemble your results into a one-page treatment report: diagnosis, proposed treatment, quantities of reagent, expected final quality of the water, and one precaution for safe operation.

14.4 Model answer

Task 1 — Ion concentrations.

By definition pH = $-\log_{10}[H_3O^+]$, so

$$[H_3O^+] = 10^{-pH} = 10^{-5.6} = 2.5 \times 10^{-6}\ \text{mol L}^{-1} \quad \text{(2 significant figures).}$$

Since $K_w = [H_3O^+][OH^-] = 1.0 \times 10^{-14}\ \text{mol}^2\ \text{L}^{-2}$ at 25 °C,

$$[OH^-] = \frac{K_w}{[H_3O^+]} = \frac{1.0 \times 10^{-14}}{2.5 \times 10^{-6}} = 4.0 \times 10^{-9}\ \text{mol L}^{-1}.$$

Because $[H_3O^+] > [OH^-]$ (and pH < 7), the water is **acidic**.

KEY POINTS **pH and K_w at 25 °C**

pH = $-\log[H_3O^+]$, pOH = $-\log[OH^-]$, pH + pOH = 14.00. **Acidic**: pH < 7 ($[H_3O^+] > [OH^-]$); **Neutral**: pH = 7; **Basic**: pH > 7.

Task 2 — Origin of the acidity.

The Fe^{3+} ion is small and highly charged (high charge density); it strongly polarises the O – H bonds in the water molecules coordinated to it in the hydration sphere, facilitating the loss of a proton:

$$Fe^{3+}(aq) + H_2O(l) \rightleftharpoons Fe(OH)^{2+}(aq) + H_3O^+(aq) \qquad \text{(acidic hydrolysis)}$$

This is **salt hydrolysis** (specifically, acidic hydrolysis of the cation of a weak base, $Fe(OH)_3$). In Brønsted–Lowry terms, the hydrated Fe^{3+} ion acts as an **acid** (proton donor). The equilibrium produces H_3O^+, lowering the pH.

When OH^- is added (from lime), it reacts with the free H_3O^+:

$$H_3O^+(aq) + OH^-(aq) \longrightarrow 2\,H_2O(l)$$

By **Le Chatelier's principle**, removing the product H_3O^+ shifts the hydrolysis equilibrium to the **right**, causing more Fe^{3+} to hydrolyse and release additional H_3O^+. This continues until most Fe^{3+} is converted to $Fe(OH)^{2+}$ and eventually to solid $Fe(OH)_3$ (see Task 4). Hence the total acidity (free H_3O^+ + hydrolysable Fe^{3+}) is much larger than the free H_3O^+ alone.

> **WARNING** **Common mistake**
>
> Do not confuse free acidity (measured by pH) with total acidity (including the H_3O^+ that will be released as hydrolysable cations react with added base). The titration curve of a Fe^{3+} solution with a strong base shows a large buffer-like region due to this hydrolysis.

Task 3 — Lime dose for free acidity only.

Neutralisation equation:

$$H_3O^+(aq) + OH^-(aq) \longrightarrow 2\,H_2O(l) \qquad \text{mole ratio } 1 : 1.$$

Volume of daily tank: $V = 10\,000$ L.

$$n(H_3O^+)_{free} = [H_3O^+] \times V = 2.5 \times 10^{-6}\ \text{mol L}^{-1} \times 10\,000\ \text{L} = 2.5 \times 10^{-2}\ \text{mol}.$$

Hence $n(OH^-)_{neutralisation} = 2.5 \times 10^{-2}$ mol. Each mole of $Ca(OH)_2$ yields 2 mol OH^-:

$$n\left(Ca(OH)_2\right)_{free} = \frac{2.5 \times 10^{-2}}{2} = 1.25 \times 10^{-2}\ \text{mol}.$$

Mass of slaked lime:

$$m = n \times M = 1.25 \times 10^{-2}\ \text{mol} \times 74\ \text{g mol}^{-1} = 0.925\ \text{g} \approx \mathbf{0.93\ g}.$$

Why much more lime is needed in practice:

1. The hydrolysis equilibrium (Task 2) shifts right as free H_3O^+ is consumed, releasing more H_3O^+ from Fe^{3+}. This continues until the Fe^{3+} is largely precipitated.
2. The precipitation of $Fe(OH)_3$ itself consumes OH^- ions (3 mol OH^- per mol Fe^{3+}).
3. A slight excess is needed to reach and maintain pH ≈ 7.

The 0.93 g calculated above covers only the free H_3O^+; the dominant consumption is for iron precipitation (Task 5).

Task 4 — Precipitation of iron.

Dissolution equilibrium of iron(III) hydroxide:

$$Fe(OH)_3(s) \rightleftharpoons Fe^{3+}(aq) + 3\,OH^-(aq)$$

Solubility product expression:

$$K_{sp} = [Fe^{3+}][OH^-]^3 = 1.0 \times 10^{-38}\ mol^4\ L^{-4} \quad (\text{at } 25\,^\circ C).$$

Adding lime increases $[OH^-]$; the ionic product $Q = [Fe^{3+}][OH^-]^3$ quickly exceeds K_{sp}, so the equilibrium shifts to the **left** and **$Fe(OH)_3$ precipitates** as a reddish-brown solid (the stains on clothes).

At target pH = 7.0:

$$[H_3O^+] = 10^{-7.0} = 1.0 \times 10^{-7}\ mol\,L^{-1}, \qquad [OH^-] = \frac{K_w}{[H_3O^+]} = \frac{1.0 \times 10^{-14}}{1.0 \times 10^{-7}} = 1.0 \times 10^{-7}\ mol\,L^{-1}.$$

Residual Fe^{3+} concentration:

$$[Fe^{3+}]_{residual} = \frac{K_{sp}}{[OH^-]^3} = \frac{1.0 \times 10^{-38}}{(1.0 \times 10^{-7})^3} = 1.0 \times 10^{-17}\ mol\,L^{-1}.$$

Comparison with guideline ($1.0 \times 10^{-6}\ mol\,L^{-1}$):

$$1.0 \times 10^{-17} \ll 1.0 \times 10^{-6} \quad \Rightarrow \quad \textbf{the iron is effectively completely removed.}$$

Mass of iron removed per day (assuming initial $c_0 = 2.0 \times 10^{-4}\ mol\,L^{-1}$ is fully precipitated):

$$n(Fe)_{removed} \approx c_0 V = 2.0 \times 10^{-4}\ mol\,L^{-1} \times 10\,000\ L = 2.0\ mol.$$

$$m(Fe) = n \times M(Fe) = 2.0\ mol \times 56\ g\,mol^{-1} = \textbf{112 g per day}.$$

(Recovered as $Fe(OH)_3$ sludge; molar mass $Fe(OH)_3 \approx 107\ g\,mol^{-1}$, so ≈ 214 g sludge/day.)

Task 5 — Total lime requirement.

1. **Neutralise free H_3O^+:** $n(OH^-)_1 = 2.5 \times 10^{-2}$ mol.
2. **Precipitate Fe^{3+}:** $Fe^{3+} + 3\,OH^- \longrightarrow Fe(OH)_3(s)$. $n(Fe^{3+})_{initial} = 2.0$ mol. $n(OH^-)_2 = 3 \times 2.0 = 6.0$ mol.
3. **Excess to reach pH 7:** At pH 7, $[OH^-] = 1.0 \times 10^{-7}$ M in 10^4 L $\Rightarrow n(OH^-)_{excess} = 1.0 \times 10^{-3}$ mol (negligible).

Total OH^- required ≈ 6.025 mol.

$$n\big(Ca(OH)_2\big)_{total} = \frac{6.025}{2} = 3.0125\ mol.$$

$$m\big(Ca(OH)_2\big)_{total} = 3.0125 \times 74 = 222.9\ g \approx \textbf{0.223 kg per day}.$$

(Add ~ 5% excess for incomplete mixing/kinetics → ≈ 0.23 kg/day.)

WARNING **Key stoichiometry**

$Ca(OH)_2 \rightarrow Ca^{2+} + 2\,OH^-$. $Fe^{3+} + 3\,OH^- \rightarrow Fe(OH)_3(s)$. The cube in $K_{sp} = [Fe^{3+}][OH^-]^3$ means a small increase in $[OH^-]$ drastically reduces $[Fe^{3+}]$.

Task 6 — Why no buffer.

A buffer solution (e.g., CH_3COOH/CH_3COO^- or NH_4^+/NH_3) contains a weak acid/base conjugate pair in comparable concentrations. It resists pH change upon addition of small amounts of acid or base, but only while both components remain present.

Here the goal is the opposite: to **permanently change** the pH from 5.6 to 7 and to **keep iron precipitated**. Buffering the water at pH 7 would require:

- Dissolving large quantities of organic reagents (or NH_3/NH_4^+ salts) in 10 m^3 of drinking water every day — costly and impractical.
- Adding unnecessary dissolved substances to water meant for human consumption.
- Fighting against the very adjustment we want: a buffer would consume the added OH^- to maintain pH, preventing the pH rise needed to precipitate $Fe(OH)_3$.

A one-off, controlled dose of cheap slaked lime achieves the treatment; a buffer would be counterproductive and uneconomical.

Task 7 — Report (summary).

BATCHAM WATER TREATMENT REPORT	Date: [Today]
Diagnosis	Borehole water: pH 5.6 (acidic), $[Fe^{3+}] = 2.0 \times 10^{-4}$ M. Acidity due to Fe^{3+} hydrolysis. Reddish stains = $Fe(OH)_3$ precipitating on exposure to air.
Proposed treatment	Daily dose of slaked lime $Ca(OH)_2$ to 10 000 L tank. Stir, allow $Fe(OH)_3$ to precipitate (15–30 min), filter/decant.
Reagent quantity	**0.23 kg $Ca(OH)_2$ per day** (covers free acidity, complete Fe^{3+} precipitation, and pH 7 target).
Expected final quality	pH ≈ 7.0; $[Fe^{3+}] \approx 1\times10^{-17}$ M (guideline: < 1×10^{-6} M); iron removed ≈ 112 g/day as $Fe(OH)_3$ sludge.
Precaution	Add lime in small portions while monitoring pH with a meter; avoid overshooting above pH 8 (causes $CaCO_3$ scaling, bitter taste). Wear gloves/goggles — $Ca(OH)_2$ is caustic.

WARNING **Common mistakes to avoid in the examination**

- Confusing free acidity ($[H_3O^+]$ from pH) with total acidity (including hydrolysable

Fe^{3+}). The latter determines the true lime demand.

- Forgetting the 1 : 2 stoichiometry of $Ca(OH)_2$ to OH^-, or the 1 : 3 stoichiometry of Fe^{3+} to OH^- in precipitation.
- Miswriting K_{sp} expression: it is $[Fe^{3+}][OH^-]^3$ (cube!), not $[Fe^{3+}][OH^-]$.
- Using K_w or K_{sp} values at the wrong temperature; always state 25 °C.
- Suggesting a buffer for a process that requires a permanent pH shift and precipitation.
- Omitting units or giving answers to unjustified significant figures (pH 5.6 → 2 s.f. in concentrations).

DID YOU KNOW? **Cameroon context**

Lateritic soils in the West Region (Bafoussam, Dschang, Bafang) are rich in iron oxides. Boreholes often yield water with high Fe^{2+}/Fe^{3+} and low pH. The standard Camwater treatment is aeration (oxidises Fe^{2+} to Fe^{3+}) followed by liming and filtration. The $Fe(OH)_3$ sludge is a disposal issue — it can be dried and used as a pigment or in brick-making.

15 METHODS AND WORKED EXAMPLES

15.1 Method 1 — Writing the Equilibrium Law and Calculating K_c

METHOD **Calculating K_c from equilibrium data**

1. Balance the equation and write it with state symbols.
2. Write the equilibrium law: for $aA + bB \rightleftharpoons cC + dD$,

$$K_c = \frac{[C]^c\,[D]^d}{[A]^a\,[B]^b}$$

   Products on top, reactants below, each concentration raised to its **stoichiometric coefficient**.
3. Build a "start–change–equilibrium" (ICE) table in moles (or directly in concentrations if the volume is given): initial amounts, change (in the ratio of the coefficients), equilibrium amounts.
4. Convert moles to concentrations: $c = \frac{n}{V}$. Never forget the volume of the container!
5. Substitute into the expression, keeping units (mol dm^{-3}) at each step, and work out the units of K_c by cancelling: units = $(\text{mol dm}^{-3})^{\Delta n}$ where $\Delta n = (c + d) - (a + b)$.
6. Quote the answer to 2–3 significant figures.

Reflexes: pure solids and pure liquids never appear in the expression; if K_c is asked for the reverse reaction, take $\frac{1}{K_c}$; if the equation is doubled, square K_c.

WORKED EXAMPLE 29 **Esterification equilibrium**

3.0 mol of ethanoic acid and 2.0 mol of ethanol are mixed and allowed to reach equilibrium at 298 K in a total volume of 2.0 dm^3:

$$CH_3COOH(l) + C_2H_5OH(l) \rightleftharpoons CH_3COOC_2H_5(l) + H_2O(l)$$

At equilibrium, 1.2 mol of ethyl ethanoate is present. Calculate K_c and state its units.

SOLUTION

Step 1 – ICE table (moles): since 1.2 mol of ester has formed, 1.2 mol of each reactant has been consumed (coefficients all equal to 1).

	CH_3COOH	C_2H_5OH	$CH_3COOC_2H_5$	H_2O
Start / mol	3.0	2.0	0	0
Change / mol	−1.2	−1.2	+1.2	+1.2
Equilibrium / mol	1.8	0.8	1.2	1.2

Step 2 – Concentrations (V = 2.0 dm^3, using $c = n/V$):

$$[CH_3COOH] = 0.90, \quad [C_2H_5OH] = 0.40, \quad [CH_3COOC_2H_5] = [H_2O] = 0.60 \text{ mol dm}^{-3}$$

Step 3 – Substitute:

$$K_c = \frac{[CH_3COOC_2H_5][H_2O]}{[CH_3COOH][C_2H_5OH]} = \frac{(0.60)(0.60)}{(0.90)(0.40)} = \frac{0.36}{0.36} = 1.0$$

Units: $\Delta n = 2 - 2 = 0$, so K_c has no units. ✓ Note how the volume V cancels here (equal numbers of moles on both sides) — a useful check on your working.

15.2 Method 2 — Calculating K_p from Partial Pressures

METHOD K_p **and partial pressures**

1. Write the equation and the expression $K_p = \dfrac{p^{\nu}_{\text{products}}}{p^{\nu}_{\text{reactants}}}$ (gaseous species only).
2. Find the mole fraction of each gas: $x_i = \dfrac{n_i}{n_{\text{total}}}$.
3. Compute each partial pressure: $p_i = x_i\,P_{\text{total}}$.
4. Substitute into K_p; units = (pressure unit)$^{\Delta n_g}$, where Δn_g counts gaseous moles only.
5. Link with K_c if needed: $K_p = K_c(RT)^{\Delta n_g}$.

Reflex: if the total pressure at equilibrium is given together with the amount dissociated, first complete the ICE table in moles, then get mole fractions, then partial pressures.

WORKED EXAMPLE 30 **Dissociation of dinitrogen tetroxide**

1.00 mol of N_2O_4 is placed in a vessel at 350 K. At equilibrium the total pressure is 2.0 atm and 0.20 mol of N_2O_4 has dissociated:

$$N_2O_4(g) \rightleftharpoons 2\,NO_2(g)$$

Calculate K_p at this temperature and give its units.

SOLUTION

Step 1 – ICE table (moles): each mole of N_2O_4 that dissociates gives two moles of NO_2.

	N_2O_4	NO_2
Start / mol	1.00	0
Change / mol	−0.20	+0.40
Equilibrium / mol	0.80	0.40

Total moles at equilibrium: $n_{tot} = 0.80 + 0.40 = 1.20$ mol. Step 2 – Partial pressures ($p_i = x_i P_{total}$):

$$p_{N_2O_4} = \frac{0.80}{1.20} \times 2.0 = 1.33 \text{ atm}; \qquad p_{NO_2} = \frac{0.40}{1.20} \times 2.0 = 0.667 \text{ atm}$$

Step 3 – K_p:

$$K_p = \frac{p_{NO_2}^2}{p_{N_2O_4}} = \frac{(0.667)^2}{1.33} = \frac{0.444}{1.33} = 0.33 \text{ atm}$$

Units: $\Delta n_g = 2 - 1 = +1$, so K_p is in atm (or Pa if pressures are in Pa). ✓

15.3 Method 3 — Heterogeneous Equilibria and the Effect of Temperature on K

METHOD **Heterogeneous equilibria and van 't Hoff reasoning**

1. Omit pure solids and pure liquids from the equilibrium expression — their concentrations are constant and are absorbed into the value of K.
2. For a thermal decomposition such as $CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$, $K_p = p_{CO_2}$ only.
3. Temperature effect: K changes only with temperature. For an **endothermic** forward reaction ($\Delta H > 0$), K increases as T rises; for an **exothermic** one ($\Delta H < 0$), K decreases as T rises.
4. Justify with Le Chatelier's principle: raising T shifts the equilibrium in the direction that absorbs heat.

Reflex: catalysts, pressure and concentration changes alter the position of equilibrium but never the value of K.

WORKED EXAMPLE 31 **Decomposition of calcium carbonate**

Limestone from a quarry near Figuil is heated in a closed kiln:

$$CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g) \qquad \Delta H^\theta = +178\ kJ\ mol^{-1}$$

(a) Write the expression for K_p. (b) At 1100 K the pressure of CO_2 at equilibrium is 0.50 atm. State K_p. (c) Explain the effect on K_p of (i) raising the temperature, (ii) adding more $CaCO_3$, (iii) pumping away CO_2.

SOLUTION

(a) Both solids are omitted:

$$K_p = p_{CO_2}$$

(b) $K_p = 0.50$ atm. (c) (i) The forward reaction is endothermic ($\Delta H > 0$). Raising T shifts the equilibrium to the right (the direction that absorbs heat), so p_{CO_2} rises and K_p increases. (ii) Adding $CaCO_3(s)$: a pure solid has a constant concentration term; the position of equilibrium and K_p are unchanged. (iii) Removing CO_2 momentarily lowers p_{CO_2}; by Le Chatelier the system decomposes more $CaCO_3$ until p_{CO_2} returns to 0.50 atm. The position shifts right, but K_p is unchanged because the temperature is constant. ✓

15.4 Method 4 — Balancing Redox Equations (Ion–Electron Method)

METHOD **Half-equation method in acid solution**

1. Assign oxidation numbers to identify what is oxidised (oxidation number rises) and what is reduced (oxidation number falls).
2. Write the two half-equations: balance the element being oxidised/reduced, then balance O with H_2O, then H with H^+, then charge with electrons e^-.
3. Multiply the half-equations so that the numbers of electrons are equal.
4. Add the half-equations and cancel electrons (and any H^+ or H_2O appearing on both sides).
5. Check: atoms balanced and total charge balanced on both sides.

Reflex: in alkaline solution, neutralise each H^+ by adding OH^- to both sides, then combine $H^+ + OH^-$ into H_2O.

WORKED EXAMPLE 32 **Permanganate and iron(II)**

Acidified potassium manganate(VII), $KMnO_4$, oxidises iron(II) ions to iron(III). Given

$$MnO_4^-(aq) + 8\,H^+(aq) + 5\,e^- \longrightarrow Mn^{2+}(aq) + 4\,H_2O(l)$$

construct the balanced ionic equation for the titration reaction used in redox analysis.

SOLUTION

Step 1 – Oxidation numbers: Mn falls from +7 to +2 (reduction, gain of 5 e^-); Fe rises from +2 to +3 (oxidation, loss of 1 e^-). Step 2 – Oxidation half-equation:

$$Fe^{2+}(aq) \longrightarrow Fe^{3+}(aq) + e^-$$

Step 3 – Equalise electrons: multiply the iron half-equation by 5:

$$5\,Fe^{2+}(aq) \longrightarrow 5\,Fe^{3+}(aq) + 5\,e^-$$

Step 4 – Add and cancel the 5 electrons:

$$MnO_4^-(aq) + 8\,H^+(aq) + 5\,Fe^{2+}(aq) \longrightarrow Mn^{2+}(aq) + 5\,Fe^{3+}(aq) + 4\,H_2O(l)$$

Step 5 – Check: Mn: 1/1; O: 4/4; H: 8/8; Fe: 5/5. Charge: left = (−1) + (+8) + 5(+2) = +17; right = (+2) + 5(+3) = +17. ✓ Balanced. This is why 1 mol of MnO_4^- reacts with exactly 5 mol of Fe^{2+} in titrations — the basis of redox titration calculations.

15.5 Method 5 — Cell emf and Predicting Spontaneous Redox Reactions

METHOD **Using standard electrode potentials**

1. Write both half-cells as reductions with their E^θ values.
2. The more positive E^θ system is reduced (cathode, positive pole); the other is reversed and acts as the oxidation half-cell (anode, negative pole).
3. Cell emf: $E^\theta_{cell} = E^\theta_{cathode} - E^\theta_{anode}$ (right-hand electrode minus left-hand electrode in the conventional cell diagram).
4. If $E^\theta_{cell} > 0$, the reaction is feasible (spontaneous) under standard conditions.
5. Never multiply E^θ when multiplying a half-equation — it is an intensive quantity (it does not depend on amounts).

WORKED EXAMPLE 33 **Copper–silver cell**

Given $E^{\theta}(Cu^{2+}/Cu) = +0.34\ V$ and $E^{\theta}(Ag^{+}/Ag) = +0.80\ V$: (a) Write the cell diagram for the standard cell. (b) Calculate E^{θ}_{cell}. (c) Write the overall cell reaction. (d) State the direction of electron flow in the external circuit.

SOLUTION

(a) The copper electrode (less positive E^{θ}) is the negative pole, written on the left by convention:

$$Cu(s)\,|\,Cu^{2+}(aq, 1\,mol\,dm^{-3})\,||\,Ag^{+}(aq, 1\,mol\,dm^{-3})\,|\,Ag(s)$$

(b)

$$E^{\theta}_{cell} = E^{\theta}_{right} - E^{\theta}_{left} = (+0.80) - (+0.34) = +0.46\ V$$

(c) Silver is reduced (cathode): $Ag^{+} + e^{-} \longrightarrow Ag$; copper is oxidised (anode): $Cu \longrightarrow Cu^{2+} + 2\,e^{-}$. Doubling the silver half-equation (without touching E^{θ}!):

$$Cu(s) + 2\,Ag^{+}(aq) \longrightarrow Cu^{2+}(aq) + 2\,Ag(s)$$

(d) Electrons flow through the external circuit from the copper electrode (anode, negative pole) to the silver electrode (cathode, positive pole). Since $E^{\theta}_{cell} > 0$, the reaction is spontaneous under standard conditions. ✓

15.6 Method 6 — pH of Strong and Weak Acids

METHOD **pH calculations**

1. Strong monoprotic acid (full dissociation): $[H_3O^{+}] = c_a$, so $pH = -\log c_a$.
2. Strong base: $[OH^{-}] = c_b$, then $[H_3O^{+}] = \dfrac{K_w}{[OH^{-}]}$ with $K_w = 1.0 \times 10^{-14}\ mol^2\,dm^{-6}$ at 298 K; equivalently $pH = 14 - pOH$.
3. Weak acid: set up $K_a = \dfrac{[H_3O^{+}]^2}{c_a - [H_3O^{+}]} \approx \dfrac{[H_3O^{+}]^2}{c_a}$, hence $[H_3O^{+}] \approx \sqrt{K_a\,c_a}$ and $pH = \tfrac{1}{2}(pK_a - \log c_a)$.
4. Check the approximation: the degree of dissociation must be less than about 5%; otherwise solve the quadratic equation.
5. Weak base: same pattern with K_b: $[OH^{-}] \approx \sqrt{K_b\,c_b}$, then convert through K_w.

WORKED EXAMPLE 34 **Comparing a strong and a weak acid**

(a) Calculate the pH of 0.010 mol dm^{-3} hydrochloric acid. (b) Calculate the pH of 0.010 mol dm^{-3} ethanoic acid, $K_a = 1.8 \times 10^{-5}$ mol dm^{-3}. (c) Comment on the difference.

SOLUTION

(a) HCl is a strong acid, fully ionised in water:

$$[H_3O^+] = 0.010 \text{ mol dm}^{-3} \quad \Rightarrow \quad \text{pH} = -\log(0.010) = 2.00$$

(b) Ethanoic acid is weak: $CH_3COOH + H_2O \rightleftharpoons CH_3COO^- + H_3O^+$

$$K_a = \frac{[H_3O^+][CH_3COO^-]}{[CH_3COOH]} \approx \frac{[H_3O^+]^2}{c_a}$$

$$[H_3O^+] = \sqrt{K_a c_a} = \sqrt{1.8 \times 10^{-5} \times 0.010} = \sqrt{1.8 \times 10^{-7}} = 4.24 \times 10^{-4} \text{ mol dm}^{-3}$$

$$\text{pH} = -\log(4.24 \times 10^{-4}) = 3.37$$

Check: degree of dissociation $= \dfrac{4.24 \times 10^{-4}}{0.010} \times 100\% = 4.2\% < 5\%$ — the approximation is valid. ✓(c) At the same concentration, the weak acid has a higher pH (3.37 against 2.00) because it is only partially ionised: $[H_3O^+]$ is about 24 times smaller than in the strong acid.

15.7 Method 7 — pH of Buffer Solutions

METHOD **Buffer pH (Henderson–Hasselbalch)**

1. Identify the buffer pair: weak acid + its conjugate base (acidic buffer) or weak base + its conjugate acid (alkaline buffer).
2. Acidic buffer: $\text{pH} = \text{p}K_a + \log \dfrac{[\text{salt}]}{[\text{acid}]}$.
3. Alkaline buffer: $\text{pOH} = \text{p}K_b + \log \dfrac{[\text{salt}]}{[\text{base}]}$, then $\text{pH} = 14 - \text{pOH}$.
4. Use the mole ratio directly when both species share the same solution volume — the volumes cancel.
5. A buffer works best when [salt] ≈ [acid], i.e. when $\text{pH} \approx \text{p}K_a$.

WORKED EXAMPLE 35 **Ethanoate buffer**

A buffer is prepared by dissolving 0.10 mol of ethanoic acid and 0.15 mol of sodium ethanoate in water to make 1.0 dm^3 of solution. Given $pK_a(CH_3COOH) = 4.74$: (a) Calculate the pH of the buffer. (b) Explain, with equations, how the buffer resists pH change when a small amount of HCl is added.

SOLUTION

(a) Both species are in the same volume, so the mole ratio may be used directly:

$$\text{pH} = pK_a + \log\frac{[CH_3COO^-]}{[CH_3COOH]} = 4.74 + \log\frac{0.15}{0.10} = 4.74 + \log(1.5) = 4.74 + 0.18 = 4.92$$

(b) The added H^+ is removed by the reservoir of ethanoate ions:

$$CH_3COO^-(aq) + H^+(aq) \longrightarrow CH_3COOH(aq)$$

The ratio $\dfrac{[CH_3COO^-]}{[CH_3COOH]}$ changes only slightly (a little salt is converted into acid, but both are present in large amounts), so log(ratio) — and hence the pH — barely changes. If a base were added instead, the acid reservoir would neutralise it: $CH_3COOH + OH^- \longrightarrow CH_3COO^- + H_2O$. ✓

15.8 Method 8 — Solubility Product and the Common Ion Effect

METHOD **K_{sp} calculations**

1. Write the dissolution equation, e.g. $AgCl(s) \rightleftharpoons Ag^+(aq) + Cl^-(aq)$.
2. Write K_{sp} as the product of the ion concentrations raised to their coefficients (the solid is omitted).
3. If the molar solubility is s, express each ion concentration in terms of s (e.g. for CaF_2: $[Ca^{2+}] = s$, $[F^-] = 2s$, so $K_{sp} = 4s^3$), then substitute.
4. Common ion effect: in a solution already containing a common ion of concentration c, write K_{sp} with $(c + s)$ for that ion and approximate $c + s \approx c$, since s becomes very small.
5. Precipitation occurs when the ionic product Q exceeds K_{sp}.

WORKED EXAMPLE 36 **Solubility of silver chloride**

$K_{sp}(AgCl) = 1.8 \times 10^{-10}$ mol^2 dm^{-6} at 298 K. (a) Calculate the solubility of AgCl in pure water in mol dm^{-3}. (b) Calculate its solubility in 0.10 mol dm^{-3} NaCl solution, and comment.

SOLUTION

(a) $AgCl(s) \rightleftharpoons Ag^+(aq) + Cl^-(aq)$; let the molar solubility be s, so $[Ag^+] = [Cl^-] = s$:

$$K_{sp} = [Ag^+][Cl^-] = s^2 \quad \Rightarrow \quad s = \sqrt{1.8 \times 10^{-10}} = 1.34 \times 10^{-5} \text{ mol dm}^{-3}$$

(b) In 0.10 mol dm^{-3} NaCl, the chloride ion is the common ion: $[Cl^-] = 0.10 + s' \approx 0.10$ mol dm^{-3}. Then:

$$K_{sp} = s'(0.10) \quad \Rightarrow \quad s' = \frac{1.8 \times 10^{-10}}{0.10} = 1.8 \times 10^{-9} \text{ mol dm}^{-3}$$

Comment: the solubility falls from 1.34×10^{-5} to 1.8×10^{-9} mol dm^{-3} — about 7 000 times smaller. The added Cl^- shifts the dissolution equilibrium to the left (Le Chatelier's principle). This is exactly why, in qualitative analysis, an excess of the precipitating reagent is used to precipitate an ion almost completely. ✓

WARNING **Frequent traps in equilibrium calculations**

- Using moles instead of concentrations in K_c — always divide by the volume of the container.
- Multiplying E^θ by stoichiometric coefficients — electrode potentials are intensive quantities and never change with scaling of the equation.
- Forgetting that K changes only with temperature; a catalyst speeds up the attainment of equilibrium but leaves both K and the equilibrium position unchanged.
- For weak acids, wrongly using pH = − log c_a (that formula applies only to strong monoprotic acids).
- In K_{sp} for salts of formula A_2B or AB_2, forgetting the factor 2 and the cube: $K_{sp} = (2s)^2(s) = 4s^3$.
- Omitting pure solids and pure liquids is correct — but do not omit gases or aqueous species from K_p or K_c expressions.

16 EXERCISES

16.1 I check my knowledge

EXERCISE 14 **Multiple choice questions**

For each question, choose the single correct answer and justify your choice briefly.

1. For the equilibrium $N_2(g) + 3\,H_2(g) \rightleftharpoons 2\,NH_3(g)$, $\Delta H^\theta = -92\ kJ\ mol^{-1}$, raising the temperature at constant pressure:

   (a) increases both the yield of NH_3 and the value of K_p;

   (b) decreases the yield of NH_3 and decreases K_p;

   (c) decreases the yield of NH_3 but increases K_p;

   (d) has no effect because a catalyst is present.

2. The units of K_c for the equilibrium $CH_3COOH(l) + C_2H_5OH(l) \rightleftharpoons CH_3COOC_2H_5(l) + H_2O(l)$ are:

   (a) $mol\ dm^{-3}$;

   (b) $mol^{-1}\ dm^3$;

   (c) $mol^2\ dm^{-6}$;

   (d) no units.

3. The pH of a $0.010\ mol\ dm^{-3}$ solution of NaOH at 25 °C ($K_w = 1.0 \times 10^{-14}\ mol^2\ dm^{-6}$) is:

   (a) 2.0;

   (b) 7.0;

   (c) 10.0;

   (d) 12.0.

4. In the cell $Zn(s)|Zn^{2+}(aq)||Cu^{2+}(aq)|Cu(s)$, the electrode at which oxidation occurs is:

   (a) the copper electrode (cathode);

   (b) the zinc electrode (anode);

   (c) the salt bridge;

   (d) neither electrode.

EXERCISE 15 **True or false**

State whether each statement is true or false, and justify your answer in one or two sentences.

1. At dynamic equilibrium, the forward and reverse reactions have both stopped.
2. Adding a catalyst to a system at equilibrium increases the equilibrium yield of the products.
3. The standard hydrogen electrode is assigned a standard electrode potential of 0.00 V under standard conditions.
4. A solution of NH_4Cl in pure water has a pH greater than 7 at 25 °C.
5. The solubility of AgCl in 0.10 mol dm^{-3} NaCl is smaller than its solubility in pure water.

EXERCISE 16 **Definitions**

Define each of the following terms precisely, giving one example in each case:

1. dynamic equilibrium;
2. Le Chatelier's principle;
3. standard electrode potential, E^{θ};
4. a Brønsted–Lowry acid and a Lewis base;
5. a buffer solution;
6. the solubility product, K_{sp}.

EXERCISE 17 **Direct calculations**

1. Calculate the pH at 25 °C of: (i) 0.050 mol dm^{-3} HCl; (ii) 0.050 mol dm^{-3} NaOH.
2. For $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$, the equilibrium mixture contains $[PCl_5] = 0.10$, $[PCl_3] = 0.40$ and $[Cl_2] = 0.40$ mol dm^{-3}. Calculate K_c and state its units.
3. Given $E^{\theta}(Zn^{2+}/Zn) = -0.76$ V and $E^{\theta}(Cu^{2+}/Cu) = +0.34$ V, calculate the standard e.m.f. of the Daniell cell and identify the positive electrode.

16.2 I practise

EXERCISE 18 **Decomposition of phosphorus pentachloride**

1.00 mol of PCl_5 is placed in a closed vessel of volume 2.00 dm^3 and heated to a temperature T. At equilibrium, 0.20 mol of PCl_5 remains.

$$PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$$

1. Construct an ICE (initial–change–equilibrium) table in moles.
2. Deduce the equilibrium concentrations of the three species.
3. Calculate K_c at temperature T and state its units.
4. State and explain the effect on the equilibrium position of decreasing the volume of the vessel at constant temperature.

EXERCISE 19 **Redox titration of iron(II)**

1. Using the half-equation method, balance the equation for the reaction between acidified potassium manganate(VII) and iron(II) sulphate:

$$MnO_4^-(aq) + Fe^{2+}(aq) + H^+(aq) \longrightarrow Mn^{2+}(aq) + Fe^{3+}(aq) + H_2O(l)$$

2. Calculate the volume of 0.0200 mol dm^{-3} $KMnO_4$ solution required to oxidise exactly the Fe^{2+} contained in a sample of 1.52 g of $FeSO_4 \cdot 7\,H_2O$ dissolved in dilute sulphuric acid. ($M(FeSO_4 \cdot 7\,H_2O) = 278$ g mol^{-1})
3. State the colour change observed at the end-point and explain why no indicator is needed.

EXERCISE 20 **pH of a buffer solution**

A buffer solution is prepared by dissolving ethanoic acid and sodium ethanoate in water so that $[CH_3COOH] = 0.10$ mol dm^{-3} and $[CH_3COONa] = 0.15$ mol dm^{-3}. Given $K_a(CH_3COOH) = 1.8 \times 10^{-5}$ mol dm^{-3} at 25 °C:

1. Write the expression for K_a of ethanoic acid and derive the buffer formula $pH = pK_a + \log \dfrac{[salt]}{[acid]}$.
2. Calculate the pH of the buffer.
3. 0.010 mol of NaOH is added to 1.00 dm^3 of this buffer (assume no volume change). Write the equation of the reaction that occurs and calculate the new pH.

4. Comment on the pH change compared with adding the same amount of NaOH to 1.00 dm^3 of pure water.

EXERCISE 21 **Solubility and common ion effect**

Given at 25 °C: $K_{sp}(AgCl) = 1.8 \times 10^{-10}$ mol^2 dm^{-6} and $K_{sp}(Mg(OH)_2) = 1.8 \times 10^{-11}$ mol^3 dm^{-9}.

1. Calculate the solubility, in mol dm^{-3}, of AgCl and of $Mg(OH)_2$ in pure water.
2. Calculate the solubility of AgCl in a 0.10 mol dm^{-3} NaCl solution, and explain the result using Le Chatelier's principle.
3. 50.0 cm^3 of 2.0×10^{-4} mol dm^{-3} $AgNO_3$ is mixed with 50.0 cm^3 of 2.0×10^{-4} mol dm^{-3} NaCl. Determine, by calculation, whether a precipitate of AgCl forms.

16.3 I prepare for the GCE

EXERCISE 22 **The Haber process**

Ammonia is manufactured industrially by the Haber process:

$$N_2(g) + 3\,H_2(g) \rightleftharpoons 2\,NH_3(g) \qquad \Delta H^\theta = -92\ kJ\ mol^{-1}$$

1. Write the expression for K_p for this equilibrium and state its units. (3 marks)
2. State and explain, using Le Chatelier's principle, the effect on the equilibrium yield of ammonia of: (i) increasing the pressure; (ii) increasing the temperature. (4 marks)
3. State the effect, if any, of (i) increasing the pressure and (ii) increasing the temperature on the value of K_p. Justify your answers. (3 marks)
4. Explain why an iron catalyst is used, and state its effect on the equilibrium position and on the rate at which equilibrium is attained. (3 marks)
5. Typical industrial conditions are about 450 °C, 200 atm and an iron catalyst. Explain why these conditions represent a compromise between yield and rate. (3 marks)
6. At a certain temperature, an equilibrium mixture contains N_2, H_2 and NH_3 at partial pressures 20 atm, 60 atm and 40 atm respectively. Calculate K_p at this temperature and give its units. (4 marks)

(Total: 20 marks)

EXERCISE 23 Electrode potentials and electrochemical cells

Use the following standard electrode potentials at 25 °C:

Half-cell	E^θ / V
$Zn^{2+}(aq) + 2e^- \rightleftharpoons Zn(s)$	-0.76
$Cu^{2+}(aq) + 2e^- \rightleftharpoons Cu(s)$	+0.34
$Fe^{3+}(aq) + e^- \rightleftharpoons Fe^{2+}(aq)$	+0.77
$Ag^+(aq) + e^- \rightleftharpoons Ag(s)$	+0.80

1. Describe, with the aid of a labelled diagram, how the standard electrode potential of the Cu^{2+}/Cu half-cell is measured using the standard hydrogen electrode. State the standard conditions of measurement. (5 marks)

2. Draw a labelled diagram of the Daniell cell, showing the electrodes, the solutions, the salt bridge and the direction of electron flow in the external circuit. Write the conventional cell notation. (4 marks)

3. Calculate the standard e.m.f. of the Daniell cell and write the overall cell reaction. (3 marks)

4. Predict, with a calculation to support your answer, whether copper metal can displace silver from a solution of silver nitrate. Write the equation for any reaction that occurs. (4 marks)

5. State and explain, using the Nernst equation qualitatively, how the electrode potential of the Cu^{2+}/Cu electrode changes when the concentration of Cu^{2+} ions is decreased. (2 marks)

6. Give two reasons why a salt bridge containing KNO_3 is preferred to one containing KCl when a solution of $AgNO_3$ is involved. (2 marks)

(Total: 20 marks)

EXERCISE 24 Acid–base equilibria and titration curves

Ethanoic acid is a weak acid with $K_a = 1.8\times10^{-5}\ \text{mol dm}^{-3}$ at 25 °C ($K_w = 1.0\times10^{-14}\ \text{mol}^2\ \text{dm}^{-6}$).

1. Define a weak acid according to the Brønsted–Lowry theory and write the equation for the ionisation of ethanoic acid in water. (2 marks)

2. Calculate the pH of a $0.100\ \text{mol dm}^{-3}$ solution of CH_3COOH, stating any approximation you make. (4 marks)

3. 25.0 cm^3 of this 0.100 mol dm^{-3} CH_3COOH solution is titrated with 0.100 mol dm^{-3} NaOH. Sketch and fully label the titration curve (pH against volume of NaOH added), marking: the initial pH, the half-equivalence point (where pH = pK_a), the equivalence point and the buffer region. (5 marks)

4. Calculate the volume of NaOH needed to reach the equivalence point, and explain why the pH at the equivalence point is greater than 7. (3 marks)

5. Choosing between methyl orange (range 3.1–4.4) and phenolphthalein (range 8.3–10.0), state which indicator is suitable for this titration and justify your choice. (2 marks)

6. Sodium carbonate is titrated with dilute hydrochloric acid. Write the equations of the two successive reactions that occur, and sketch the shape of the titration curve showing the two equivalence points. (4 marks)

(Total: 20 marks)

17 SOLUTIONS TO THE EXERCISES

SOLUTION **Exercise 1 — Multiple choice questions**

1. Answer: (b) decreases the yield of NH_3 and decreases K_p.

   Justification: The reaction $N_2(g) + 3\,H_2(g) \rightleftharpoons 2\,NH_3(g)$ is exothermic ($\Delta H^\theta = -92\ kJ\,mol^{-1}$). According to Le Chatelier's principle, increasing the temperature shifts the equilibrium to the left (endothermic direction), decreasing the yield of NH_3. The equilibrium constant K_p for an exothermic reaction decreases with increasing temperature (van't Hoff equation: $\frac{d \ln K_p}{dT} = \frac{\Delta H^\theta}{RT^2} < 0$).

2. Answer: (d) no units.

   Justification: For the reaction $CH_3COOH(l) + C_2H_5OH(l) \rightleftharpoons CH_3COOC_2H_5(l) + H_2O(l)$, all species are pure liquids. Their molar concentrations (density/molar mass) are constant at constant temperature and are incorporated into the equilibrium constant. The thermodynamic equilibrium constant K (and the concentration quotient K_c for this homogeneous liquid system) is therefore dimensionless.

3. Answer: (d) 12.0.

   Justification: NaOH is a strong base, fully dissociated: $[OH^-] = 0.010\ mol\,dm^{-3}$. $pOH = -\log_{10}(0.010) = 2.00$. At 25 °C, $K_w = 1.0 \times 10^{-14}$, so $pH + pOH = 14.00$, giving $pH = 14.00 - 2.00 = 12.00$.

4. Answer: (b) the zinc electrode (anode).

   Justification: In the cell notation $Zn(s)|Zn^{2+}(aq)||Cu^{2+}(aq)|Cu(s)$, the left-hand electrode is the anode where oxidation occurs: $Zn(s) \longrightarrow Zn^{2+}(aq) + 2\,e^-$. The right-hand electrode is the cathode where reduction occurs: $Cu^{2+}(aq) + 2\,e^- \longrightarrow Cu(s)$.

SOLUTION **Exercise 2 — True or false**

1. False. At dynamic equilibrium, the forward and reverse reactions continue to occur at equal rates; they have not stopped. The concentrations of reactants and products remain constant because the rates are equal, not because the reactions have ceased.

2. False. A catalyst increases the rate of both the forward and reverse reactions equally by providing an alternative reaction pathway with lower activation energy. It does not change the equilibrium constant or the equilibrium yield; it only helps the system reach equilibrium faster.

3. True. By international convention (IUPAC), the standard hydrogen electrode (SHE)

$Pt|H_2(g, 1\,bar)|H^+(aq, 1\,mol\,dm^{-3})$ is assigned a standard electrode potential of exactly 0.00 V at 298 K (25 °C).

4. False. NH_4Cl is the salt of a weak base (NH_3, $K_b = 1.8 \times 10^{-5}$) and a strong acid (HCl). In water, NH_4^+ hydrolyses: $NH_4^+(aq) + H_2O(l) \rightleftharpoons NH_3(aq) + H_3O^+(aq)$, producing an acidic solution with pH < 7 at 25 °C.

5. True. The common ion effect: adding NaCl provides a common Cl^- ion, shifting the equilibrium $AgCl(s) \rightleftharpoons Ag^+(aq) + Cl^-(aq)$ to the left (Le Chatelier's principle), decreasing the solubility of AgCl compared to its solubility in pure water.

SOLUTION **Exercise 3 — Definitions**

1. Dynamic equilibrium: A state in a reversible reaction occurring in a closed system at constant temperature where the rate of the forward reaction equals the rate of the reverse reaction, so the concentrations (or partial pressures) of reactants and products remain constant over time. Example: $N_2(g) + 3\,H_2(g) \rightleftharpoons 2\,NH_3(g)$ in a closed vessel at constant temperature and pressure.

2. Le Chatelier's principle: If a system at equilibrium is subjected to a change in concentration, temperature, pressure (or volume), the system will shift its equilibrium position to counteract the effect of the change. Example: Increasing the total pressure on $N_2(g) + 3\,H_2(g) \rightleftharpoons 2\,NH_3(g)$ (4 mol gas → 2 mol gas) shifts the equilibrium to the right, increasing the yield of NH_3.

3. Standard electrode potential, E^θ: The potential difference (electromotive force) of a half-cell relative to the standard hydrogen electrode (SHE) when all species are in their standard states: temperature 298 K (25 °C), pressure 1 bar for gases, concentration 1 mol dm^{-3} for solutions. Example: $E^\theta(Cu^{2+}/Cu) = +0.34$ V.

4. Brønsted–Lowry acid: A proton (H^+) donor. Example: HCl donates a proton to water: $HCl + H_2O \longrightarrow H_3O^+ + Cl^-$. Lewis base: An electron-pair donor. Example: NH_3 donates its lone pair to H^+ to form NH_4^+.

5. Buffer solution: A solution that resists changes in pH when small amounts of acid or base are added, or upon dilution. It typically consists of a weak acid and its conjugate base (acidic buffer, e.g., CH_3COOH/CH_3COONa) or a weak base and its conjugate acid (alkaline buffer, e.g., NH_3/NH_4Cl).

6. Solubility product, K_{sp}: The equilibrium constant for the dissolution of a sparingly soluble ionic solid in water, expressed as the product of the equilibrium concentrations of the ions raised to the power of their stoichiometric coefficients in the balanced dissolution equation. Example: For $AgCl(s) \rightleftharpoons Ag^+(aq) + Cl^-(aq)$, $K_{sp} = [Ag^+][Cl^-]$; for $Mg(OH)_2(s) \rightleftharpoons Mg^{2+}(aq) + 2\,OH^-(aq)$, $K_{sp} = [Mg^{2+}][OH^-]^2$.

SOLUTION **Exercise 4 — Direct calculations**

1. (i) 0.050 mol dm^{-3} HCl: Strong acid, fully dissociated: $[H^+]$ = 0.050 mol dm^{-3}. pH = $-\log_{10}(0.050) = 1.30$.

   (ii) 0.050 mol dm^{-3} NaOH: Strong base, fully dissociated: $[OH^-]$ = 0.050 mol dm^{-3}. pOH = $-\log_{10}(0.050) = 1.30$. At 25 °C, pH = 14.00 − pOH = 12.70.

2. For the equilibrium $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$ at a certain temperature:

$$K_c = \frac{[PCl_3][Cl_2]}{[PCl_5]} = \frac{(0.40)(0.40)}{0.10} = 1.6$$

   Units: $\frac{(\text{mol dm}^{-3})(\text{mol dm}^{-3})}{\text{mol dm}^{-3}} = \text{mol dm}^{-3}$.

3. Daniell cell: $Zn(s)|Zn^{2+}(aq)||Cu^{2+}(aq)|Cu(s)$. Standard e.m.f. $E^\theta_{cell} = E^\theta_{cathode} - E^\theta_{anode} = E^\theta(Cu^{2+}/Cu) - E^\theta(Zn^{2+}/Zn) = 0.34 - (-0.76) = +1.10$ V. Positive electrode (cathode): copper electrode (where reduction $Cu^{2+} + 2\,e^- \longrightarrow Cu$ occurs).

SOLUTION **Exercise 5 — Decomposition of phosphorus pentachloride**

1. ICE table (amounts in moles):

	$PCl_5(g)$	$PCl_3(g)$	$Cl_2(g)$
Initial	1.00	0	0
Change	−0.80	+0.80	+0.80
Equilibrium	0.20	0.80	0.80

2. Volume of vessel = 2.00 dm^3. Equilibrium concentrations:

$$[PCl_5] = \frac{0.20}{2.00} = 0.10\,\text{mol dm}^{-3}, \quad [PCl_3] = \frac{0.80}{2.00} = 0.40\,\text{mol dm}^{-3}, \quad [Cl_2] = \frac{0.80}{2.00} = 0.40\,\text{mol dm}^{-3}.$$

3. $K_c = \frac{[PCl_3][Cl_2]}{[PCl_5]} = \frac{(0.40)(0.40)}{0.10} = 1.6\,\text{mol dm}^{-3}$.

4. Decreasing the volume at constant temperature increases the total pressure. According to Le Chatelier's principle, the equilibrium shifts to the side with fewer moles of gas to counteract the pressure increase. Here, the left side has 1 mole of gas, the right side has 2 moles of gas. Therefore, the equilibrium shifts to the left (towards PCl_5), decreasing the yield of PCl_3 and Cl_2. The value of K_c remains unchanged (temperature constant).

SOLUTION **Exercise 6 — Redox titration of iron(II) with permanganate**

1. Half-equations: Oxidation: $Fe^{2+}(aq) \longrightarrow Fe^{3+}(aq) + e^-$ (×5) Reduction: $MnO_4^-(aq) + 8\,H^+(aq) + 5\,e^- \longrightarrow Mn^{2+}(aq) + 4\,H_2O(l)$

   Overall balanced ionic equation:

   $$MnO_4^-(aq) + 5\,Fe^{2+}(aq) + 8\,H^+(aq) \longrightarrow Mn^{2+}(aq) + 5\,Fe^{3+}(aq) + 4\,H_2O(l)$$

2. Molar mass of $FeSO_4 \cdot 7\,H_2O = 56 + 32 + 4 \times 16 + 7 \times (2 + 16) = 278\ g\,mol^{-1}$. Amount of $FeSO_4 \cdot 7\,H_2O = \frac{1.52\ g}{278\ g\,mol^{-1}} = 5.468 \times 10^{-3}$ mol. Amount of $Fe^{2+} = 5.468 \times 10^{-3}$ mol. Stoichiometry: 1 mol $MnO_4^- \equiv 5$ mol Fe^{2+}. Amount of MnO_4^- required $= \frac{5.468 \times 10^{-3}}{5} = 1.0936 \times 10^{-3}$ mol. Volume of 0.0200 mol dm^{-3} $KMnO_4 = \frac{1.0936 \times 10^{-3}\ mol}{0.0200\ mol\,dm^{-3}} = 0.05468\ dm^3 =$ **54.7** cm^3.

3. Colour change at end-point: The intense purple colour of MnO_4^- disappears (solution becomes colourless or very pale pink due to Mn^{2+}). The first permanent faint pink colour (due to a tiny excess of MnO_4^-) signals the equivalence point.

   Indicator: No external indicator is needed; MnO_4^- is intensely coloured and acts as its own indicator (self-indicating).

SOLUTION **Exercise 7 — pH of a buffer solution**

1. K_a expression for CH_3COOH: $K_a = \frac{[H^+][CH_3COO^-]}{[CH_3COOH]}$.

   Derivation of Henderson–Hasselbalch equation:

   $$[H^+] = K_a \frac{[CH_3COOH]}{[CH_3COO^-]}$$

   Taking $-\log_{10}$:

   $$pH = pK_a + \log_{10} \frac{[CH_3COO^-]}{[CH_3COOH]}$$

   Since CH_3COONa is a strong electrolyte, it fully dissociates: $[CH_3COO^-] \approx$ [salt]. The weak acid is only slightly dissociated, so $[CH_3COOH] \approx$ [acid].

2. $pK_a = -\log_{10}(1.8 \times 10^{-5}) = 4.7447$.

   $$pH = 4.7447 + \log_{10}\left(\frac{0.15}{0.10}\right) = 4.7447 + \log_{10}(1.5) = 4.7447 + 0.1761 = \mathbf{4.92}$$

3. Reaction with added NaOH: $CH_3COOH(aq) + OH^-(aq) \longrightarrow CH_3COO^-(aq) + H_2O(l)$. Amount of NaOH added = 0.010 mol in 1.00 dm^3. New amounts: acid = 0.10 − 0.010 =

0.090 mol; salt = 0.15 + 0.010 = 0.16 mol. Concentrations (volume ≈ constant): [acid] = 0.090 M, [salt] = 0.16 M.

$$\text{pH} = 4.7447 + \log_{10}\left(\frac{0.16}{0.090}\right) = 4.7447 + \log_{10}(1.7778) = 4.7447 + 0.2499 = \mathbf{4.99}$$

4. Adding 0.010 mol NaOH to 1.00 dm^3 pure water gives $[OH^-]$ = 0.010 M, pOH = 2.00, pH = 12.00. The buffer pH changes only from 4.92 to 4.99 (ΔpH = 0.07), demonstrating its capacity to resist pH change effectively.

SOLUTION **Exercise 8 — Solubility and common ion effect**

1. AgCl: $AgCl(s) \rightleftharpoons Ag^+(aq) + Cl^-(aq)$. Let s = molar solubility in mol dm^{-3}. $K_{sp} = s^2 = 1.8 \times 10^{-10} \Rightarrow s = \sqrt{1.8 \times 10^{-10}} = \mathbf{1.34 \times 10^{-5}}$ mol dm^{-3}.

   $Mg(OH)_2$: $Mg(OH)_2(s) \rightleftharpoons Mg^{2+}(aq) + 2\,OH^-(aq)$. Let s = molar solubility. $K_{sp} = [Mg^{2+}][OH^-]^2 = s(2s)^2 = 4s^3 = 1.8 \times 10^{-11}$. $s^3 = 4.5 \times 10^{-12} \Rightarrow s = \sqrt[3]{4.5 \times 10^{-12}} = \mathbf{1.65 \times 10^{-4}}$ mol dm^{-3}.

2. In 0.10 M NaCl, $[Cl^-] \approx 0.10$ M (from complete dissociation of NaCl; contribution from AgCl negligible). Let s' = solubility in 0.10 M NaCl. $K_{sp} = [Ag^+][Cl^-] = s' \times 0.10 = 1.8 \times 10^{-10} \Rightarrow s' = \mathbf{1.8 \times 10^{-9}}$ mol dm^{-3}.

   Explanation: The common ion Cl^- from NaCl increases $[Cl^-]$, shifting the equilibrium $AgCl(s) \rightleftharpoons Ag^+ + Cl^-$ to the left (Le Chatelier's principle), decreasing the solubility of AgCl by a factor of ≈ 7500.

3. Mixing equal volumes halves all concentrations: $[Ag^+] = \frac{2.0\times10^{-4}}{2} = 1.0 \times 10^{-4}$ M. $[Cl^-] = \frac{2.0\times10^{-4}}{2} = 1.0 \times 10^{-4}$ M. Ionic product $Q = [Ag^+][Cl^-] = (1.0 \times 10^{-4})^2 = 1.0 \times 10^{-8}$. Since $Q = 1.0 \times 10^{-8} > K_{sp} = 1.8 \times 10^{-10}$, the solution is supersaturated and a precipitate of AgCl will form until $Q = K_{sp}$.

SOLUTION **Exercise 9 — The Haber process**

Mark scheme and examiner expectations:

1. Expression for K_p: $K_p = \frac{(p_{NH_3})^2}{(p_{N_2})(p_{H_2})^3}$.

   Units: Pressure^{-2} (e.g., atm^{-2} or Pa^{-2}). (3 marks: 1 for expression, 1 for correct powers, 1 for units)

2. (i) Increasing pressure: Equilibrium shifts to the side with fewer moles of gas (4 moles reactants → 2 moles products). Yield of NH_3 increases. (2 marks: shift direction +

reason)

(ii) Increasing temperature: Reaction is exothermic (ΔH^θ = −92 kJ mol^{-1}). Equilibrium shifts to the left (endothermic direction). Yield of NH_3 decreases. (2 marks: shift direction + reason)

3. (i) Effect on K_p of increasing pressure: No effect. K_p depends only on temperature. (1 mark)

   (ii) Effect on K_p of increasing temperature: Decreases K_p. For an exothermic reaction (ΔH^θ < 0), the van't Hoff equation shows $\frac{d \ln K_p}{dT}$ < 0, so K_p decreases as temperature increases. (2 marks: direction + justification)

4. Iron catalyst: Used to increase the rate of reaction (lowers activation energy). Effect on equilibrium position: None (catalyst does not change K_p or equilibrium composition). Effect on rate: Increases the rate at which equilibrium is attained, allowing lower temperatures to be used for a given rate. (3 marks: 1 for purpose, 1 for equilibrium position, 1 for rate)

5. Compromise conditions (industrial practice): - High pressure favours yield but is expensive/dangerous; 200–250 atm is a practical limit. - Low temperature favours yield (exothermic) but slows rate; 400–450 °C gives a reasonable rate with acceptable yield (≈ 15–20%). - Iron catalyst (with promoters K_2O, Al_2O_3) allows operation at lower temperature for same rate, improving yield.

   Examiner expects: Mention of yield vs rate trade-off, cost/safety of high pressure, catalyst role. (3 marks)

6. Calculation of K_p at equilibrium: p_{N_2} = 20 atm, p_{H_2} = 60 atm, p_{NH_3} = 40 atm.

$$K_p = \frac{(40)^2}{(20)(60)^3} = \frac{1600}{20 \times 216\,000} = \frac{1600}{4\,320\,000} = \mathbf{3.70 \times 10^{-4}}\ \text{atm}^{-2}$$

(4 marks: 1 for substitution, 1 for arithmetic, 1 for correct value, 1 for units)

Total: 20 marks.

SOLUTION **Exercise 10 — Electrode potentials and electrochemical cells**

Mark scheme and examiner expectations:

1. Diagram description for measuring $E^\theta(Cu^{2+}/Cu)$: - SHE: Platinum electrode in 1 mol dm^{-3} H^+ (e.g., H_2SO_4) with H_2 gas at 1 bar bubbling over it. - Cu^{2+}/Cu half-cell: Copper rod in 1 mol dm^{-3} Cu^{2+} (e.g., $CuSO_4$). - Salt bridge (e.g., saturated KNO_3 in agar gel) connecting the two solutions to complete the circuit and minimise liquid junction potential. - High-resistance voltmeter (or potentiometer) connecting the two electrodes.

Standard conditions: 298 K (25 °C), 1 bar for gases, 1 mol dm^{-3} for all solutions. (5 marks: 2 for diagram labels, 2 for description, 1 for conditions)

2. Daniell cell diagram: – Left half-cell (anode): Zn electrode in Zn^{2+} (oxidation: $Zn \longrightarrow Zn^{2+} + 2\,e^-$). – Right half-cell (cathode): Cu electrode in Cu^{2+} (reduction: $Cu^{2+} + 2\,e^- \longrightarrow Cu$). – Salt bridge (e.g., KNO_3) connecting solutions. – External circuit with voltmeter; electrons flow from Zn (negative) to Cu (positive).

   Cell notation (IUPAC): $Zn(s)|Zn^{2+}(aq, 1\,M)||Cu^{2+}(aq, 1\,M)|Cu(s)$. (4 marks: 2 for diagram, 1 for electron flow, 1 for notation)

3. Standard e.m.f.: $E^\theta_{cell} = E^\theta_{cathode} - E^\theta_{anode} = E^\theta(Cu^{2+}/Cu) - E^\theta(Zn^{2+}/Zn) = 0.34 - (-0.76) = +1.10\ V$.

   Overall cell reaction: $Zn(s) + Cu^{2+}(aq) \longrightarrow Zn^{2+}(aq) + Cu(s)$. (3 marks: 1 for calculation, 1 for sign, 1 for equation)

4. Can Cu displace Ag from Ag^+ solution? Compare standard reduction potentials: $E^\theta(Ag^+/Ag) = +0.80\ V$, $E^\theta(Cu^{2+}/Cu) = +0.34\ V$. For the reaction $Cu(s) + 2\,Ag^+(aq) \longrightarrow Cu^{2+}(aq) + 2\,Ag(s)$: $E^\theta_{cell} = E^\theta(Ag^+/Ag) - E^\theta(Cu^{2+}/Cu) = 0.80 - 0.34 = +0.46\ V > 0$. Since $E^\theta_{cell} > 0$, the reaction is thermodynamically spontaneous. Yes, copper displaces silver.

   Equation: $Cu(s) + 2\,Ag^+(aq) \longrightarrow Cu^{2+}(aq) + 2\,Ag(s)$. (4 marks: 1 for comparison, 1 for E^θ_{cell} calculation, 1 for conclusion, 1 for equation)

5. Effect of decreasing $[Cu^{2+}]$ on $E(Cu^{2+}/Cu)$: Nernst equation at 298 K: $E = E^\theta - \frac{0.059}{2}\log\frac{1}{[Cu^{2+}]} = E^\theta + \frac{0.059}{2}\log[Cu^{2+}]$. Decreasing $[Cu^{2+}]$ makes $\log[Cu^{2+}]$ more negative, so E becomes less positive (decreases). (2 marks: 1 for direction, 1 for explanation)

6. Reasons for using KNO_3 (not KCl) as salt bridge with $AgNO_3$ half-cell: 1. Cl^- from KCl would react with Ag^+ in the half-cell or salt bridge to form insoluble AgCl(s), blocking the bridge and altering the electrode potential. 2. K^+ and NO_3^- have very similar ionic mobilities, minimising the liquid junction potential at the salt bridge/solution interfaces. (2 marks: 1 each)

Total: 20 marks.

SOLUTION **Exercise 11 — Acid–base equilibria and titration curves**

Mark scheme and examiner expectations:

1. Weak acid (Brønsted–Lowry): A substance that partially donates protons to water, establishing an equilibrium with its conjugate base. (1 mark)

   Ionisation equation: $CH_3COOH(aq) + H_2O(l) \rightleftharpoons CH_3COO^-(aq) + H_3O^+(aq)$ (or sim–

plified: $CH_3COOH(aq) \rightleftharpoons H^+(aq) + CH_3COO^-(aq)$). (1 mark)

2. Calculation of pH of 0.100 M CH_3COOH ($K_a = 1.8 \times 10^{-5}$): Approximation: $[H^+] \ll c$, so $c - [H^+] \approx c$. $K_a = \frac{[H^+]^2}{c} \Rightarrow [H^+] = \sqrt{K_a c} = \sqrt{(1.8 \times 10^{-5})(0.100)} = \sqrt{1.8 \times 10^{-6}} = 1.34 \times 10^{-3}$ mol dm^{-3}. pH = $-\log_{10}(1.34 \times 10^{-3})$ = **2.87**.

   Justification of approximation: $\frac{[H^+]}{c} = \frac{1.34 \times 10^{-3}}{0.100} = 1.34\% < 5\%$, so approximation is valid. (4 marks: 1 for approximation statement, 1 for formula, 1 for $[H^+]$, 1 for pH)

3. Titration curve sketch (25.0 cm^3 0.100 M CH_3COOH vs 0.100 M NaOH): - Axes: pH (vertical, 0–14) vs Volume NaOH added (horizontal, 0–50 cm3). - Initial point (V = 0): pH = 2.87. - Buffer region: gradual rise. - Half-equivalence point: V = 12.5 cm^3, pH = pK_a = 4.74. - Equivalence point: V = 25.0 cm^3, pH ≈ 8.7 (vertical jump ≈ pH 7–10). - After equivalence: gradual rise approaching pH of excess NaOH (e.g., pH ≈ 12.6 at V = 50 cm^3).

   Examiner expects: Correct S-shape with asymmetric vertical jump, labelled axes, initial pH, half-equivalence point, equivalence point volume and pH. (5 marks)

4. Equivalence volume: Moles CH_3COOH = 0.0250 dm^3 × 0.100 mol dm^{-3} = 0.00250 mol. 1 : 1 stoichiometry ⇒ moles NaOH = 0.00250 mol. Volume = $\frac{0.00250 \text{ mol}}{0.100 \text{ mol dm}^{-3}}$ = 0.0250 dm^3 = **25.0** cm^3. (1 mark)

   pH > 7 at equivalence: Salt formed is CH_3COONa, the conjugate base of a weak acid. CH_3COO^- hydrolyses: $CH_3COO^-(aq) + H_2O(l) \rightleftharpoons CH_3COOH(aq) + OH^-(aq)$, making solution alkaline. (2 marks)

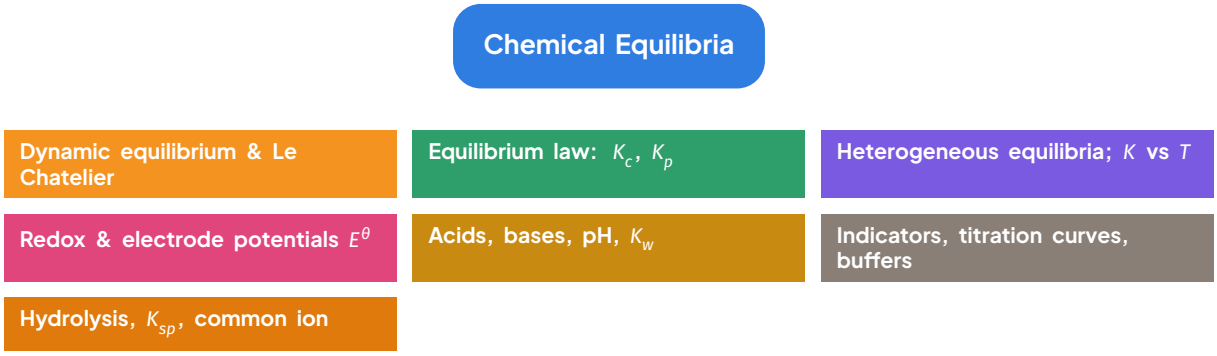
5. Suitable indicator: Phenolphthalein (transition range 8.3–10.0, colourless to pink). Justification: The pH jump at equivalence (approx 7–10) falls entirely within phenolphthalein's range. Methyl orange (3.1–4.4) changes colour in the buffer region, well before equivalence, giving a large titration error. (2 marks)

6. Sodium carbonate (Na_2CO_3) titration with HCl (diprotic base): $CO_3^{2-}(aq) + H^+(aq) \longrightarrow HCO_3^-(aq)$ (1st equivalence point, pH ≈ 8.3) $HCO_3^-(aq) + H^+(aq) \longrightarrow H_2CO_3(aq) \longrightarrow CO_2(g) + H_2O(l)$ (2nd equivalence point, pH ≈ 3.7)

   Curve sketch: Two distinct steps (two buffer regions). First vertical jump at V_1 (1st eq. pt., pH ≈ 8.3), second vertical jump at $V_2 = 2V_1$ (2nd eq. pt., pH ≈ 3.7). (4 marks: 2 for equations, 2 for sketch with two equivalence points labelled)

Total: 20 marks.

18 SUMMARY SHEET



Mind map of the chapter Equilibria: seven interconnected themes, from gaseous and heterogeneous equilibria through redox systems to acid–base equilibria in aqueous solution.

FICHE BILAN **L'essentiel à connaître pour l'examen**

Key definitions

- **Reversible reaction**: a reaction that can proceed in both the forward and the reverse directions, denoted by ⇌; it never goes to completion.
- **Dynamic equilibrium**: the state reached in a closed system when the forward and reverse reactions proceed at equal rates; macroscopic properties (concentration, colour, pressure) are constant, but particles keep reacting in both directions.
- **Le Chatelier's principle**: when a constraint (change of concentration, pressure or temperature) is imposed on a system at equilibrium, the system adjusts itself so as to oppose (annul the effect of) the constraint.
- **Equilibrium constant**: for $aA + bB \rightleftharpoons cC + dD$,

$$K_c = \frac{[C]^c[D]^d}{[A]^a[B]^b} \quad \text{and} \quad K_p = \frac{p_C^c\, p_D^d}{p_A^a\, p_B^b},$$

related by $K_p = K_c(RT)^{\Delta n}$, where $\Delta n = (c + d) - (a + b)$ is the change in the number of moles of gas.

- **Standard electrode potential** E^θ: the emf of a half-cell measured against the standard hydrogen electrode (SHE) under standard conditions: temperature 298 K, ion concentration 1 mol dm^{-3}, gas pressure 1 atm (101 kPa).
- **pH**: pH = $-\log_{10}[H_3O^+]$; the ionic product of water is $K_w = [H_3O^+][OH^-] = 1.0 \times 10^{-14}$ mol^2 dm^{-6} at 298 K, so that pH + pOH = 14 at this temperature.

- **Buffer solution**: a solution that resists a change of pH on addition of small amounts of acid or base; it consists of a weak acid and one of its salts (acidic buffer) or a weak base and one of its salts (alkaline buffer).
- **Solubility product** K_{sp}: the product of the concentrations of the ions of a sparingly soluble salt, each raised to its stoichiometric power, in a saturated solution at a given temperature.

Laws and formulas to know

- Reaction quotient Q (same expression as K, at any instant): if $Q < K$ the reaction proceeds in the forward direction; if $Q > K$ it proceeds in the reverse direction; if $Q = K$ the system is at equilibrium.
- Effect of temperature on K: K increases with temperature for an endothermic forward reaction and decreases for an exothermic one. Quantitatively (van 't Hoff equation):

$$\ln \frac{K_2}{K_1} = -\frac{\Delta H^\theta}{R}\left(\frac{1}{T_2} - \frac{1}{T_1}\right).$$

  A catalyst changes neither K nor the position of equilibrium; it only increases the rate at which equilibrium is attained.
- Cell emf: $E^\theta_{\text{cell}} = E^\theta_{\text{cathode}} - E^\theta_{\text{anode}}$ (right-hand electrode minus left-hand electrode in the conventional cell diagram); a reaction is feasible (spontaneous) under standard conditions if $E^\theta_{\text{cell}} > 0$.
- Strong acid (concentration c, fully ionised): $\text{pH} = -\log_{10} c$. Strong base at 298 K: $\text{pH} = 14 + \log_{10} c$.
- Weak acid: $K_a = \dfrac{[H_3O^+][A^-]}{[HA]} = \dfrac{[H_3O^+]^2}{c - [H_3O^+]} \approx \dfrac{[H_3O^+]^2}{c}$, giving $\text{pH} = \frac{1}{2}\left(\text{p}K_a - \log_{10} c\right)$. Weak base analogously with K_b, and $\text{p}K_a + \text{p}K_b = 14$ for a conjugate pair at 298 K.
- Acidic buffer (Henderson–Hasselbalch): $\text{pH} = \text{p}K_a + \log_{10}\dfrac{[\text{salt}]}{[\text{acid}]}$; alkaline buffer: $\text{pOH} = \text{p}K_b + \log_{10}\dfrac{[\text{salt}]}{[\text{base}]}$.
- Salt hydrolysis: a salt of a weak acid and a strong base gives $\text{pH} > 7$; a salt of a weak base and a strong acid gives $\text{pH} < 7$; a salt of a strong acid and a strong base gives $\text{pH} = 7$ (at 298 K).
- K_{sp} for A_mB_n of molar solubility s: $K_{sp} = (ms)^m(ns)^n = m^m n^n s^{m+n}$. The common ion effect decreases the solubility of a sparingly soluble salt.

Methods

- K_c/K_p calculations: write the balanced equation with state symbols; set up an ICE table (Initial moles, Change, Equilibrium moles); convert to concentrations (divide by the volume V) or to partial pressures ($p_i = x_i P_{\text{total}}$, where x_i is the mole fraction); substitute into the expression; state the units from Δn.

- Balancing redox equations (ion–electron method): split into oxidation and reduction half-equations; balance atoms other than O and H; balance O with H_2O and H with H^+; balance charge with electrons; multiply so that electrons cancel; add the half-equations and simplify.
- Titration curves: identify the four regions (initial pH, buffer region where relevant, equivalence point, excess of titrant); the vertical jump brackets the equivalence point; choose an indicator whose colour-change range falls within the vertical jump.

Common mistakes

- Including pure solids or pure liquids in K_c/K_p expressions: their "concentration" is constant and is incorporated into K.
- Forgetting the units of K_c/K_p, which depend on Δn; K is unitless only when $\Delta n = 0$.
- Claiming that a catalyst shifts the position of equilibrium: it affects only the rate of attainment, never K nor the position.
- Writing $E^{\theta}_{\text{cell}} = E^{\theta}_{\text{anode}} - E^{\theta}_{\text{cathode}}$ (wrong sign order), or multiplying E^{θ} values by stoichiometric coefficients (E^{θ} is an intensive quantity).
- Using $\text{pH} = -\log_{10} c$ for a weak acid, or forgetting that dilution of a weak acid increases its degree of ionisation (Ostwald's dilution law).
- Assuming the equivalence point of a weak acid–strong base titration is at pH 7: it lies above 7 because the anion of the weak acid is hydrolysed.
- Confusing solubility s (in mol dm^{-3}) with K_{sp}, and forgetting to raise each ion concentration to its stoichiometric power.

GCE tips

- Always write the balanced equation with state symbols before any equilibrium calculation; examiners award marks for a clearly laid-out ICE table.
- Quote Le Chatelier's principle precisely: state the imposed change, the direction of the shift, and why (which side has fewer moles of gas; which direction is endothermic).
- In electrode-potential questions, state the standard conditions, draw the cell diagram with the salt bridge, and remember that the half-cell with the more negative E^{θ} is the anode (oxidation).
- For buffers and K_{sp}, state the approximation used (e.g. $[H_3O^+] \ll c$) and check that it is valid.
- Give pH to 2 decimal places, and K values with sensible significant figures and correct units.

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